

Probing Viterbo's Conjecture

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2026

Abstract

Haim–Kislev and Ostrover disproved Viterbo's conjecture with a Lagrangian product of two regular pentagons. We investigate the geometry near their example and use finite computation to look for further structure in the systolic ratio of four-dimensional convex polytopes.

Our main theorem shows that the HKO body is a local maximizer among polytopes with exactly ten facets. Nearby equality occurs precisely under translations, positive dilations and linear symplectic transformations.

We implement, optimize and extend two capacity algorithms, based on the quadratic program of Haim–Kislev and the flow graph of Chaidez and Hutchings. We also study the first-order variation of capacity. As a side theorem, we prove an explicit capacity formula for the relative-rotation family of two regular pentagons. The proof uses an exhaustive exact computation and shows that the HKO body is a global maximizer of the systolic ratio in this family.

An empirical investigation looks for statistical patterns in geometric measurements of sampled polytopes, with the aim of turning those patterns into mathematical conjectures. We find patterns among bodies with systolic ratio below one, but do not establish that these patterns extend to the regime above one. This leaves open whether further progress requires different statistical methods or new geometrically motivated measurements.

Disclosure of AI Use

Generative artificial intelligence was used materially throughout the research and production of this thesis, principally through Codex and Claude Code agents. Agents were not limited to language polishing or executing fully specified tasks. They participated in choosing questions and methods; developing mathematical arguments, software, experiments, and prose; interpreting and criticizing results; revising work; and planning, delegating, and reviewing work by other agents. This breadth of AI use was approved by my advisor.

There was no stable division of labor by category. I am not aware of any kind of work I performed on this thesis that I did not also delegate to agents at some point. Agents and I both planned, produced, interpreted, corrected, and reviewed work, often through repeated handoffs. This overlap does not imply equal labor shares or identical contributions to every result. The research and production process, and many retained components, therefore have mixed human–AI provenance. I do not claim sole production or sole intellectual origin of every part of the thesis. Because work often passed between me and agents, I cannot reliably say who first originated every idea, passage, program, or decision.

As of 13 July 2026, I had proofread the then-current thesis and personally worked to verify its mathematical and factual claims and the overall message it presented. The manuscript was revised substantially afterwards. This statement therefore does not assert that I have completed a final review of the current candidate.

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1 Introduction

How large can a convex body's symplectic capacity be if its volume is fixed? Viterbo's conjecture proposed that a ball gives the largest value [Vit00]. For the Ekeland–Hofer–Zehnder capacity in dimension four, the proposed inequality was

$$\text{sys}(K) := \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)} \leq 1.$$

The quantity on the left is called the systolic ratio. Capacity scales by the square of a dilation and four-dimensional volume by its fourth power, so this ratio compares shape independently of overall size. Translations and linear symplectic maps also preserve it.

The inequality is false. Haim–Kislev and Ostrover found a counterexample formed from two regular pentagons in complementary Lagrangian planes [HO26]. Writing the coordinates as $(q, p) \in \mathbb{R}^2 \oplus \mathbb{R}^2$, one representative is

$$K_{\text{HKO}} = P_5 \times R(-\pi/2)P_5, \quad \text{sys}(K_{\text{HKO}}) = \frac{3 + \sqrt{5}}{5} > 1.$$

Here P_5 is a centred regular pentagon with a vertex on the positive horizontal axis, and $R(\theta)$ is planar rotation. A product placed in the q - and p -planes is called a Lagrangian product. The relative position of its factors affects capacity even though it does not affect volume.

The counterexample leaves several geometric questions. Is the HKO body a local maximum of the systolic ratio among convex bodies? Are there other families with large ratios? Can the computations reveal geometric relationships involving capacity? This thesis investigates these questions through local analysis, finite algorithms and data science. This is the sense in which we continue to probe Viterbo's conjecture: although its proposed bound is false, the geometric relationship between symplectic capacity and volume remains to be understood.

The contributions grew out of a broad computational investigation. We developed methods for evaluating capacity and used them to explore examples, following up on questions that the computations suggested. This led to several separate results, from a local maximality theorem to algorithms and statistical observations. The following overview explains what each part contributes to that investigation.

1.1 Capacities, dynamics and convex geometry

Symplectic capacities measure constraints on embeddings that volume alone cannot detect. Two basic examples are the Gromov width c_G , defined by the largest symplectic ball that embeds into a domain, and the cylindrical capacity c_Z , defined by the smallest symplectic cylinder into which the domain embeds. With the usual normalization, every symplectic capacity lies between these two [Edt24]. The EHZ capacity has a different description: on convex bodies it is the least action of a closed characteristic. Thus comparing capacities connects an embedding question to a question about Hamiltonian trajectories.

The conjecture that all normalized capacities agree on convex domains was stronger than Viterbo's volume bound. Indeed, the Gromov width satisfies that bound directly by volume preservation. The HKO example therefore also separates the EHZ capacity from the Gromov width [HO26]. The equality of EHZ and cylindrical capacity on general convex domains in four dimensions is still listed as open in [HO26, Remark 1.4(v)]. Edtmair instead characterizes cylindrical capacity on smooth star-shaped domains in $\mathbb{R}^4$ with dynamically convex boundary by the least area of a disk-like global surface of section, and proves coincidence of all normalized capacities for convex domains sufficiently C^3 -close to the ball [Edt24]. A global surface of section is a surface whose interior is transverse to the flow and is met in both forward and backward time by every trajectory outside its boundary. These results illustrate how information about the flow can determine an embedding obstruction.

There is also a direct connection to a central problem in convex geometry. For a centrally symmetric convex body $L \subset \mathbb{R}^n$ and its polar L° , Artstein–Avidan, Karasev and Ostrover prove $c_{\text{EHZ}}(L \times L^\circ) = 4$ [AKO14]. Viterbo’s proposed inequality on these particular products is consequently equivalent to the lower bound $\text{vol}(L) \text{vol}(L^\circ) \geq 4^n/n!$ in the symmetric Mahler conjecture. The general counterexample does not invalidate this equivalence for the restricted family. Capacity–volume questions therefore retain significance beyond the original universal bound.

Local maximality has a dynamical precedent as well. Abbondandolo and Benedetti prove that Zoll contact forms, whose Reeb orbits are all closed with the same minimal period, locally maximize the contact systolic ratio [AB23]. Their results include the volume–capacity inequality near the ball. The HKO body presents a different setting: it has corners and a ratio larger than the ball’s. Our local theorem studies how such a nonsmooth example behaves under perturbations of its supporting facets.

1.2 A local theorem around the counterexample

Our main result partially answers the first question: every sufficiently Hausdorff-close ten-facet polytope has ratio at most that of K_{HKO} . In a smaller neighborhood, equality occurs precisely for translations, positive dilations and linear symplectic images of K_{HKO} ; see Theorem 7.1. Thus a small change of shape that is not one of these symmetries lowers the ratio. Allowing facets to appear or disappear is a different local problem, which the theorem leaves open.

The proof uses upper bounds that agree with the systolic ratio at HKO. For each nearby direction, at least one of these bounds decreases. It follows that the ratio itself decreases. The finite task is to construct enough such bounds and check their derivatives. Feasible weights in Haim–Kislev’s capacity formula provide the bounds. An algebraic computation establishes the derivative conditions; the proof explains why those conditions imply the geometric local maximum.

1.3 From boundary dynamics to finite calculations

For a smooth convex body, the EHZ capacity is the least action of a closed characteristic on its boundary. Polytopes have corners, so the relevant curves are generalized characteristics. Their velocities are governed by the normals of the facets they meet. Haim–Kislev’s theorem makes the capacity computable by a finite optimization over facet orders and nonnegative weights [Hai19].

We expand the proof in Haim–Kislev’s original paper into a self-contained presentation for a master’s-level reader. We explain the dual action principle and the existence of a minimizing curve with a finite list of distinct facet directions.

For Lagrangian products, we derive a further reduction from the bilinear structure of the objective: some maximizing word uses at most three facets from each factor. This gives a smaller finite search for the capacity, without classifying every minimizing orbit.

We pursued the flow-graph method of Chaidez and Hutchings [CH21] as an alternative way to compute capacity. Its performance in our tests made it less useful for large searches, but it provides a reference implementation for cross-checking the quadratic-program method. Agreement between the two gives additional evidence that the implementations compute the intended quantity.

The same finite methods yield a side result for the family $P_5 \times R(\theta)P_5$: its systolic ratio is

$$\frac{5 + 2\sqrt{5}}{10} \sec^2 d(\theta),$$

where $d(\theta)$ is the distance to the nearest multiple of $\pi/5$. An exhaustive comparison proves the formula, and its maximum is the HKO value. This determines the whole relative-rotation

profile while keeping both factors regular.

1.4 Looking for mathematical structure in data

We also use data science to seek geometric relationships involving capacity. The aim is to discover statistical patterns, interpret them geometrically, and turn them into conjectures that can be proved or refuted. The experiments find patterns below systolic ratio one, but do not establish their extension above one. One geometric interpretation yields an exact edge-width formula for a measured quantity on Lagrangian products; relating it to capacity remains open.

1.5 How to read the thesis

The order follows the passage from geometry to computation and then to applications. Sections 2–4 explain how nonsmooth boundary dynamics become a finite quadratic program. Section 5 presents the alternative flow-graph method, and Section 6 develops first-order variation. These foundations prepare the local theorem in Section 7 and the further investigations in Sections 8 and 9.

A reader primarily interested in the HKO theorem can start with its statement and proof strategy in Section 7, returning to Sections 2–4 for the capacity formula and to Section 6 for variation. The flow-graph and data-science chapters are not prerequisites for that proof. Section 8 describes the empirical investigation and can be read independently of the HKO certificate; Section 9 gives the separate theorem for rotated pentagons. The later sections document visualizations, numerical methods, code and data, and reflect on the use of AI. The appendices contain computational detail that can be consulted when checking or reproducing the results.

2 Preliminaries

A polytope has edges and vertices where the smooth definition of a characteristic no longer applies. We first extend that definition to convex bodies with nonsmooth boundaries. We then introduce a variational description of capacity in terms of closed curves that need not lie on the boundary. This will allow us to modify curves when deriving the finite capacity formula.

All convex bodies below are compact convex subsets of $\mathbb{R}^4$ with non-empty interior. Unless stated otherwise, the origin lies in the interior.

2.1 Convex and symplectic notation

We write $\mathbb{R}^4$ with coordinates (q_1, q_2, p_1, p_2) . The standard complex structure is

$$J_0 = \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix},$$

and the standard symplectic form is

$$\omega_0(u, v) = \langle J_0 u, v \rangle = \sum_{i=1}^2 (u_{q_i} v_{p_i} - u_{p_i} v_{q_i}).$$

We use the standard primitive

$$\lambda_0 = \frac{1}{2} \sum_{i=1}^2 (q_i dp_i - p_i dq_i).$$

Thus $d\lambda_0 = \omega_0$ and $\lambda_0|_x(v) = \frac{1}{2}\langle -J_0v, x \rangle$. For an absolutely continuous closed curve $\gamma: [0, T] \rightarrow \mathbb{R}^4$, its action is

$$A(\gamma) := \int_{\gamma} \lambda_0 = \frac{1}{2} \int_0^T \langle -J_0\dot{\gamma}(t), \gamma(t) \rangle dt. \quad (1)$$

This action is invariant under orientation-preserving reparametrization. If γ bounds a smooth map $u: D^2 \rightarrow \mathbb{R}^4$, then Stokes' theorem gives

$$A(\gamma) = \int_{D^2} u^* \omega_0.$$

Thus action is the signed symplectic area enclosed by the curve. For Reeb orbits this is the quantity that the EHZ capacity minimizes.

Let $K \subset \mathbb{R}^4$ be a smooth convex body with $0 \in \text{int}(K)$. At $x \in \partial K$, the restriction of ω_0 to the three-dimensional space $T_x\partial K$ has a one-dimensional kernel,

$$\ell_x = \{v \in T_x\partial K : \omega_0(v, w) = 0 \text{ for all } w \in T_x\partial K\}.$$

This is the characteristic line. A *closed characteristic* is a closed curve on ∂K whose velocity lies in the positive characteristic direction. We choose the positive direction by requiring $\lambda_0(\dot{\gamma}) > 0$.

The restriction $\alpha = \lambda_0|_{\partial K}$ is a contact form. Its Reeb vector field R is the contact-normalized characteristic direction,

$$\alpha(R_x) = 1, \quad \omega_0(R_x, v) = 0 \quad \text{for all } v \in T_x\partial K.$$

A Reeb orbit is a periodic integral curve of R . It is the same geometric object as a closed characteristic, with the parametrization fixed by $\lambda_0(\dot{\gamma}) = 1$. Consequently, if γ is a Reeb orbit on $[0, T]$, then

$$A(\gamma) = \int_0^T \lambda_0(\dot{\gamma}(t)) dt = T.$$

This is the sense in which the action of a contact-normalized Reeb orbit is also its period.

2.2 Convex Hamiltonians

The smooth characteristic line field has a convex-analytic formulation that continues to make sense when ∂K is not smooth. For a convex body K containing the origin, its support and gauge functions are

$$h_K(y) := \max_{x \in K} \langle x, y \rangle, \quad g_K(x) := \inf\{r > 0 : x \in rK\}.$$

The boundary is the level set $g_K = 1$. We use the convex, 2-homogeneous Hamiltonian

$$H_K(x) := g_K(x)^2.$$

For a convex function H , its subdifferential at x is

$$\partial H(x) = \{y : H(z) \geq H(x) + \langle y, z - x \rangle \text{ for every } z\}.$$

When H is differentiable, this set consists only of $\nabla H(x)$.

Definition 2.1 (Contact-normalized generalized characteristic). *A contact-normalized generalized characteristic on ∂K is a curve $\gamma \in W^{1,2}([0, T], \mathbb{R}^4)$, with $T > 0$ and $\gamma(0) = \gamma(T)$, whose image lies in ∂K and which satisfies*

$$-J_0\dot{\gamma}(t) \in \partial H_K(\gamma(t)) \quad \text{for almost every } t \in [0, T]. \quad (2)$$

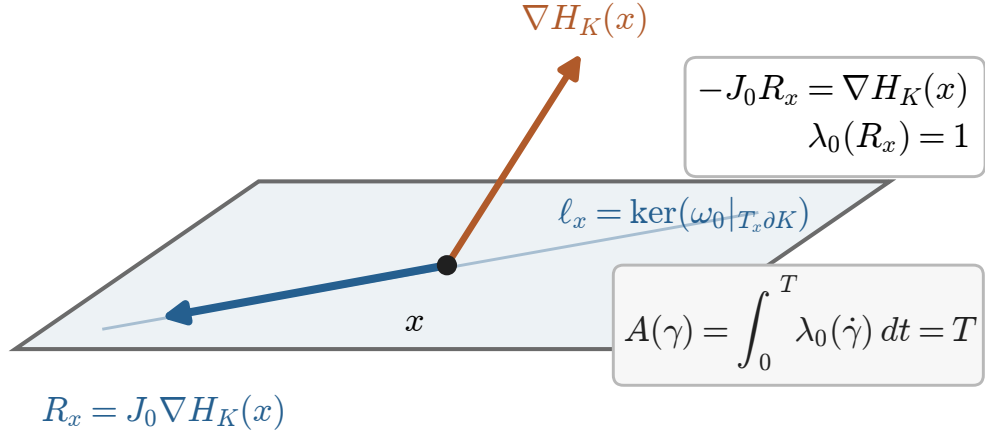


Figure 1: The characteristic and contact normalizations at a smooth boundary point. This is a dimension-reduced tangent-space schematic: it records that $R_x = J_0 \nabla H_K(x)$ spans the characteristic line inside $T_x \partial K$, but projected angles and lengths carry no meaning. The contact condition $\lambda_0(R_x) = 1$ turns elapsed time into action.

For smooth K , the subdifferential in (2) is the singleton $\{\nabla H_K\}$. The resulting Hamiltonian vector is tangent to the characteristic line on ∂K . Thus the definition recovers the Reeb orbits described above, including their parametrization.

The normalization also persists without smoothness. Indeed, if $y \in \partial H_K(x)$, then the 2-homogeneity of H_K gives

$$\langle y, x \rangle = 2H_K(x).$$

Along a generalized characteristic, $H_K(\gamma) = 1$, and hence

$$\lambda_0|_{\gamma(t)}(\dot{\gamma}(t)) = \frac{1}{2} \langle -J_0 \dot{\gamma}(t), \gamma(t) \rangle = 1 \quad \text{for almost every } t.$$

Consequently every contact-normalized generalized characteristic satisfies

$$A(\gamma) = T. \tag{3}$$

This formulation separates the general nonsmooth dynamical object from its finite description on a polytope. The latter will follow by calculating the subdifferential of H_K from the active facets.

2.3 EHZ capacity

For smooth convex bodies, the Ekeland–Hofer and Hofer–Zehnder capacities coincide. Their common value is the minimum action among closed characteristics on the boundary. We use the resulting EHZ capacity in the form

$$c_{\text{EHZ}}(K) = \min\{A(\gamma) : \gamma \text{ is a closed characteristic on } \partial K\}. \tag{4}$$

This is the smooth least-action characterization of [AO14, Theorem 2.2]. Equivalently, the minimum may be taken over Reeb orbits. The contact normalization only chooses the parametrization of each characteristic, while the action is a line integral and is invariant under orientation-preserving reparametrization. For a contact-normalized representative, $A(\gamma)$ is also the period. The same least-action theorem gives a positive attained minimum.

For a nonsmooth convex body, we define capacity as the limit of the capacities of smooth convex approximations in the Hausdorff metric. This limit is independent of the approximating

sequence. The least-action description remains valid when closed characteristics are replaced by the generalized characteristics of Definition 2.1 [AO14, Proposition 2.7]. Their finite facet description on a polytope is derived in Section 3.

As a symplectic capacity on convex bodies, c_{EHZ} is monotone under inclusion, invariant under symplectomorphisms, and conformal:

$$c_{\text{EHZ}}(\lambda K) = \lambda^2 c_{\text{EHZ}}(K) \quad (\lambda > 0).$$

It is normalized by

$$c_{\text{EHZ}}(B^4(r)) = c_{\text{EHZ}}(Z^4(r)) = \pi r^2,$$

where $B^4(r)$ is the radius- r ball and $Z^4(r) = B^2(r) \times \mathbb{R}^2$ is the symplectic cylinder.

Throughout the thesis, vol_4 denotes four-dimensional Lebesgue measure in the coordinate order (q_1, q_2, p_1, p_2) . In dimension four we define the systolic ratio by

$$\text{sys}(K) := \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)}. \quad (5)$$

With this normalization, Viterbo's proposed inequality for four-dimensional convex bodies was

$$\text{sys}(K) \leq 1, \quad \text{equivalently} \quad c_{\text{EHZ}}(K)^2 \leq 2 \text{vol}_4(K)$$

[Vit00].

Definition 2.2 (Systolic-ratio symmetries). *Let*

$$G_{\text{sys}} = \{x \mapsto b + \rho Mx : b \in \mathbb{R}^4, \rho > 0, M \in \text{Sp}(4, \mathbb{R})\}.$$

This is the group of affine transformations generated by translations, positive scalings, and linear symplectic transformations.

Lemma 2.3 (Systolic-ratio invariance). *For every convex body $K \subset \mathbb{R}^4$ and every $g \in G_{\text{sys}}$,*

$$\text{sys}(gK) = \text{sys}(K).$$

Proof. Translations and linear symplectic transformations are symplectomorphisms, so they preserve c_{EHZ} . They also preserve four-dimensional volume, because translations do not change volume and every symplectic matrix has determinant 1. A positive scaling by ρ gives

$$c_{\text{EHZ}}(\rho K) = \rho^2 c_{\text{EHZ}}(K), \quad \text{vol}_4(\rho K) = \rho^4 \text{vol}_4(K).$$

Substituting these identities in (5) shows that the ratio is unchanged. $\square$

Proposition 2.4 (Continuity). *Let K_t be a family of four-dimensional convex bodies which is continuous in the Hausdorff topology. Then $c_{\text{EHZ}}(K_t)$ is continuous in t . If, moreover, $t \mapsto \text{vol}_4(K_t)$ is positive, finite, and continuous, then the systolic ratio of the family is continuous.*

Proof. The EHZ capacity is C^0 -continuous on convex bodies with respect to the Hausdorff metric [CH21]. The assertion for sys is then ordinary continuity of the quotient in (5). In the applications below the volume hypothesis is usually automatic, since four-volume is continuous under Hausdorff convergence of convex bodies; for the rotated-polygon family it is even constant. $\square$

2.4 Polytope input language

The finite methods represent polytopes by normalized halfspaces. Let $K \subset \mathbb{R}^4$ be full-dimensional with $0 \in \text{int}(K)$. We write

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, i = 1, \dots, F\}. \quad (6)$$

If n_i is the outward unit normal to the corresponding supporting hyperplane and $h_i = h_K(n_i) > 0$ is its height, then

$$a_i = \frac{n_i}{h_i}.$$

For an irredundant description, the vectors a_i are precisely the vertices of the polar body

$$K^\circ = \text{conv}\{a_1, \dots, a_F\}.$$

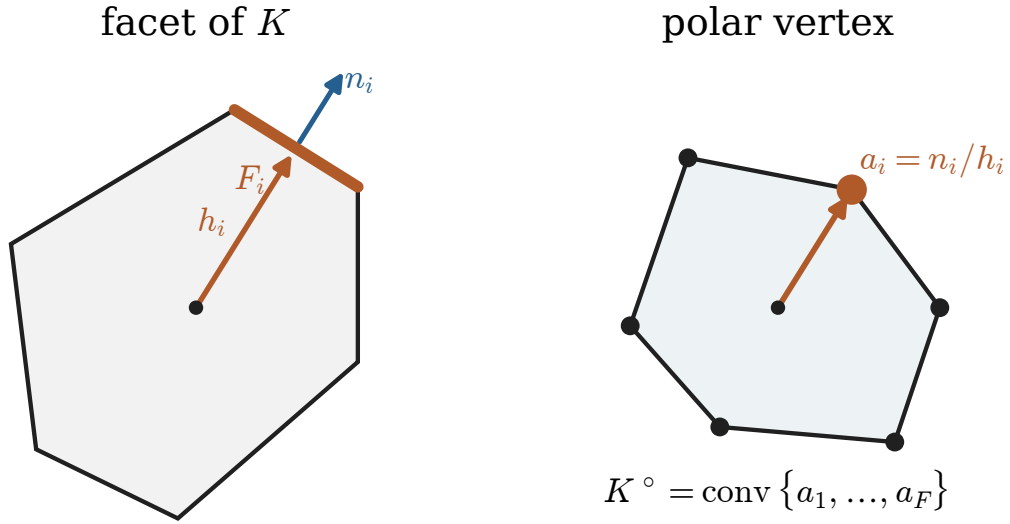


Figure 2: Facet normalization and polarity in a planar model. The supporting line $\langle n_i, x \rangle = h_i$ becomes $\langle a_i, x \rangle = 1$ after setting $a_i = n_i/h_i$, and that row is a vertex of the polar polygon. The same facet–polar-vertex correspondence holds in four dimensions; the planar metric proportions are specific to this example and are not part of the duality statement.

Conversely, the normalized halfspaces define a bounded full-dimensional polytope containing the origin in its interior exactly when $0 \in \text{int conv}\{a_i\}$. The description is irredundant exactly when the listed vectors are pairwise distinct and every a_i is an extreme point of this convex hull.

The ordered list $(a_1, \dots, a_F)$ is a labelled presentation. Relabelling the same inequalities does not change the unlabelled convex body, but labels matter when an algorithm records facet words or when a local calculation uses the a_i as coordinates.

2.5 Lagrangian products

Write

$$\mathbb{R}^4 = \mathbb{R}_q^2 \oplus \mathbb{R}_p^2 = \{(q_1, q_2, p_1, p_2)\}.$$

Definition 2.5 (Lagrangian product). *For planar convex bodies $K_q, K_p \subset \mathbb{R}^2$, their Lagrangian product is*

$$K_q \times_L K_p = \{(q_1, q_2, p_1, p_2) : (q_1, q_2) \in K_q, (p_1, p_2) \in K_p\}. \quad (7)$$

Thus K_q occupies the Lagrangian q -plane and K_p occupies the Lagrangian p -plane. This is different from taking a product of planar domains in the symplectic coordinate planes (q_1, p_1) and (q_2, p_2) .

The two decompositions pair the same four coordinates in different ways:

$$\underbrace{(q_1, q_2)}_{K_q} \mid \underbrace{(p_1, p_2)}_{K_p} \quad \text{versus} \quad \underbrace{(q_1, p_1)}_{\text{first symplectic plane}} \mid \underbrace{(q_2, p_2)}_{\text{second symplectic plane}}.$$

In block notation $J_0(u, v) = (-v, u)$. Consequently, the Reeb direction associated later with a supporting row in one Lagrangian factor moves in the other factor. This exchange, rather than a Cartesian-product decoupling of the dynamics, is what makes Lagrangian products useful here.

The central examples of the thesis are Lagrangian products of polygons; their lifted facet rows and the resulting restricted word enumeration are introduced with the Haim–Kislev formula, where they are first used.

2.6 Clarke’s dual action principle

A curve constrained to lie on ∂K is difficult to modify: changing one velocity can move the rest of it off the boundary. We will instead work with closed curves in $\mathbb{R}^4$ and a functional that depends only on velocity. The dual action principle says that minimizing this functional recovers the capacity, and that a minimizer can be placed back on the boundary. This is the freedom used in the next section to rearrange facet velocities.

Recall $H_K = g_K^2$, and define its convex conjugate by $H_K^*(y) = \sup_x (\langle x, y \rangle - H_K(x))$. For a closed curve $z \in W^{1,2}([0, T], \mathbb{R}^4)$ put

$$I_K(z) = \int_0^T H_K^*(-J_0 \dot{z}(t)) dt = \frac{1}{4} \int_0^T h_K(-J_0 \dot{z}(t))^2 dt. \quad (8)$$

The last identity will be proved below. Our normalization is

$$d(K) = \inf \{ I_K(z) : T > 0, z(0) = z(T), A(z) = T \}. \quad (9)$$

The infimum is over Sobolev curves as above. Their positions are unrestricted; the action constraint fixes their size relative to their period.

Theorem 2.6 (Dual action principle). *For a convex body $K \subset \mathbb{R}^4$ with $0 \in \text{int } K$, the infimum in (9) is attained and*

$$d(K) = c_{\text{EHZ}}(K) = \min_{\gamma \subset \partial K} A(\gamma),$$

where the last minimum is over contact-normalized generalized characteristics. Every such characteristic satisfies $I_K(\gamma) = A(\gamma)$. Every dual minimizer can be rescaled and translated into a minimizing generalized characteristic, without changing its value of I_K . If a dual minimizer already has $I_K(z) = T$, only a translation is needed.

We use the nonsmooth variational results underlying Clarke’s principle [AO14, Proposition 2.5 and Lemmas 5.1–5.2]; for polytopes the correspondence is also stated in [Hai19, Lemma 2.1]. The following argument explains the conjugacy, normalization and reconstruction used here.

The conjugate and its equality condition.

Lemma 2.7. *One has $H_K^*(y) = h_K(y)^2/4$, and*

$$H_K(x) + H_K^*(y) \geq \langle x, y \rangle.$$

Equality holds exactly when $y \in \partial H_K(x)$, equivalently $x \in \partial H_K^(y)$.*

Proof. Write $x = ru$ with $r \geq 0$ and $g_K(u) = 1$. For a fixed r , the largest value of $\langle x, y \rangle$ is $rh_K(y)$. Thus

$$H_K^*(y) = \sup_{r \geq 0} (rh_K(y) - r^2) = \frac{1}{4}h_K(y)^2.$$

The inequality follows from the definition of the supremum. Equality says that x maximizes $\langle x, y \rangle - H_K(x)$, which is precisely the subgradient inequality defining $y \in \partial H_K(x)$. Applying conjugacy in the other direction gives the equivalent condition. $\square$

The variational equation. A minimizer satisfies a weak Euler–Lagrange equation. Its useful form is

$$\nu z(t) + b \in \partial H_K^*(-J_0 \dot{z}(t)) \quad \text{for almost every } t, \quad (10)$$

for a constant vector b and a real multiplier ν .

Definition 2.8 (Weak dual critical curve). *A feasible curve satisfying this equation is called weak dual critical.*

The constant b reflects the freedom to translate the curve. Conjugacy rewrites the same equation as $-J_0 \dot{z} \in \partial H_K(\nu z + b)$, bringing it close to the characteristic equation.

To see the form of the variational equation, choose $p(t) \in \partial H_K^*(-J_0 \dot{z}(t))$. The derivative of action in a periodic variation v is $DA(z)[v] = \int \langle -J_0 \dot{z}, v \rangle$. The cited nonsmooth multiplier theorem provides a measurable L^2 choice of p such that, for every smooth periodic variation v ,

$$\int_0^T \langle p, -J_0 \dot{v} \rangle dt = \nu \int_0^T \langle -J_0 \dot{z}, v \rangle dt.$$

The left side pairs the chosen subgradient with the change in velocity; the right side is the variation of the action constraint. Integration by parts gives $-J_0 \dot{p} = -\nu J_0 \dot{z}$ in distributions, hence $\dot{p} = \nu \dot{z}$ and $p = \nu z + b$. The action constraint is regular because $DA(z)[z] = 2T \neq 0$. These are the variational assertions invoked from the cited nonsmooth theorem; the calculation fixes their signs and constants in our convention.

For any weak critical curve, the multiplier is determined by its value. The two-homogeneity of H_K^* gives $\langle p, y \rangle = 2H_K^*(y)$ when $p \in \partial H_K^*(y)$. Pairing (10) with $-J_0 \dot{z}$ and integrating, the constant vector contributes zero because the curve is closed. We obtain

$$2I_K(z) = 2\nu A(z) = 2\nu T, \quad \nu = \frac{I_K(z)}{T} > 0. \quad (11)$$

Positivity follows since a feasible curve has positive action and is nonconstant, while H_K^* is positive away from zero.

Reconstructing a boundary curve. Let z be weak critical and set $x = \nu z + b$ and $y = -J_0 \dot{z}$. Then $y \in \partial H_K(x)$. The convex chain rule gives

$$\frac{d}{dt} H_K(x(t)) = \langle y(t), \dot{x}(t) \rangle = \nu \langle -J_0 \dot{z}(t), \dot{z}(t) \rangle = 0.$$

Thus $H_K(x)$ is constant. Since $y \in \partial H_K(x)$, Fenchel equality gives $H_K(x) + H_K^*(y) = \langle x, y \rangle$. Two-homogeneity gives $\langle x, y \rangle = 2H_K(x)$, so $H_K^*(y) = H_K(x)$ almost everywhere. Integrating the constant value yields

$$I_K(z) = \int_0^T H_K^*(y) dt = TH_K(x), \quad H_K(x) = I_K(z)/T = \nu.$$

Put $c = \sqrt{\nu}$ and define

$$\gamma(t) = cz(t/c^2) + b/c, \quad 0 \leq t \leq c^2 T.$$

Since $\gamma = x(t/c^2)/c$, it satisfies $H_K(\gamma) = 1$. Subgradient homogeneity also gives

$$-J_0\dot{\gamma}(t) = y(t/c^2)/c \in \partial H_K(\gamma(t)).$$

This is the contact-normalized characteristic equation. Its period and action are $c^2T = I_K(z)$. Direct substitution in (8) gives $I_K(\gamma) = I_K(z)$. In particular, when $I_K(z) = T$ we have $c = 1$, so reconstruction only translates the curve.

Conversely, a contact-normalized characteristic satisfies $H_K(\gamma) = 1$ and $-J_0\dot{\gamma} \in \partial H_K(\gamma)$. Fenchel equality therefore gives $H_K^*(-J_0\dot{\gamma}) = 1$ almost everywhere. Consequently $I_K(\gamma) = T = A(\gamma)$, and it is a feasible dual curve. An attained dual minimum reconstructs to a characteristic with the same value; every characteristic supplies a feasible dual value. This proves equality of the minima and, by the least-action characterization, their equality with capacity.

Period convention and attainment. The cited fixed-period formulation uses $w : [0, 1] \rightarrow \mathbb{R}^4$ with $A(w) = 1/2$. If z is feasible in our convention, then

$$w(s) = \frac{z(Ts)}{\sqrt{2T}}, \quad I_K(w) = \frac{1}{2}I_K(z).$$

Conversely, $z(t) = \sqrt{2T}w(t/T)$ is feasible for any chosen $T > 0$ and has $I_K(z) = 2I_K(w)$. If the fixed-period formulation imposes mean zero, subtract the mean of w ; translation changes neither its action nor I_K . Thus the fixed-period attainment theorem applies to our infimum, with its factor of two accounted for. Choosing $T = 2I_K(w)$ gives an attained representative with $I_K(z) = T$.

The distinction needed below is now concrete. Feasibility allows us to compare functional values, but does not place a curve on the boundary. Once a modified curve is shown to minimize I_K , the variational equation and reconstruction justify placing it there. This is why rearranging velocities will be done in the dual problem first.

3 Generalized Reeb Orbits On Polytopes

The boundary of a polytope is not smooth, so the smooth Reeb vector field is replaced by a differential inclusion. The inclusion still has contact normalization, but it allows convex combinations of several facet directions at a nonsmooth face. The purpose of this section is to show why the capacity problem nevertheless contains one minimizer described by finitely many pure facet directions. It first separates the finite velocity data from the additional base-point condition that places the path on the prescribed facets, and then proves the simple-minimizer theorem that supplies the input to the Haim–Kislev quadratic program.

3.1 Definition

Let

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, i = 1, \dots, F\}$$

be a full-dimensional polytope with $0 \in \text{int}(K)$. We assume that this H -representation is irredundant, so the facet corresponding to a_i is

$$F_i = \{x \in K : \langle a_i, x \rangle = 1\}.$$

The normalized halfspace description gives

$$g_K(x) = \max_{1 \leq i \leq F} \langle a_i, x \rangle.$$

If $x \in \partial K$, let $I(x) = \{i : x \in F_i\}$ be its active facets. The subdifferential rules for a finite maximum and for the square on the level set $g_K = 1$ give

$$\partial H_K(x) = 2 \operatorname{conv}\{a_i : i \in I(x)\}. \quad (12)$$

Thus Definition 2.1 becomes a finite differential inclusion. In this normalization the pure contact-normalized Reeb direction on F_i is

$$R_i = 2J_0 a_i. \quad (13)$$

Indeed R_i is tangent to F_i , because $\langle a_i, J_0 a_i \rangle = 0$, and for $x \in F_i$,

$$\lambda_0|_x(R_i) = \frac{1}{2} \langle -J_0 R_i, x \rangle = \langle a_i, x \rangle = 1.$$

For a Lagrangian product, this formula makes the exchange between the two factors explicit. A row $a_i = (u, 0)$ from a facet of the q -factor and a row $a_j = (0, v)$ from a facet of the p -factor give

$$R_i = 2J_0(u, 0) = (0, 2u), \quad R_j = 2J_0(0, v) = (-2v, 0).$$

Thus a segment on a q -facet moves only in the p -coordinates, while a segment on a p -facet moves only in the q -coordinates.

A *generalized Reeb orbit* on K is a contact-normalized generalized characteristic in the sense of Definition 2.1. By (12), this means precisely that $\gamma \in W^{1,2}([0, T], \mathbb{R}^4)$ is closed, has image in ∂K , and satisfies

$$\dot{\gamma}(t) \in \operatorname{conv}\{R_i : \gamma(t) \in F_i\} \quad \text{for a.e. } t \in [0, T]. \quad (14)$$

The general normalization already shows that such an orbit has

$$A(\gamma) = T.$$

A generalized Reeb orbit is *simple* if its periodic parametrization can be started so that $\dot{\gamma}$, after changing values at finitely many breakpoints if necessary, is constant on finitely many intervals and each constant value is one of the pure facet directions R_i . In addition, each facet direction occurs in one connected time interval, possibly not at all. Thus a simple orbit uses pure facet directions rather than nontrivial convex combinations, and it does not return to the same facet direction in separated time intervals.

3.2 Words, dwell times, and closure

The finite computation records a simple orbit by its facet directions, dwell times, and starting point. We use the *active-word convention*: the word contains exactly the distinct facet directions that occur with positive duration, and the base point is chosen at a velocity breakpoint. A cyclic word

$$\sigma = (\sigma_1, \dots, \sigma_m)$$

with distinct entries, together with dwell times $\tau_k > 0$ and a base point $b = \gamma(0)$, describes the velocity data

$$\dot{\gamma}(t) = R_{\sigma_k} \quad \text{for a time interval of length } \tau_k.$$

Such data can represent a closed orbit only if the closing equation holds:

$$\sum_{k=1}^m \tau_k R_{\sigma_k} = 0. \quad (15)$$

Closure of the velocity polygon is only the closedness equation for the broken path. To turn the word and dwell times into an actual simple Reeb orbit on ∂K , one must choose a base point b

closure in displacement space one feasible translated path

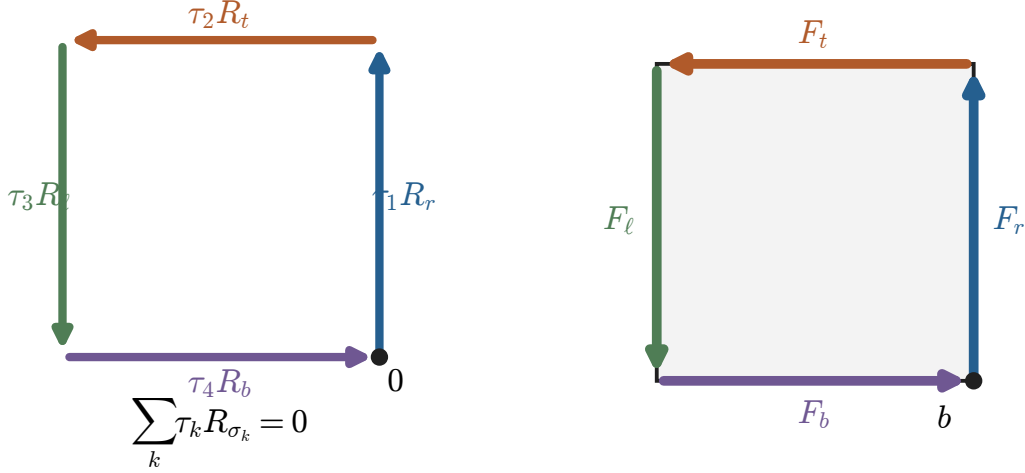


Figure 3: The two layers of finite orbit data in a two-dimensional Hamiltonian analogue. The ordered displacements first form a closed velocity polygon. A translation by a feasible base point then places every segment on its prescribed facet. This is a planar example, not a projection of the four-dimensional orbit; it illustrates only closure and facet realizability.

so that each segment lies on its prescribed facet; this is the base-point recovery construction of Lemma 3.2.

Closure does not imply that the second layer is feasible. For example, in the planar square $[-1, 1]^2$, the right and left facet rows give opposite directions $R_r = (0, 2)$ and $R_\ell = (0, -2)$. Equal dwell times satisfy $\tau R_r + \tau R_\ell = 0$, but the breakpoint between the two segments would have to lie in the disjoint facets F_r and F_ℓ . No translation can realize this closed two-segment path on the prescribed facets.

The period and action of a closed pure-word orbit are

$$T = \sum_{k=1}^m \tau_k, \quad A(\gamma) = T.$$

This convention is the free-period convention used in Theorem 2.6.

The same geometric orbit has no preferred starting segment. Cyclically shifting the (facet, duration)-pairs gives the same orbit, provided the base point is shifted to the corresponding breakpoint on the curve. If an orbit is instead started inside a constant-velocity segment, that segment is split at the start time and then merged again when the active word is recovered. If one uses a padded convention with all facets in a permutation and dwell times $\tau_k \geq 0$, then zero-duration entries are first deleted to recover the active word.

The action of a finite word can be calculated directly from its velocities and dwell times.

Lemma 3.1 (Symplectic shoelace formula). *Let $z: [0, T] \rightarrow \mathbb{R}^4$ be a closed piecewise affine curve. Suppose that z has constant velocity w_k on consecutive intervals of length ℓ_k , for $k = 1, \dots, m$. Then*

$$2A(z) = \sum_{1 \leq j < k \leq m} \ell_j \ell_k \omega_0(w_j, w_k). \quad (16)$$

In particular, the action of a closed pure-word Reeb orbit depends only on the cyclic velocity word and the dwell times, not on the base point.

Proof. Write $t_k = \sum_{j < k} \ell_j$. On the k -th interval,

$$z(t) = z(0) + \sum_{j < k} \ell_j w_j + (t - t_k) w_k.$$

The contribution of $z(0)$ to $\int_0^T \langle -J_0 \dot{z}, z \rangle dt$ vanishes because z is closed, so $\sum_k \ell_k w_k = 0$. The self-term $\langle -J_0 w_k, w_k \rangle$ vanishes by skew-symmetry. Hence

$$2A(z) = \sum_k \ell_k \left\langle -J_0 w_k, \sum_{j < k} \ell_j w_j \right\rangle = \sum_{j < k} \ell_j \ell_k \omega_0(w_j, w_k),$$

using $\omega_0(u, v) = \langle J_0 u, v \rangle$. $\square$

Breakpoints of $\dot{\gamma}$ are velocity changes, not necessarily changes in the minimal face containing $\gamma(t)$: an orbit may move inside a nonsmooth face while switching from one active facet direction to another.

Lemma 3.2 (Base-point recovery). *Fix an active word $\sigma = (\sigma_1, \dots, \sigma_m)$ and dwell times $\tau_k > 0$ satisfying the closure equation (15). Put*

$$v_1 = 0, \quad v_k = \sum_{j=1}^{k-1} \tau_j R_{\sigma_j} \quad (k = 2, \dots, m+1).$$

Then $v_{m+1} = 0$. A point $b \in \mathbb{R}^4$ recovers a simple Reeb orbit with these velocity data, by

$$\gamma(t) = b + v_k + s R_{\sigma_k}, \quad 0 \leq s \leq \tau_k$$

on the k -th segment, if and only if

$$\langle a_{\sigma_k}, b + v_k \rangle = 1 \quad \text{for } k = 1, \dots, m, \quad (17)$$

and

$$\langle a_i, b + v_k \rangle \leq 1 \quad \text{for } i = 1, \dots, F \text{ and } k = 1, \dots, m. \quad (18)$$

Equivalently, the equalities have the form

$$N_\sigma b = r_\sigma, \quad (N_\sigma)_{k\bullet} = a_{\sigma_k}^\top, \quad (r_\sigma)_k = 1 - \langle a_{\sigma_k}, v_k \rangle.$$

When this system is nonempty, its solution space is an affine space of dimension $4 - \text{rank } N_\sigma$. The inequalities cut out the valid base points as a convex polytope inside this affine space. In particular, the set of valid base points is bounded, because $v_1 = 0$ and (18) imply $b \in K$.

Proof. On the k -th segment, the prescribed facet equation is

$$\langle a_{\sigma_k}, b + v_k + s R_{\sigma_k} \rangle = 1.$$

The s -term vanishes because $\langle a_{\sigma_k}, R_{\sigma_k} \rangle = 2 \langle a_{\sigma_k}, J_0 a_{\sigma_k} \rangle = 0$. Hence the whole segment lies on F_{σ_k} exactly when (17) holds.

The segment lies in K exactly when

$$\langle a_i, b + v_k + s R_{\sigma_k} \rangle \leq 1 \quad \text{for all } i \text{ and all } s \in [0, \tau_k].$$

For fixed i, k this expression is affine in s , so it is enough to check the endpoints. The endpoint at $s = 0$ is $b + v_k$, and the endpoint at $s = \tau_k$ is $b + v_{k+1}$. As k runs cyclically and $v_{m+1} = v_1 = 0$, these endpoint checks are exactly (18). With these equalities and inequalities, the resulting closed broken path lies on ∂K and has velocity R_{σ_k} on the k -th segment, hence is a simple Reeb orbit with the prescribed data. $\square$

Remark 3.3 (Non-uniqueness and computation). *The base point is not part of the finite action data. If $\text{rank } N_\sigma = 4$, the facet equations determine it uniquely when they are consistent. If $\text{rank } N_\sigma < 4$, the inequalities may leave a positive-dimensional set of base points. This non-uniqueness does not affect the action, which is determined by the closed velocity polygon via Lemma 3.1. Computationally, base-point recovery is therefore a finite linear feasibility check after the word and dwell times have been chosen.*

3.3 Existence of simple minimizers

Haim–Kislev prove that every convex polytope has a minimum-action generalized Reeb orbit with a simple finite description [Hai19, Theorem 1.5]. In their statement the velocities are positive multiples of $J_0 n_i$ for the unit facet normals n_i . Here we reparametrize each segment by its contact-normalized facet direction $R_i = 2J_0 a_i$, absorbing the positive multiples into the dwell times. The proof is important for the finite computation, so we record the argument in the conventions of this thesis. It starts from an arbitrary minimizer, passes to the dual formulation from Theorem 2.6, and constructs simpler dual curves whose values converge back to the minimum.

Theorem 3.4 (Simple minimizer). *Let $K \subset \mathbb{R}^4$ be a full-dimensional convex polytope with $0 \in \text{int}(K)$. Among all generalized Reeb orbits on ∂K , there exists a minimum-action orbit that is simple in the contact-normalized convention above. Hence*

$$c_{\text{EHZ}}(K) = \min\{A(\gamma) : \gamma \text{ is a simple Reeb orbit on } \partial K\}.$$

The proof uses five elementary operations on dual curves: approximate the minimizer, split mixed velocities into pure facet velocities, merge repeated facet velocities, rescale to restore the dual constraint, and take a limit. The middle two operations may increase the action while keeping the period fixed. After rescaling, dual minimality forces this increase to vanish in the limit; compactness then preserves the resulting simple word. We record the operations separately because the same normalization and rearrangement ideas also explain the quadratic program below.

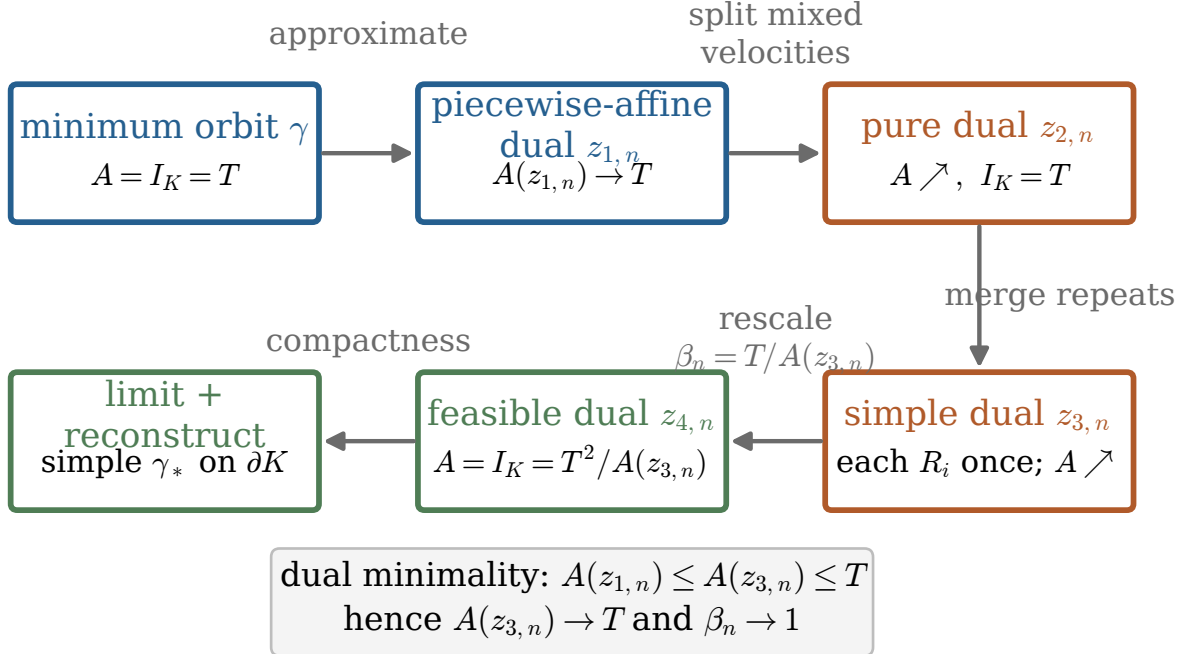


Figure 4: The five-operation proof spine. Approximation, splitting, merging, and rescaling are performed on position-unconstrained dual loops, not on curves already placed on ∂K . The lower inequality is the mechanism that makes the action increases disappear. Only the final dual minimizer is reconstructed as a boundary orbit.

In the proof, a *simple pure-velocity dual loop* means a closed piecewise-affine loop whose velocities lie in $\{R_1, \dots, R_F\}$, with each R_i occurring in at most one cyclic interval. This term asserts neither that the loop lies on ∂K nor that its segments are already realized on the facets carrying the corresponding labels.

Lemma 3.5 (Piecewise affine approximation). *Let $z \in W^{1,2}([0, T], \mathbb{R}^4)$ be closed and satisfy*

$$\dot{z}(t) \in \text{conv}\{R_1, \dots, R_F\} \quad \text{for a.e. } t.$$

For every $\varepsilon > 0$, there is a closed piecewise affine curve z' with

$$\dot{z}'(t) \in \text{conv}\{R_1, \dots, R_F\}$$

on each affine piece and $\|z' - z\|_{W^{1,2}} < \varepsilon$. Consequently $A(z') \rightarrow A(z)$ and $I_K(z') \rightarrow I_K(z)$ as $\varepsilon \rightarrow 0$.

Proof. Let $C = \text{conv}\{R_1, \dots, R_F\}$. Conditional expectations onto refining finite partitions converge to $\dot{z}$ in L^2 . Choose one such partition $0 = t_0 < \dots < t_m = T$ so that the conditional averages of $\dot{z}$ on the partition intervals are L^2 -close to $\dot{z}$. On $[t_k, t_{k+1}]$, put

$$w_k = \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} \dot{z}(s) ds.$$

Because C is convex and closed, each w_k lies in C . Define z' by $z'(0) = z(0)$ and $\dot{z}' = w_k$ on $[t_k, t_{k+1}]$. Then

$$z'(T) - z'(0) = \int_0^T \dot{z}'(t) dt = \int_0^T \dot{z}(t) dt = 0,$$

so z' is closed. Since $\dot{z}' \rightarrow \dot{z}$ in L^2 , integration gives $z' \rightarrow z$ uniformly, hence in $W^{1,2}$. The convergence of A follows from bilinearity in $(z, \dot{z})$. Both $-J_0 \dot{z}'$ and $-J_0 \dot{z}$ take values in the compact set $-J_0 C$, on which h_K^2 is Lipschitz; hence the L^2 -convergence also gives $I_K(z') \rightarrow I_K(z)$. $\square$

Lemma 3.6 (Splitting mixed velocities). *Let z be a closed piecewise affine curve of period T whose velocities lie in $\text{conv}\{R_1, \dots, R_F\}$. Then there is a closed piecewise affine curve z' , with the same period, whose velocity on each segment is one pure Reeb vector R_i , such that $A(z') \geq A(z)$ and*

$$I_K(z') = T.$$

Proof. On a segment with velocity $v = \sum_i \theta_i R_i$, $\theta_i \geq 0$ and $\sum_i \theta_i = 1$, replace the segment by subsegments with velocities R_i and lengths θ_i times the original length. The total displacement and period are unchanged. The only change in the action is the sum of the new internal cross-terms from Lemma 3.1. Reversing the order of the inserted subsegments changes the sign of this internal contribution, so one of the two orders does not decrease the action.

For a pure Reeb velocity $R_i = 2J_0 a_i$,

$$-J_0 R_i = 2a_i, \quad h_K(-J_0 R_i) = 2h_K(a_i) = 2,$$

because $F_i \neq \emptyset$ and $\max_{x \in K} \langle a_i, x \rangle = 1$. Thus $h_K^2(-J_0 R_i) = 4$ on every pure segment, and

$$I_K(z') = \frac{1}{4} \int_0^T 4 dt = T.$$

$\square$

Lemma 3.7 (Merging repeated velocities). *Let z be a closed piecewise affine curve whose velocity on each segment is one of the R_i . If some R_i occurs in two non-adjacent segments, then there is a rearrangement z' that merges those two occurrences into one contiguous segment and satisfies*

$$A(z') \geq A(z), \quad I_K(z') = I_K(z).$$

Repeating this operation, and merging adjacent equal-velocity intervals, gives a closed piecewise affine curve in which each pure Reeb vector occurs in at most one interval. Here adjacency is understood cyclically, since the curve is closed.

Proof. Suppose a velocity v occurs in two segments I_r and I_s , with middle segments $M = \{r+1, \dots, s-1\}$. Merging I_s forward into I_r changes only the action terms involving v and the middle velocities, and the change is

$$\Delta A_{\text{fwd}} = |I_s| \sum_{i \in M} |I_i| \omega_0(v, w_i).$$

Merging I_r backward into I_s gives

$$\Delta A_{\text{bwd}} = -|I_r| \sum_{i \in M} |I_i| \omega_0(v, w_i).$$

The two changes have opposite signs, or both vanish, so one merge has nonnegative action change. Closure is preserved because the multiset of velocity-duration pairs is unchanged, and I_K is unchanged because it depends only on the velocities and their durations. $\square$

Lemma 3.8 (Rescaling). *Let z be a closed piecewise affine curve with pure Reeb velocities and period T . If all durations are multiplied by $\beta > 0$, the resulting curve z' satisfies*

$$A(z') = \beta^2 A(z), \quad I_K(z') = \beta I_K(z).$$

Proof. In Lemma 3.1, every action term contains a product of two durations, so A scales by β^2 . The integrand in I_K is unchanged and the time intervals are all multiplied by β , so I_K scales by β . $\square$

Lemma 3.9 (Compactness of simple pure-velocity dual loops). *Let z_n be simple pure-velocity dual loops whose periods satisfy $T_n \rightarrow T > 0$. After translating the curves and passing to a subsequence, z_n converges to a simple pure-velocity dual loop z_* with period T , and*

$$A(z_*) = \lim_n A(z_n), \quad I_K(z_*) = T.$$

Proof. Choose the initial point of each periodic curve at a breakpoint, and insert zero dwell times for unused facet directions. Each curve is then determined up to translation by a permutation of the finitely many facet directions and a nonnegative dwell-time vector of total mass T_n . Pass first to a subsequence with a fixed permutation. Since $T_n \rightarrow T$, the dwell-time vectors lie in a common compact set and have a convergent subsequence whose entries sum to T . Translating the starting points gives uniform convergence after identifying each domain with $[0, 1]$ by linear time rescaling. Closure passes to the limit because $\sum_i \tau_i R_{\sigma(i)} = 0$ for every curve. Some limiting dwell times may be zero; deleting those zero-duration entries recovers the active word of the limiting simple curve. The action is continuous in the dwell-time data by Lemma 3.1. Since the remaining positive-duration segments still have pure Reeb velocities, the same computation as in Lemma 3.6 gives $I_K(z_*) = T$. $\square$

Proof of Theorem 3.4. Choose a capacity-realizing generalized Reeb orbit γ . By Theorem 2.6, γ is also a dual minimizer in the thesis convention, with

$$I_K(\gamma) = A(\gamma) = T = c_{\text{EHZ}}(K).$$

Moreover $\dot{\gamma}(t) \in \text{conv}\{R_1, \dots, R_F\}$ for almost every t .

Choose, by Lemma 3.5, a sequence of closed piecewise affine curves $z_{1,n} \rightarrow \gamma$ in $W^{1,2}$, with velocities in the convex hull of the R_i . Apply Lemma 3.6 to obtain closed pure-velocity curves $z_{2,n}$, and then Lemma 3.7 to obtain closed simple pure-velocity curves $z_{3,n}$, such that

$$A(z_{3,n}) \geq A(z_{2,n}) \geq A(z_{1,n}), \quad I_K(z_{3,n}) = T.$$

Since $A(z_{1,n}) \rightarrow A(\gamma) = T$, discard finitely many terms so that $A(z_{3,n}) > 0$. Rescale all durations of $z_{3,n}$ by

$$\beta_n = \frac{T}{A(z_{3,n})},$$

and denote the resulting curve by $z_{4,n}$. It satisfies

$$A(z_{4,n}) = \frac{T^2}{A(z_{3,n})}, \quad I_K(z_{4,n}) = \frac{T^2}{A(z_{3,n})}.$$

Thus $z_{4,n}$ is feasible for the dual problem, with period $T^2/A(z_{3,n})$. Since γ is a dual minimizer, $I_K(z_{4,n}) \geq T$, and therefore $A(z_{3,n}) \leq T$. Together with $A(z_{3,n}) \geq A(z_{1,n}) \rightarrow T$, this gives

$$A(z_{3,n}) \rightarrow T, \quad \beta_n \rightarrow 1, \quad \text{period}(z_{4,n}) \rightarrow T.$$

By Lemma 3.9, a subsequence of the $z_{4,n}$ converges up to translation to a simple pure-velocity dual loop z_* with $A(z_*) = T$ and $I_K(z_*) = T$. This curve is a dual minimizer. Applying Theorem 2.6 again gives a contact-normalized generalized Reeb orbit on ∂K in the same translate/rescale family. In the reconstruction part of that theorem, the corresponding characteristic has the form

$$t \mapsto c z_*(t/c^2) + b/c$$

for some $c > 0$, with

$$c^2 = \nu = \frac{I_K(z_*)}{T} = 1$$

by (11). Thus the reconstruction only translates z_* . Its velocities remain the pure facet directions R_i , each used in one connected interval, so the resulting orbit is simple and has minimum action. The translation also recovers the prescribed facet labels. Indeed, on an interval with $\dot{\gamma} = R_i$, the reconstructed inclusion gives

$$2a_i = -J_0 R_i \in 2 \operatorname{conv}\{a_j : \gamma(t) \in F_j\}.$$

Because the irredundant row a_i is an extreme point of K° , this forces F_i to be active along that interval. The reconstructed curve lies in K , so its starting point $\gamma(0)$ is a valid base point in Lemma 3.2; closure supplies the remaining displacement equation. $\square$

The theorem proves existence of at least one simple capacity minimizer. It is not a classification of all minimum-action generalized Reeb orbits. Other minimizers may pass through nonsmooth faces and may use convex combinations of active facet directions. Thus the finite Haim–Kislev quadratic program is justified as a capacity search because its search family contains a simple capacity minimizer, not because every minimizer or every word has this form. Infeasible words and duplicate descriptions can still occur.

4 The Haim–Kislev Quadratic Program

The preceding section showed that some capacity-minimizing generalized Reeb orbit uses a finite list of distinct facet directions. We now express its action in terms of that list and its dwell times. Haim–Kislev’s formula turns the result into a finite quadratic program [Hai19].

The chapter has two tasks. First, it explains the variables and how to compute the global optimum by solving linear stationarity systems and comparing their values. Second, it proves reductions for Lagrangian products of planar polygons. Those reductions make the later family calculations practical.

4.1 The finite quadratic program

Let

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, i = 1, \dots, F\}$$

be a full-dimensional polytope with $0 \in \text{int}(K)$, written as in Section 3. Thus $a_i = n_i/h_i$ if n_i is the outward unit normal to the i -th facet and $h_i = h_K(n_i) > 0$ is its height. Haim–Kislev state their formula in the (n_i, h_i) -variables. We use the equivalent dual-vertex variables

$$\beta_i^{\text{here}} = \beta_i^{\text{HK}} h_i.$$

For a contact-normalized orbit represented by a pure facet word, β_i is the fraction of its period spent with velocity $2J_0 a_i$. The weights therefore sum to one. Closure of the curve says that the weighted sum of its velocities vanishes; since J_0 is invertible, this is equivalent to $\sum_i \beta_i a_i = 0$. Thus the constraints are

$$\beta_i \geq 0, \quad \sum_i \beta_i = 1, \quad \sum_i \beta_i a_i = 0.$$

For an ordered facet word $\sigma = (\sigma_1, \dots, \sigma_m)$ with distinct entries, define

$$M_\sigma = \left\{ \beta \in \mathbb{R}_{\geq 0}^m : \sum_{k=1}^m \beta_k = 1, \sum_{k=1}^m \beta_k a_{\sigma_k} = 0 \right\}.$$

The quadratic objective for this active word is

$$Q_\sigma(\beta) = \sum_{1 \leq j < k \leq m} \beta_j \beta_k \omega_0(a_{\sigma_j}, a_{\sigma_k}). \quad (19)$$

Here σ is written in the Reeb traversal order used in Section 3. Haim–Kislev’s displayed formula uses the reversed ordered word after translating their symplectic convention to ω_0 . Since word reversal is a bijection on the ordered candidate family, the maximum is unchanged; the convention matters only when a single fixed word is interpreted.

Let

$$Q_{\max} = \max_{\sigma} \max_{\beta \in M_\sigma} Q_\sigma(\beta),$$

where σ ranges over ordered words with distinct facets and $M_\sigma \neq \emptyset$. Equivalently, one may range over all permutations of all facets and allow zero weights; the zero-weight entries are then deleted to recover the active word. This is only an enumeration convention: HK’s theorem is the full-permutation statement, and the active-word form is obtained by deleting unused zero-weight facets and reindexing the remaining entries. This formulation intentionally permits duplicates of the same geometric orbit, for example from changing the starting segment.

Theorem 4.1 (Haim–Kislev quadratic program). *For a full-dimensional convex polytope $K \subset \mathbb{R}^4$ with $0 \in \text{int}(K)$, written in the facet normalization above, $Q_{\max} > 0$ and*

$$c_{\text{EHZ}}(K) = \frac{1}{2Q_{\max}}.$$

Proof. This is Haim–Kislev’s finite capacity formula [Hai19, Theorem 1.1], rewritten from unit normals and heights to the dual vertices $a_i = n_i/h_i$. The variable change above sends

$$\sum_i \beta_i^{\text{HK}} h_i = 1, \quad \sum_i \beta_i^{\text{HK}} n_i = 0$$

to the constraints defining M_σ . Their displayed objective uses the opposite word orientation from (19); replacing each ordered word by its reverse sends it to Q_σ . Since reversal is a bijection on the words in the maximum, their formula reads

$$c_{\text{EHZ}}(K) = \frac{1}{2} \left[\max_{\sigma} \max_{\beta \in M_\sigma} Q_\sigma(\beta) \right]^{-1}.$$

□

The following construction explains what the finite variables represent and where global optimality enters.

Proposition 4.2 (From finite weights to a dual curve). *Let $\beta \in M_\sigma$ and put $q = Q_\sigma(\beta)$. If $q > 0$, define*

$$T = \frac{1}{2q}, \quad \tau_k = T\beta_k.$$

Traversing the velocities R_{σ_k} for the durations τ_k , and deleting zero-duration pieces, gives a closed simple dual curve z with

$$A(z) = I_K(z) = T.$$

If (σ, β) attains $Q_{\max}$ across all words, then $T = c_{\text{EHZ}}(K)$, and a translation of z is a minimum-action simple generalized Reeb orbit on ∂K , with the active word obtained from σ by deleting its zero-weight entries.

Conversely, if a simple generalized Reeb orbit has period T , active word σ , and positive dwell times τ_k , then $\beta_k = \tau_k/T$ lies in M_σ and $Q_\sigma(\beta) = 1/(2T)$.

Proof. The closure constraint gives

$$\sum_k \tau_k R_{\sigma_k} = 2T J_0 \sum_k \beta_k a_{\sigma_k} = 0,$$

while $\sum_k \tau_k = T$. The shoelace formula gives

$$2A(z) = 4T^2 Q_\sigma(\beta) = 2T.$$

Every positive-duration piece has velocity R_i , for which

$$H_K^*(-J_0 R_i) = \frac{1}{4} h_K^2(2a_i) = 1.$$

Hence $I_K(z) = T$, and z is feasible for the free-period dual problem.

If (σ, β) attains $Q_{\max}$, the Haim–Kislev formula gives $T = 1/(2Q_{\max}) = c_{\text{EHZ}}(K)$. Thus z is a dual minimizer. Its multiplier in (11) is $\nu = I_K(z)/T = 1$. The reconstruction in Theorem 2.6 therefore has scale one: it only translates z onto ∂K . Translation preserves the ordered pure-velocity pieces. On a piece with velocity R_i , the reconstructed inclusion is

$$2a_i = -J_0 R_i \in \partial H_K(\gamma) = 2 \operatorname{conv}\{a_j : F_j \text{ is active at } \gamma\}.$$

The irredundant rows a_i are extreme points of K° , so this convex combination forces F_i to be active there. Thus, after zero durations are deleted, the resulting minimum-action orbit is simple and has the reduced active word.

For the converse, closure of the orbit gives $\sum_k \beta_k a_{\sigma_k} = 0$, and positivity and $\sum_k \beta_k = 1$ are immediate. Since $A(\gamma) = T$, the shoelace formula yields

$$2T = 4T^2 \sum_{1 \leq r < s \leq m} \frac{\tau_r}{T} \frac{\tau_s}{T} \omega_0(a_{\sigma_r}, a_{\sigma_s}) = 4T^2 Q_\sigma(\beta), \quad \text{so} \quad Q_\sigma(\beta) = \frac{1}{2T}.$$

□

4.2 Solving one word and comparing the results

For a fixed word, the constraints are linear and the objective is quadratic. Write

$$C_\sigma = \begin{pmatrix} a_{\sigma_1} & \cdots & a_{\sigma_m} \\ 1 & \cdots & 1 \end{pmatrix}, \quad e = (0, 0, 0, 0, 1)^T, \quad Q_\sigma(\beta) = \frac{1}{2} \beta^T H_\sigma \beta.$$

Here H_σ is symmetric with zero diagonal and, for $j < k$, $(H_\sigma)_{jk} = (H_\sigma)_{kj} = \omega_0(a_{\sigma_j}, a_{\sigma_k})$. Thus the problem is to maximize this quadratic form subject to $C_\sigma \beta = e$ and $\beta \geq 0$.

First suppose that the maximizing weights are all positive. Small perturbations in $\ker C_\sigma$ remain feasible, so the first derivative must vanish on that kernel. Equivalently, there is a vector of multipliers λ such that

$$H_\sigma \beta = C_\sigma^T \lambda, \quad C_\sigma \beta = e.$$

These are the equality-constrained Karush–Kuhn–Tucker (KKT) equations:

$$\begin{pmatrix} H_\sigma & -C_\sigma^T \\ C_\sigma & 0 \end{pmatrix} \begin{pmatrix} \beta \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ e \end{pmatrix}. \quad (20)$$

They are a linear system even though the objective is quadratic. After solving it, one checks whether the weights are positive and evaluates Q . A stationary point can be a minimum or a saddle, so this calculation does not yet identify the maximum.

Example 4.3 (A four-facet calculation). Take $K = [-1, 1]^4$ and the ordered facet rows $(e_1, e_3, -e_1, -e_3)$, where the coordinate order is (q_1, q_2, p_1, p_2) . Closure forces the weights to have the form (x, y, x, y) , and normalization gives $x + y = 1/2$. The ordered symplectic pairings give

$$Q = 2xy = 2x(1/2 - x), \quad 0 \leq x \leq 1/2.$$

Thus stationarity reduces to $1 - 4x = 0$. The second derivative is -4 , so $x = y = 1/4$ maximizes this word, with $Q = 1/8$; both boundary choices have value zero. The corresponding capacity upper bound is 4. Identifying the capacity still requires a global comparison or another argument giving the matching lower bound.

If a maximum has zero weights, delete those entries from the word. The remaining weights are positive, and deleting zeros changes neither closure nor the objective. It follows that solving all active words covers maxima on every face of the nonnegative feasible set. Conversely, a shorter word can be padded with zero weights to recover a full-facet permutation. Comparing the feasible stationary values over this complete family therefore gives the global maximum in Haim–Kislev’s formula.

A singular system requires care, but not a search over infinitely many objective values. Suppose (β, λ) and $(\beta + z, \lambda + \mu)$ solve (20). Then $C_\sigma z = 0$ and $H_\sigma z = C_\sigma^T \mu$, whence

$$Q_\sigma(\beta + z) - Q_\sigma(\beta) = z^T H_\sigma \beta + \frac{1}{2} z^T H_\sigma z = \lambda^T C_\sigma z + \frac{1}{2} \mu^T C_\sigma z = 0.$$

Thus all solutions of the same stationary system have the same quadratic value. One must still decide whether that affine solution set contains positive weights. A singular matrix by itself means neither infeasibility nor failure of the capacity formula. This observation will also be used in the pentagon classification.

A feasible positive value q gives a dual curve with value $1/(2q)$. Capacity is the infimum of the dual values, so $c_{\text{EHZ}}(K) \leq 1/(2q)$. Once q is proved to be the largest value across all words, that curve is a dual minimizer and can be translated onto the boundary. For a nonglobal candidate, boundary placement instead requires the conditions in Lemma 3.2.

The full enumeration can be reduced when a geometric argument proves that a smaller family still contains a capacity minimizer. For example, an actual transition from facet F_i to F_j requires an intersection of those facets and the directed sign $\omega_0(a_i, a_j) \geq 0$. These necessary conditions can exclude words before solving their systems. They need not hold for every feasible dual curve; what makes them valid for computing capacity is the existence of a boundary-realized global minimizer. A stronger reduction for Lagrangian products is proved next.

4.3 Finite enumeration for Lagrangian products

Let $K_q, K_p \subset \mathbb{R}^2$ be nondegenerate convex polygons containing the origin in their interiors, and let

$$K = K_q \times_L K_p \subset \mathbb{R}_q^2 \oplus \mathbb{R}_p^2$$

be their Lagrangian product. A facet of K is either a q -facet, coming from an edge of K_q , or a p -facet, coming from an edge of K_p . We call a single facet a one-facet block, and an ordered adjacent pair of facets of the same type a two-facet block.

Theorem 4.4 (Finite enumeration for Lagrangian products). *For a Lagrangian product $K = K_q \times_L K_p$ of nondegenerate planar convex polygons containing the origin in their interiors, $c_{\text{EHZ}}(K)$ is obtained by maximizing Q_σ over the Haim–Kislev quadratic programs associated to cyclic words formed by k alternating q -blocks and p -blocks, where $k \in \{2, 3\}$. Each block contains one or two facets from its factor, distinct blocks use disjoint facets, and a two-facet block must use adjacent polygon edges. Thus every word in the family uses at most twelve facets. This is an exhaustive candidate family for the capacity value; it may contain overcounting or infeasible words, and it is not a classification of all minimum-action simple Reeb orbits.*

Proof. Put $T = c_{\text{EHZ}}(K)$. Rudolf’s nonsmooth billiard theorem gives a minimizing Minkowski billiard in K_q , together with its polygonal dual in K_p , having at most three bounce points and length T [Rud24, Theorem 1]. Write its vertices as q_j and its dual vertices as p_j , with indices understood cyclically. In the notation needed here, write

$$N_C(x) = \{u : \langle u, y - x \rangle \leq 0 \text{ for every } y \in C\}$$

for the outward normal cone. Rudolf calls this a strong billiard because the paired polygons satisfy the following normal-cone relations:

$$q_{j+1} - q_j \in N_{K_p}(p_j), \quad p_{j+1} - p_j \in -N_{K_q}(q_{j+1}),$$

Its length is

$$\ell_{K_p}(q) = \sum_j h_{K_p}(q_{j+1} - q_j) = \sum_j \langle p_j, q_{j+1} - q_j \rangle.$$

The last equality follows from the first normal-cone relation and $p_j \in \partial K_p$. These relations say precisely that the alternating broken line through

$$(q_j, p_j), (q_{j+1}, p_j), (q_{j+1}, p_{j+1})$$

is a generalized characteristic for Rudolf’s complex-structure convention. That convention is opposite to ours, so reverse the loop. Since λ_0 differs from $-p dq$ by an exact form, its action in our convention is

$$-\oint_{\text{rev}} p dq = \oint_{\text{orig}} p dq = \sum_j \langle p_j, q_{j+1} - q_j \rangle = \ell_{K_p}(q) = T.$$

Thus the reversed lift is a capacity-realizing generalized characteristic on $\partial(K_q \times_L K_p)$ whose legs alternately move only one factor. Parametrize it in the contact normalization. Its period is then T , and, as a dual curve z , it satisfies

$$A(z) = I_K(z) = T.$$

The following surgery is performed on z as a competitor in the dual variational problem, not as an orbit already constrained to ∂K . In particular, splitting a normal-cone direction into pure facet directions does not assert that the corresponding broken path, drawn directly in

K_q or K_p , would remain inside that polygon. Boundary placement is recovered only after the surgery has been shown to preserve dual minimality.

Apply the splitting–merging–rescaling part of the proof of Theorem 3.4 to z . Approximation and compactness are unnecessary here because the billiard lift is already a finite broken line. The only extra point to retain from the billiard is its block bound. On each leg, the polygonal normal cone is generated by one facet normal or by the two adjacent normals at a vertex, so splitting replaces that leg by one or two pure directions inside the same block. When a repeated direction is merged, it is removed from one same-type block and absorbed into another block that already contains it. Thus merging cannot increase the number of blocks of either factor, although it may decrease them. If the source block is a singleton, deleting it coalesces its two neighboring opposite-type blocks and reduces both block counts by one. This possible loss is harmless: only the inherited upper bound of three blocks of each type is needed below.

The rescaling argument in the cited proof uses dual minimality to show that the resulting distinct-facet curve is again a dual minimizer. Indeed, write $A' \geq T$ for its action before rescaling; its dual value is still T . Multiplying every duration by T/A' gives a dual-feasible curve with

$$A = I_K = \frac{T^2}{A'}.$$

The dual minimum is T , so $T^2/A' \geq T$, hence $A' \leq T$. Therefore $A' = T$: no rescaling is needed, and the distinct-facet curve z' is a dual minimizer with $A(z') = I_K(z') = T$. By the preceding observations, its cyclic word still has at most three blocks of each type.

Only now do we return to the primal problem. Normalize the positive dwell times of z' by their sum. The resulting finite weights have QP value $1/(2T) = Q_{\max}$, so Proposition 4.2 translates z' to a simple generalized Reeb orbit with the same active word on $\partial(K_q \times_L K_p)$. Its factor coordinates therefore remain in K_q and K_p ; this containment comes from dual-minimizer reconstruction, not from the splitting operation. A simple orbit cannot contain three consecutive facets of the same type. Indeed, along three consecutive q -facet segments the q -coordinate is fixed; the two breakpoints would place it at both endpoints of the middle edge of K_q , a contradiction. The same holds for p -facets. Hence every maximal block in the final word has length one or two, and a two-facet block uses adjacent edges.

If only one factor type occurred, the whole word would be a single block of length at most two, whose positive closure equation would require one outward normal, or a positive combination of two adjacent outward normals, to vanish. This is impossible. Thus both types occur. Their maximal blocks alternate cyclically, so there is a common number $k \leq 3$ of q - and p -blocks. One q -block and one p -block cannot close, by the same argument in each factor, so $k \neq 1$. Hence $k \in \{2, 3\}$. Since each block contains at most two facets, the resulting simple word uses at most $4k \leq 12$ distinct facets. This proves that the restricted family contains a capacity maximizer. It does not exclude other minimum-action orbits with more blocks. $\square$

Algorithm 4.5 (Lagrangian-product capacity enumeration). *For each factor, enumerate all one-facet blocks and all ordered adjacent two-facet blocks. For $k = 2, 3$, choose k non-overlapping q -blocks and k non-overlapping p -blocks, order them cyclically in the alternating pattern q -block, p -block, q -block, p -block, and so on. Read the facet labels in that order, discard infeasible words, and solve the corresponding Haim–Kislev quadratic program. The capacity is $1/(2Q_{\max})$, where $Q_{\max}$ is the largest objective value among the feasible words in this candidate family.*

4.4 A six-facet reduction for the product capacity

The product structure gives a reduction that is unavailable for a general polytope. The q - and p -coordinate planes are Lagrangian, so pairings between two q -facets or two p -facets vanish. After separating the total weight in each factor from its normalized distribution, the objective is therefore linear in either distribution when the other is fixed. Maximizing these two linear

functions at vertices is the mechanism behind both the six-facet bound and the following algorithm.

Let $u_1, \dots, u_r$ be the planar dual vertices of K_q , and let $v_1, \dots, v_s$ be those of K_p . Define the normalized closure polytopes

$$C_q = \left\{ \alpha \in \mathbb{R}_{\geq 0}^r : \sum_i \alpha_i = 1, \sum_i \alpha_i u_i = 0 \right\},$$

$$C_p = \left\{ \gamma \in \mathbb{R}_{\geq 0}^s : \sum_j \gamma_j = 1, \sum_j \gamma_j v_j = 0 \right\}.$$

Theorem 4.6 (Six-facet product maximizer). *For a Lagrangian product $K_q \times_L K_p$ of nondegenerate planar convex polygons containing the origin in their interiors, the complete Haim–Kislev family has a global maximizer using at most three q -facets and at most three p -facets. The total weight in each factor is $1/2$. Consequently the capacity is represented by a word of length at most six.*

Algorithm 4.7 (Product capacity from closure vertices). *Enumerate the vertices of C_q and C_p . For each pair, enumerate the cyclic orders of the union of their supports, give each factor total weight $1/2$, and evaluate Q_σ . The largest value is $Q_{\max}$. Each support union has at most six facets, so fixing one cyclic starting point leaves at most $5! = 120$ orders.*

an arbitrary global maximizer (σ, β)
 $\downarrow$ fix γ , maximize the linear function in α
 α is a vertex of C_q , hence $|\text{supp } \alpha| \leq 3$
 $\downarrow$ fix α , maximize the linear function in γ
 γ is a vertex of C_p , hence $|\text{supp } \gamma| \leq 3$
 $\downarrow$ maximize $t(1-t)$ and delete zero weights
 a global maximizer with $t = \frac{1}{2}$ and at most six facets

Figure 5: The proof mechanism for the six-facet reduction. Linearity in each factor distribution is the product-specific step.

Proof. Choose a global maximizer and let I_q and I_p be its q - and p -positions. Its objective is positive, hence both factor weights are nonzero. Put

$$t = \sum_{k \in I_q} \beta_k, \quad \alpha_k = \frac{\beta_k}{t} \ (k \in I_q), \quad \gamma_k = \frac{\beta_k}{1-t} \ (k \in I_p).$$

Closure splits between the factors, so the zero-extended vectors α and γ lie in the faces

$$C_q(\sigma) = \{\alpha \in C_q : \alpha_i = 0 \text{ if } q\text{-facet } i \text{ is omitted from } \sigma\},$$

and likewise $C_p(\sigma) \subset C_p$. Since the symplectic form vanishes on each factor, there is a bilinear function

$$B_\sigma : C_q(\sigma) \times C_p(\sigma) \longrightarrow \mathbb{R}$$

such that

$$Q_\sigma(\beta) = t(1-t)B_\sigma(\alpha, \gamma). \tag{21}$$

Its coefficients are the signed pairings $\langle u_i, v_j \rangle$, with signs determined by the order in σ .

With γ fixed, the bilinear factor is linear in α , so it has a maximizer at a vertex of $C_q(\sigma)$. Replacing α by that vertex preserves global maximality. Fixing the new α and repeating the

argument in $C_p(\sigma)$ gives a global maximizer for which both normalized factor weights are vertices of the support-restricted closure polytopes. Coordinate restrictions define faces of C_q and C_p , and a vertex of a face is a vertex of the ambient polytope.

A vertex of

$$\left\{ x \geq 0 : \sum_i x_i = 1, \sum_i x_i w_i = 0 \right\}, \quad w_i \in \mathbb{R}^2,$$

has at most three positive coordinates. Indeed, more than three positive columns in the three scalar equality constraints have a nontrivial linear dependence, which gives a feasible line through the point and contradicts extremality. Thus each factor uses at most three facets.

For the selected vertices the bilinear value is positive. Varying only the total factor weight gives

$$\tau(1 - \tau)B_\sigma(\alpha, \gamma), \quad 0 \leq \tau \leq 1,$$

which is maximal at $\tau = 1/2$. Delete the remaining zero-weight positions. The relative cyclic order and objective are unchanged, and the complete Haim–Kislev family contains the resulting word of length at most six.

The same argument proves Algorithm 4.7: the selected vertices occur in the enumeration, while every enumerated candidate is feasible for the complete Haim–Kislev family and therefore cannot exceed $Q_{\max}$. $\square$

Remark 4.8 (Why the reduction is product-specific). *The argument does not extend directly to a general polytope. It uses that each factor is Lagrangian: same-factor pairings vanish, leaving a bilinear function which is linear in one normalized closure distribution after the other is fixed. Without such a splitting, the objective remains quadratic in the full weight vector, and moving to a vertex of the closure polytope need not preserve or improve it.*

The theorem asserts the existence of one sparse capacity maximizer. It neither classifies all minimizing orbits nor says that every maximizer has support at most six. The preceding twelve-facet block family remains useful when an argument must compare a specified family of geometric branches. No genericity assumption is needed: the closure polytopes are closed, and degeneracy can only lower the rank used in the support bound.

4.5 Use in later certificates

The later computer-assisted proofs use these results in two ways. The HKO local-maximum proof uses selected active words to build smooth upper bounds. The rotated-pentagon proof uses the specialized Lagrangian-product enumeration from Theorem 4.4, then checks the algebraic inequalities needed to compare every surviving branch with the displayed one.

The numerical implementation follows the same distinction between general polytopes and products. In the general case it examines the candidate words and their stationary points, using proved exclusion criteria to reduce the search. It returns an interval containing the capacity. For a recognized Lagrangian product it instead enumerates the closure-polytope vertices from Algorithm 4.7 and compares the resulting objective values. This route returns the exact capacity of the supplied input, whose floating-point coordinates are treated as exact rational numbers. Both routes use interval bounds to control numerical error and exact arithmetic when a comparison cannot otherwise be decided. Section 11 describes those checks. The HKO and rotated-pentagon proofs use separate SageMath programs over the relevant number fields.

The QP is therefore the appropriate language when the global objective value, its competing branches, or inequalities between them matter. The next section keeps the same simple facet words but asks a different geometric question: it propagates actual breakpoints through facet-pair sections and detects a closed orbit as a fixed point of an affine return map. This second route is conditional on explicit regularity assumptions, but it exposes the trajectory geometry and provides an independent implementation comparison.

5 Computing closed trajectories with a flow graph

The quadratic program computes capacity by comparing possible facet orders and weights. A second approach follows trajectories through the facets themselves. In the relative interior of a facet, the contact-normalized velocity is constant. Following a chosen facet direction carries a point on an incoming two-dimensional section to an outgoing section by an affine map. Composing these maps around a facet word produces a return map; a fixed point gives a closed trajectory.

We use the local passage geometry of Chaidez and Hutchings [CH21]. The exhaustive search below is justified by the simple-minimizer theorem: some capacity-minimizing trajectory uses each of its facet directions once. It is therefore enough to search finitely many cyclic words, retaining the conditions that make each passage possible. We state the regularity hypotheses before the correctness theorem and discuss the rational implementation after its proof.

5.1 Affine tubes

Let

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, i = 1, \dots, F\}$$

be a bounded full-dimensional polytope with $0 \in \text{int } K$, and assume that all listed inequalities define genuine facets F_i . Put

$$R_i = 2J_0 a_i.$$

As in Section 3, a trajectory on F_i with velocity R_i has action equal to its elapsed time.

For each ordered pair (i, j) for which a_i, a_j are linearly independent, choose affine coordinates

$$\psi_{ij} : \mathbb{R}^2 \longrightarrow H_i \cap H_j, \quad H_i = \{x : \langle a_i, x \rangle = 1\},$$

and let

$$P_{ij} = \{u \in \mathbb{R}^2 : \psi_{ij}(u) \in K\}.$$

The set P_{ij} is allowed to be empty or lower-dimensional. Thus the construction also permits a breakpoint to lie on additional facets; it does not assume that every breakpoint lies in the relative interior of a two-face.

Consider a passage from facet F_p along F_i to F_n , with three distinct facet labels. We consider the case where the incoming direction approaches H_i and the direction along F_i approaches H_n :

$$\langle R_p, a_i \rangle > 0, \quad \langle R_i, a_n \rangle > 0.$$

For $x \in F_p \cap F_i$, the time at which the line $x + tR_i$ reaches H_n is

$$\tau_{pin}(x) = \frac{1 - \langle a_n, x \rangle}{\langle a_n, R_i \rangle}. \tag{22}$$

The denominator is positive. Restrict the starting section to those points for which $x \in K$, $x + \tau_{pin}(x)R_i \in K$, and $\tau_{pin}(x) \geq 0$. On this closed polygonal domain, the endpoint map

$$\varphi_{pin}(x) = x + \tau_{pin}(x)R_i$$

and the action contribution τ_{pin} are affine. Convexity of K ensures that the whole segment lies in $K \cap H_i$.

We describe a tube by a closed starting polygon P , an affine endpoint map Φ , and an affine total-action function A . If (P_1, Φ_1, A_1) and (P_2, Φ_2, A_2) are consecutive tubes in matching coordinates, their concatenation is

$$(P_1 \cap \Phi_1^{-1}(P_2), \Phi_2 \circ \Phi_1, A_1 + A_2 \circ \Phi_1). \tag{23}$$

Thus an empty intermediate intersection excludes every longer word containing that partial tube.

A *simple cyclic facet word* is a cyclic word $w = [s_1, \dots, s_k]$ with distinct entries. To close it, concatenate the primitive tubes for

$$(s_1, \dots, s_k, s_1, s_2).$$

Write the resulting data as (P_w, Φ_w, A_w) . A point $u \in P_w$ is a closed search candidate precisely when $\Phi_w(u) = u$. It represents the displayed word as a strict generalized Reeb orbit precisely when every reconstructed segment time is positive. In that case its action is $A_w(u)$.

Lemma 5.1 (Tube semantics). *For every simple cyclic word whose primitive denominators are positive, the fixed points of Φ_w in P_w are exactly its closed nonnegative-time tube realizations. The fixed points with all segment times positive are exactly the strict closed generalized Reeb orbits represented by w , and their action is A_w .*

Proof. Equation (22) gives the unique next breakpoint and elapsed time for one passage. The endpoint and the elapsed time are affine, and the domain inequalities state exactly that the endpoints and the passage belong to K . Formula (23) says that two passages can be concatenated exactly when their common breakpoint agrees; it also composes their endpoint maps and adds their elapsed times. Induction over the word proves the assertion for the closed tube. Its starting and ending sections have the same coordinates, so closure is equivalent to $\Phi_w(u) = u$. Finally, the closed tube was defined with nonnegative times, whereas the active word records exactly the positive-duration pieces. This gives the strict-time condition. Since $\lambda_0(R_i) = 1$ on F_i , the sum of the elapsed times is the Reeb action. $\square$

Figure 6 follows one word through the two-dimensional polygons P_{ij} . Within each panel, the coordinate axes are orthonormal for the Euclidean metric induced from $\mathbb{R}^4$. Their affine origins and orientations are chosen independently for distinct sections. Each main panel is cropped to the surviving tube cross-section; its inset shows that crop inside the full section P_{ij} . The black cross is the breakpoint of the closed orbit. The start and return panels use the same facet pair, frame, and plot limits. Thus the return-map fixed-point equation is visible as the return of the black cross to its initial local coordinates. The figure is regenerated from exact rational tube geometry; rationals are converted to finite floating-point values only at the JSON/rendering boundary. The JSON distinguishes the orthonormalized plotting coordinates from the raw affine construction-chart vertices and inequalities; those fields must not be read as coordinates in a common chart.

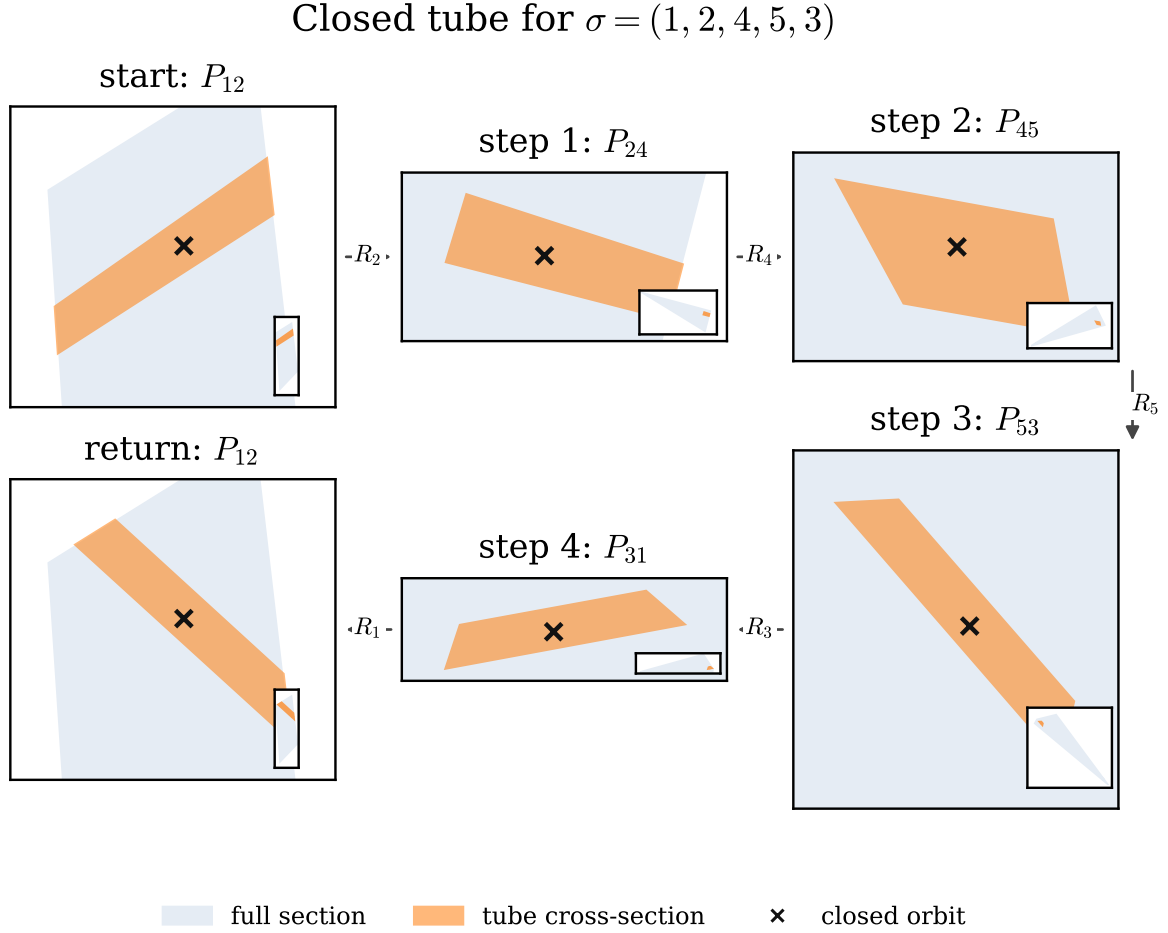


Figure 6: The same word in successive local sections. Each panel plots in a Euclidean-orthonormal basis for the corresponding affine two-plane. The affine origins and orientations are chosen independently for distinct sections; the underlying exact vertices and inequalities remain in the raw construction chart and are not plotted as if they used the orthonormal coordinates. Main panels are tube-focused; each inset locates the colored tube cross-section inside the full blue section. The start and return panels share a frame and plot limits. The arrows show the affine passages, and the black cross returns to its initial coordinates.

5.2 Exhaustive search and correctness

We now state the regularity boundary of the algorithm.

Definition 5.2 (Flow-graph regular presentation). *The facet presentation above is flow-graph regular if:*

1. *for distinct i, j , $F_i \cap F_j \neq \emptyset$ implies $\omega_0(a_i, a_j) \neq 0$;*
2. *every sublist of at most four of the vectors $a_1, \dots, a_F$ is linearly independent; and*
3. *for every simple cyclic word of length at least five which survives the facet-intersection and positive-transition tests and has nonempty closed tube domain, the linear part M_w of Φ_w satisfies $\det(I - M_w) \neq 0$.*

The last condition is called finite-orbit regularity.

Algorithm 5.3 (Idealized exhaustive flow-graph search). *Enumerate every simple cyclic facet word once. For each word:*

1. *discard it if an adjacent facet intersection is empty or an adjacent transition sign is not positive;*
2. *discard it if its length is at most four: under the regularity hypothesis, these facet directions are linearly independent and cannot form a closed positive-time path;*
3. *construct its closed tube (P_w, Φ_w, A_w) ; discard the word if $P_w = \emptyset$, and otherwise solve $\Phi_w(u) = u$;*
4. *reject a singular fixed-point equation as outside the regular case, and otherwise record the unique solution only if it lies in P_w and every reconstructed segment time is positive.*

Return the least positive recorded action.

We justify both pruning steps. If a strict trajectory changes from R_i to R_j at $x \in F_i \cap F_j$, then the inequalities for K immediately before the breakpoint give

$$\langle a_j, R_i \rangle \geq 0.$$

The first condition in Definition 5.2 excludes equality. Thus every trajectory transition has the positive sign used by the search, even when x lies on additional facets. Second, a strict closed orbit with active word $(s_1, \dots, s_k)$ satisfies

$$\sum_{r=1}^k \tau_r R_{s_r} = 0, \quad \tau_r > 0.$$

If $k \leq 4$, this contradicts the linear independence in condition 2.

Theorem 5.4 (Idealized real-arithmetic flow-graph correctness). *Let $K \subset \mathbb{R}^4$ be a bounded full-dimensional convex polytope with $0 \in \text{int } K$, equipped with a flow-graph regular ordered normalized irredundant facet presentation. Then the finite procedure in Algorithm 5.3, with its algebraic and order operations performed exactly over $\mathbb{R}$, returns*

$$c_{\text{EHZ}}(K).$$

Proof. There are finitely many simple cyclic words. By Lemma 5.1, every recorded point is a strict closed generalized Reeb orbit with the recorded action. The two arguments preceding the theorem show that the transition and short-word pruning cannot remove such an orbit. Empty tubes contain no orbit and are discarded before solving a fixed-point equation. Finite-orbit regularity makes every remaining fixed-point equation nonsingular, so no long word reaches the rejection branch and each word has at most one fixed point.

By Theorem 3.4, a capacity-minimizing orbit can be chosen with an active cyclic word in which every used facet direction occurs in one positive-duration interval. Its breakpoints lie in the corresponding facet-pair sections, its transitions have the required positive signs, and its passages satisfy the primitive tube equations. Lemma 5.1 therefore represents it by a strict fixed point with action $c_{\text{EHZ}}(K)$. The returned minimum is no larger than this action, while soundness of every recorded orbit makes it no smaller. $\square$

Corollary 5.5 (Exact rational-arithmetic flow-graph correctness). *Under the hypotheses of Theorem 5.4, assume in addition that $a_i \in \mathbb{Q}^4$ for every facet row. Then rational affine coordinates can be chosen for every facet-pair plane, and Algorithm 5.3 is a finite exact rational-arithmetic algorithm returning $c_{\text{EHZ}}(K)$.*

Proof. Rational Gaussian elimination gives a rational base point and basis for every facet-pair plane. In these coordinates, the section inequalities, primitive times and maps, tube intersections and compositions, and nonsingular fixed-point systems all have rational coefficients. Their feasibility, signs, solutions, and action comparisons can therefore be decided in exact rational arithmetic. The conclusion follows from Theorem 5.4. $\square$

5.3 Why the regular class is generic

The regularity assumptions are explicit algebraic nonvanishing conditions. For fixed facet count there are only finitely many of them. Pairwise symplectic degeneracy and short-word dependence are ordinary polynomial conditions. The only non-immediate point is that the long-word determinant is not the zero rational function.

When all cyclic primitive denominators of a word are nonzero, let M_w^{alg} denote the composition of the primitive linear maps, regardless of their signs or whether the geometric tube domain is nonempty. For a searched tube, this is the linear part M_w of its affine return map.

Proposition 5.6 (Presentation-chamber genericity). *Fix F , and let $\mathcal{C} \subset (\mathbb{R}^4)^F$ be a nonempty Euclidean-open set of ordered normalized dual-row lists defining bounded full-dimensional polytopes with 0 in the interior and all F inequalities irredundant. An open dense subset of $\mathcal{C}$ satisfies the stronger conditions*

1. $\omega_0(a_i, a_j) \neq 0$ for every $i \neq j$;
2. every sublist of at most four rows is linearly independent; and
3. $\det(I - M_w^{\text{alg}}) \neq 0$ for every simple cyclic word w of length at least five.

Consequently every presentation in this subset satisfies the idealized real-arithmetic Theorem 5.4. A presentation in this subset with rational facet rows also satisfies Corollary 5.5; the open-dense statement itself does not assert rational representability or executable coverage for arbitrary real rows.

Proof. The first two families are finite collections of nonzero polynomial conditions. For a fixed word $w = (s_1, \dots, s_k)$, each primitive linear map is

$$\xi \mapsto \xi - \frac{\langle a_{s_{r+2}}, \xi \rangle}{\langle a_{s_{r+2}}, 2J_0 a_{s_{r+1}} \rangle} 2J_0 a_{s_{r+1}}.$$

On the domain where the denominators are nonzero, coordinate bases for the facet-pair planes and hence $\det(I - M_w)$ depend rationally on the rows.

To see that this rational function is nonzero for every $k \geq 5$, use

$$\begin{aligned} A &= (2, 3, -4, -3), & B &= (0, 1, 2, -1), & C &= (-4, -3, 0, -1), \\ D &= (4, -1, 1, 1), & E &= (-4, 2, -4, -4). \end{aligned}$$

All pairings needed below are nonzero. Direct substitution gives

$$\det(I - M_{ABCDE}) = -\frac{10643}{600}, \quad \det(I - M_{ACDE}) = \frac{1}{168}.$$

At the linear-map level, replacing a cyclic subword x, A, B, A, y by x, A, y leaves the surrounding return map unchanged: the middle transition is the identity on $\ker a_A$, and the two flows in the A -direction combine algebraically. Hence the specializations $(A, B)^m, C, D, E$ cover every odd length at least five, while $(A, B)^m, A, C, D, E$ reduce every even length at least six to the second displayed value. Thus no long-word determinant is identically zero.

The common denominator domain is a nonempty Zariski-open subset of an irreducible affine parameter space. Thus a nonzero rational function remains nonzero on every nonempty coordinate-minor chart. The complement of each nonvanishing condition is therefore a proper algebraic set. There are only finitely many sublists and simple words for fixed F , so their common nonzero locus is open dense. Intersecting it with the Euclidean-open chamber $\mathcal{C}$ preserves density. The all-pairs condition is stronger than the nonempty-pair condition in Definition 5.2, and the remaining conditions imply finite-orbit regularity. $\square$

This is chamber-relative genericity for ordered normalized irredundant presentations. It is not the conjectural CH2021 claim that generic polytopes have no Type 2 orbit below an action bound.

5.4 Relation to CH2021

Chaidez–Hutchings study a finite model that compares Reeb dynamics on a four-dimensional polytope with Reeb dynamics on its smoothings [CH21]. Their setting and proof route differ from the general Haim–Kislev formula. They assume that the polytope is *symplectic*; in particular, its two-faces are not Lagrangian, so the relevant combinatorial Reeb direction on each two-face is one-dimensional.

A Chaidez–Hutchings combinatorial Reeb orbit is a cyclic sequence of oriented line segments in the boundary. Their symplectic flow graph records the orbits which avoid the one-skeleton: its vertices are open two-faces, an edge is a Reeb passage through a three-face, and periodic graph orbits correspond to these Type 1 combinatorial Reeb orbits with the same action [CH21, Proposition 2.14]. Their smoothing theory relates this model to Reeb orbits on smooth convex approximations and yields a capacity formula involving Type 1 and Type 2 combinatorial orbits [CH21, Theorems 1.11 and 1.12, Corollary 1.15].

The rational-arithmetic algorithm developed here uses part of the same local geometry but a different global argument. Its nodes are closed facet-pair sections, and it searches simple facet words whose affine return maps have fixed points. The simple-minimizer theorem supplies completeness on the stated regular class. Consequently this algorithm neither implements the CH2021 rotation pruning nor imports the Type 1/Type 2/Type 3 classification as its capacity proof. In particular, an action cutoff alone would not control arbitrarily long helical combinatorial words in the CH2021 model; the finite search here instead comes from simple words with no repeated facet direction.

5.5 Rational implementation and evidence

The current executable implementation targets Corollary 5.5: it represents the normals and all tube arithmetic by rational numbers. It enumerates transition-pruned simple directed cycles, checks the short-word independence hypothesis, constructs and composes affine tubes, solves fixed-point equations in rational arithmetic, and accepts an orbit only after reconstructing strictly positive segment times.

This implementation does not validate an arbitrary raw halfspace description. It assumes that the caller supplies a bounded irredundant presentation in dimension four, together with matching rational matrices for facet intersections and symplectic signs. It rejects a caller-marked nonempty distinct facet pair with zero symplectic pairing, failure of the short-word independence hypothesis, and unsupported singular fixed sets. An inconsistent singular fixed equation, or one whose fixed set misses the tube, is instead a certified no-orbit outcome. Thus the implementation is intended to realize the corollary on rational inputs satisfying both these input requirements and the regularity hypotheses; correspondence is tested, not formally verified.

The earlier binary64 flow-graph prototype was approximate and lacked a sound way to distinguish predicates proved true, proved false, or left undecided. It was retired and supplies no current flow-graph capacity evidence.

On selected deterministic rational F5, F6, and F7 cases, rational flow-graph search agrees with the certified Haim–Kislev/QP scalar aggregate. These comparisons, the retained words, and the targeted resolver tests are implementation evidence for the stated rational inputs; they do not prove the idealized algorithm or the CH2021 capacity scope. Targeted tests also exercise tube construction, fixed-point reconstruction, and rejection of unsupported inputs. These checks are falsification and correspondence evidence on selected examples; they do not prove tube semantics or broad input coverage.

6 First-Order Variation in Facet Coordinates

We now let the supporting hyperplanes move and ask how the systolic ratio changes. The volume has a direct derivative in a neighborhood with fixed simple face structure. Capacity is more delicate: the Haim–Kislev formula expresses it through a maximum over words, and the maximizing word can change.

There are two useful calculations. A smooth feasible choice of weights gives a differentiable upper bound for the ratio. If the constrained optimizer is nondegenerate, its weights also vary smoothly, and we can differentiate its optimized value. The first construction needs less: it will supply the upper bounds in the HKO proof even at singular stationarity systems. For local search, the second construction gives branch gradients that can be compared before proposing a step.

6.1 Row coordinates and the volume variation

Fix a labelled presentation

$$K(a) = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1, i = 1, \dots, F\}, \quad a = (a_1, \dots, a_F). \quad (24)$$

The vectors a_i are the dual vertices used in Section 4. A row perturbation is a tuple $h = (h_1, \dots, h_F) \in (\mathbb{R}^4)^F$, and we write $a(t) = a + th$. Throughout this section we work in a neighborhood in which $K(a)$ remains a bounded full-dimensional polytope, the listed rows remain irredundant, and the same simple face combinatorics persists. In particular, $V(a) > 0$, and the Haim–Kislev formula applies. These are open conditions at a simple labelled polytope.

For the facet F_i , let S_i be its three-dimensional Euclidean volume and let $\bar{x}_i$ be its centroid. The volume derivative has a particularly simple expression in row coordinates.

Proposition 6.1 (Volume first variation). *In the stable simple chart above, with $V(a) = \text{vol}_4(K(a))$,*

$$DV(a)[h] = - \sum_{i=1}^F \frac{S_i}{\|a_i\|} \langle \bar{x}_i, h_i \rangle. \quad (25)$$

Proof. At a point $x \in F_i$, differentiating the moving hyperplane equation $\langle a_i(t), x(t) \rangle = 1$ gives

$$\langle a_i, \dot{x} \rangle = -\langle h_i, x \rangle.$$

Thus the outward normal velocity of this facet is $-\langle h_i, x \rangle / \|a_i\|$. The first variation of volume is the integral of the outward normal velocity over the boundary. Integrating on each facet and using $\int_{F_i} x dS = S_i \bar{x}_i$ gives (25). Intersections of facets have three-dimensional measure zero and do not contribute to the boundary integral. $\square$

The formula displays both parts of a row perturbation at once: changing $\|a_i\|$ moves the supporting hyperplane, while changing its direction tilts that hyperplane. It is also convenient computationally, because it needs only facet volumes and centroids once the incidence data are known.

6.2 Feasible sections and smooth maxima

Fix an ordered word $\sigma = (\sigma_1, \dots, \sigma_m)$ with distinct facets. Write

$$C_\sigma(a)\beta = \left(\sum_{k=1}^m \beta_k a_{\sigma_k}, \sum_{k=1}^m \beta_k \right), \quad e = (0, 0, 0, 0, 1),$$

so that the Haim–Kislev feasibility equations are $C_\sigma(a)\beta = e$ and $\beta \geq 0$. We use the quadratic objective from (19):

$$Q_\sigma(a, \beta) = \sum_{1 \leq j < k \leq m} \beta_j \beta_k \omega_0(a_{\sigma_j}, a_{\sigma_k}). \quad (26)$$

For later differentiation, its explicit row variation at fixed β is

$$D_a Q_\sigma(a, \beta)[h] = \sum_{1 \leq j < k \leq m} \beta_j \beta_k (\omega_0(h_{\sigma_j}, a_{\sigma_k}) + \omega_0(a_{\sigma_j}, h_{\sigma_k})). \quad (27)$$

First consider any smooth feasible choice, without imposing stationarity.

Lemma 6.2 (Feasible-section upper function). *Suppose that $\beta(a)$ is smooth near a_0 ,*

$$C_\sigma(a)\beta(a) = e, \quad \beta(a) > 0, \quad Q_\sigma(a, \beta(a)) > 0.$$

Define

$$A_\sigma(a) = \frac{1}{2Q_\sigma(a, \beta(a))}, \quad U_\sigma(a) = \frac{A_\sigma(a)^2}{2V(a)}.$$

Then

$$\text{sys}(K(a)) \leq U_\sigma(a) \quad (28)$$

throughout this neighborhood. If the chosen weights attain the global Haim–Kislev maximum at a_0 , then equality holds in (28) at a_0 .

Proof. The chosen $\beta(a)$ is feasible for the global maximum defining $Q_{\max}(a)$. Hence $Q_{\max}(a) \geq Q_\sigma(a, \beta(a))$. The Haim–Kislev formula gives

$$c_{\text{EHZ}}(K(a)) = \frac{1}{2Q_{\max}(a)} \leq A_\sigma(a).$$

Squaring and dividing by the positive number $2V(a)$ proves the upper bound. A globally maximizing choice gives equality at the base point. $\square$

The lemma requires no KKT system. For example, if $C_\sigma(a_0)$ has rank five and $\beta_0 > 0$ is feasible, choose an invertible five-column minor and hold the other $m - 5$ coordinates of β fixed. Solving the five feasibility equations gives a smooth local section through β_0 ; positivity persists after shrinking the neighborhood. This elementary construction is the one used for the singular seven-facet data in the HKO certificate.

At a nondegenerate constrained maximum, the KKT equations instead choose a canonical optimizing branch and remove the derivative of β from the value derivative. Let $H_\sigma(a)$ denote the Hessian of $Q_\sigma(a, \cdot)$ with respect to β .

Definition 6.3 (Nondegenerate Haim–Kislev branch). *A feasible pair (σ, β_0) at a_0 is a nondegenerate branch if*

1. $C_\sigma(a_0)$ has rank five;
2. $\beta_0 > 0$ and $Q_\sigma(a_0, \beta_0) > 0$;
3. β_0 is a constrained critical point of $Q_\sigma(a_0, \cdot)$; and
4. $H_\sigma(a_0)$ is negative definite on $\ker C_\sigma(a_0)$.

For $m = 5$, the kernel is zero and the last condition is vacuous.

Proposition 6.4 (Derivative of a nondegenerate branch). *Let (σ, β_0) be a nondegenerate branch at a_0 . Then, after shrinking the row chart, there is a unique smooth constrained maximizer $\beta_\sigma(a) > 0$ near β_0 . Put*

$$q_\sigma(a) = Q_\sigma(a, \beta_\sigma(a)), \quad U_\sigma(a) = \frac{1}{2V(a)} \left(\frac{1}{2q_\sigma(a)} \right)^2.$$

If $\lambda_\sigma \in \mathbb{R}^5$ is chosen so that the KKT equation is

$$D_\beta Q_\sigma(a_0, \beta_0) = C_\sigma(a_0)^\top \lambda_\sigma,$$

then

$$Dq_\sigma(a_0)[h] = D_a Q_\sigma(a_0, \beta_0)[h] - \langle \lambda_\sigma, D_a C_\sigma(a_0)[h] \beta_0 \rangle. \quad (29)$$

Consequently,

$$\frac{DU_\sigma(a_0)[h]}{U_\sigma(a_0)} = -2 \frac{Dq_\sigma(a_0)[h]}{q_\sigma(a_0)} - \frac{DV(a_0)[h]}{V(a_0)}. \quad (30)$$

Proof. Full rank and negative definiteness make the saddle-point KKT matrix nonsingular. The implicit function theorem therefore gives a unique smooth critical point near β_0 . Negative definiteness and positivity are open conditions, so this point remains the positive constrained maximum.

Differentiate $q_\sigma(a) = Q_\sigma(a, \beta_\sigma(a))$. The term containing $D\beta_\sigma$ is

$$D_\beta Q_\sigma D\beta_\sigma = \lambda_\sigma^\top C_\sigma D\beta_\sigma.$$

Differentiating $C_\sigma(a)\beta_\sigma(a) = e$ gives $C_\sigma D\beta_\sigma = -(D_a C_\sigma)\beta_\sigma$, which proves (29). The logarithmic derivative of $U_\sigma = (2V)^{-1}(2q_\sigma)^{-2}$ is (30). $\square$

Equations (27), (25), and (29) are the practical first-order calculation. They use the current rows, the word, its positive KKT weights and multipliers, and facet volumes and centroids. They do not require finite differences of the full capacity computation.

6.3 From branches to the systolic ratio

A nondegenerate branch always gives the upper function U_σ , but it gives the actual systolic ratio only while it realizes the global Haim–Kislev maximum.

Suppose that finitely many nondegenerate branches have fixed-word maxima $q_1, \dots, q_r$ on a neighborhood of a_0 , and that they cover the global finite problem there in the explicit sense

$$Q_{\max}(a) = \max_{1 \leq j \leq r} q_j(a).$$

For example, this identity follows from a complete retained word list and a persistent value gap for every discarded word. Then

$$\text{sys}(K(a)) = \min_{1 \leq j \leq r} U_j(a). \quad (31)$$

If $I(a_0) = \{j : U_j(a_0) = \text{sys}(K(a_0))\}$, the one-sided directional derivative is

$$D^+ \text{sys}(a_0; h) = \min_{j \in I(a_0)} DU_j(a_0)[h]. \quad (32)$$

Indeed, this is the first-order expansion of a finite minimum of smooth functions. In particular, a unique active nondegenerate branch gives an ordinary gradient. At a tie of nondegenerate branches, the directional derivative is generally only the lower envelope of the tied branch slopes.

The hypothesis preceding (31) contains real mathematical work. A list returned by one numerical solve, or a tolerance window around its smallest actions, is not by itself a certificate of the displayed identity.

6.4 What the first-order model contributes to search

On a nondegenerate branch patch with active branch rows $r_j = DU_j(a_0)$, a common first-order ascent direction in a chosen linear slice S can be sought through the maximin problem

$$\max_{h \in S, \|h\| \leq 1} \min_{j \in I(a_0)} r_j(h). \quad (33)$$

The slice removes directions generated by the systolic-ratio symmetries from Definition 2.2. The true ratio is constant on the corresponding group orbits, so its derivative vanishes on their tangent space; S is chosen as a complementary linear slice. A positive value in (33) predicts improvement of every active nondegenerate branch. The prediction still has to be checked by recomputing the capacity after a finite step.

At finite distance, currently inactive branches can matter before they are active. A useful semi-local model therefore retains their base gaps:

$$\min_{j \in W} (U_j(a_0) - \text{sys}(K(a_0)) + DU_j(a_0)[h]), \quad (34)$$

where W is a chosen branch window. Formula (34) explains why a single active gradient can miss a nearby branch switch, but it is a prediction model rather than an infinitesimal theorem. Its reliability depends on the window containing the branches that become relevant, on the branches remaining feasible, and on the linearization error at the proposed step.

One retained finite experiment illustrates this distinction. On a fixed panel of twelve random ten-facet starts, the tested multi-direction method accepted a positive move at each of eight configured trace iterations, but all runs stopped at their iteration cap and every endpoint scan still found a further above-threshold move. Thus the evidence supports finite-step progress and a measured computation cost on that panel, not endpoint local maximality. The branch-window model has not yet received a corresponding reproducible finite-distance validation and is not used as thesis evidence here.

6.5 The non-generic boundary

The smooth branch calculation can break down in four ways.

- If a maximizing weight is zero, deleting that coordinate is valid for the capacity value at the base point. It need not preserve the nearby slope: the longer word may acquire a positive weight on one side of the perturbation and define a different branch there.
- If the constraint rank drops or the KKT matrix is singular, the implicit function theorem does not provide a unique smooth optimizing branch. A continuum of optimizers may nevertheless have a well-defined value.
- A tie is described by (32) only when all relevant tied branches extend smoothly and the stated coverage and gap hypotheses hold. Base-point branch enumeration alone does not prove this.
- When facet labels become redundant or the face combinatorics changes, both the branch family and the smooth volume formula require a new local chart. Hausdorff continuity of capacity and volume across that boundary does not imply differentiability there.

All constraints and objectives are semialgebraic. A possible route to a fully general one-sided theory would organize germs of feasible optimizing branches by a finite decomposition of direction space. Such a construction would have to record support appearances, singular optimizer sets, and dominance of every discarded germ. Obtaining a complete first-order description at an arbitrary non-generic polytope remains open here.

6.6 Application at HKO

The HKO argument uses the feasible-section calculation. It constructs twenty-six upper bounds that agree with the true ratio at the base body. Together their derivatives force at least one bound to decrease in every direction transverse to the symmetries. Section 7 gives the construction and proves that this directional statement yields a strict local maximum. No continuation of an optimizing branch is required.

6.7 Implementation and validation status

The implementation evaluates the branch and volume formulas in dual-vertex coordinates. Regression tests compare the volume derivative with central finite differences and compare a fixed-word capacity derivative with central finite differences on known fixtures. An exact rational implementation of the capacity-gradient formula is also compared with the floating-point formula on a rational simplex fixture. These checks test the correspondence between the formulas and their implementations; they do not certify branch coverage, finite-step prediction, or local maximality.

For the HKO theorem, the evidence standard is stronger and different. The SageMath verifier checks the selected feasible sections and their derivative rows exactly in the relevant algebraic number field. The mathematical use of those rows is the upper-bound and transverse-slice argument in Section 7.

7 Local maximality of the HKO pentagon product

Let P be the regular pentagon of circumradius one, centred at the origin and with vertex $(1, 0)$. In coordinates (q_1, q_2, p_1, p_2) put $K_0 = P \times R(-\pi/2)P$, with symplectic form $\omega_0 = \sum_j dq_j \wedge dp_j$. The Haim–Kislev–Ostrover capacity formula [HO26] for this body gives

$$c_{\text{EHZ}}(K_0) = 2 \cos(\pi/10)(1 + \cos(\pi/5)), \quad \text{sys}(K_0) = \frac{3 + \sqrt{5}}{5} > 1.$$

Here $\text{sys}(K) = c_{\text{EHZ}}(K)^2 / (2 \text{vol}_4 K)$. The counterexample itself is due to Haim–Kislev and Ostrover; the result proved here concerns its local maximality when the number of facets stays ten.

Theorem 7.1. *There is a Hausdorff neighborhood $\mathcal{U}$ of K_0 among convex four-dimensional polytopes with exactly ten facets such that $\text{sys}(K) \leq \text{sys}(K_0)$ for $K \in \mathcal{U}$. Equality holds precisely for those bodies in $\mathcal{U}$ obtained from K_0 by translation, positive dilation and a linear symplectic map.*

7.1 The proof strategy

The ten supporting hyperplanes may move in all four coordinate directions, so the nearby bodies include polytopes that are no longer products. Write

$$K(a) = \{x \in \mathbb{R}^4 : a_i \cdot x \leq 1, i = 1, \dots, 10\}, \quad a = (a_1, \dots, a_{10}),$$

and let a_0 be the row list of K_0 .

We construct twenty-six smooth upper bounds U_j for the systolic ratio, all equal to it at a_0 . If one of these bounds decreases, the ratio decreases with it. Different bounds handle different directions. The proof first identifies their derivatives and the symmetry directions. A linear-algebra lemma then gives a uniform negative slope transverse to symmetry. Only at the end do we pass from these derivatives to a finite neighborhood and recover the equality cases.

7.2 Local coordinates and volume

Lemma 7.2 (A simple row neighborhood). *There is a neighborhood $\mathcal{N}$ of a_0 in $(\mathbb{R}^4)^{10}$ on which $K(a)$ is bounded and simple, with the same vertex–facet incidences as K_0 . Every sufficiently Hausdorff-close ten-facet polytope has a labelled presentation with $a \in \mathcal{N}$, and its rows tend to a_0 as the body tends to K_0 .*

Proof. The origin stays in the interior under sufficiently small Hausdorff perturbations of $K_0 = K(a_0)$. One way to see this is through support functions: $h_K(u) = \max_{x \in K} u \cdot x$. Write B for the Euclidean unit ball. If $rB \subset K_0$ and the Hausdorff distance from K to K_0 is less than $\epsilon < r$, then $h_K(u) \geq h_{K_0}(u) - \epsilon \geq r - \epsilon$ for every unit vector u . Thus $(r - \epsilon)B \subset K$. On such a neighborhood polarity is continuous: the radial function, the distance from zero to the boundary along a unit direction, of K° is $1/h_K$ on the unit sphere, and the common positive lower bound on h_K makes these reciprocals converge uniformly. The polars are uniformly bounded convex bodies, so this also gives Hausdorff convergence.

The polar K_0° has the ten vertices $a_{0,i}$. Choose disjoint small balls about them. Each vertex is uniquely exposed by some linear functional ℓ_i . For polars sufficiently close to K_0° , every maximizer of ℓ_i lies in the corresponding ball. Otherwise a convergent sequence of maximizers outside that ball would limit to a maximizer on K_0° different from $a_{0,i}$. A maximizing face contains a vertex. A polytope K with exactly ten facets has a polar with exactly ten vertices; hence there is exactly one polar vertex in each ball and none elsewhere. Labelling these vertices a_i gives

$$K = \{x : a_i \cdot x \leq 1, i = 1, \dots, 10\}, \quad a_i \longrightarrow a_{0,i} \quad \text{as } K \longrightarrow K_0.$$

For completeness, the face structure is stable in this neighbourhood. The base body is a product of two polygons and is simple: precisely four facets meet at each vertex, with independent normals. Their four equations therefore determine a smoothly varying intersection under a small change of a . All the remaining inequalities have strict slack at that intersection, so it remains a vertex with exactly those incident facets.

No new vertex can appear. The convex hull of the base normals contains a ball about zero; the convex hull of sufficiently close normals contains a smaller common ball. Consequently the bodies $K(a)$ are uniformly bounded. If arbitrarily close parameters had a new vertex, choose four independent active constraints there. There are finitely many four-element subsets, so along a subsequence the same subset is active and the vertices converge to a point of K_0 where those four constraints are equalities. Simplicity forces them to be the incident facets of one of the original vertices. Their intersection is the continuation already constructed, contradicting newness. This argument also excludes a new vertex arising from a subset whose normals are dependent at the base point; no invertibility assumption for such a subset is needed.

The vertex–facet incidences therefore remain unchanged. In particular each inequality still defines a facet. Conversely, small changes of the ten normals give Hausdorff-small changes of $K(a)$: their convex hulls converge, contain a common ball about zero, and polarity is continuous there. □

Lemma 7.3 (Smooth volume and its derivative). *On a sufficiently small neighborhood $\mathcal{N}$ as above, $V(a) = \text{vol}_4 K(a)$ is smooth and positive. If $F_i(a)$ has centroid $\bar{x}_i(a)$, then*

$$DV(a)[h] = - \sum_{i=1}^{10} \frac{\text{vol}_3(F_i(a))}{\|a_i\|} \langle h_i, \bar{x}_i(a) \rangle.$$

Proof. The fixed incidences express each vertex smoothly in the rows. A fixed boundary triangulation then expresses volume smoothly in those vertices. On the moving facet, differentiating $\langle a_i + \epsilon h_i, x(\epsilon) \rangle = 1$ gives outward normal velocity $-\langle h_i, x \rangle / \|a_i\|$. Integrating this velocity over the facets and using their centroids gives the formula. □

The concrete base point and its volume

Put $t = \tan(\pi/5)$, $s = (5 - t^2)/2 = \sqrt{5}$, and

$$t^4 - 10t^2 + 5 = 0, \quad 0 < t < 1, \quad \alpha = (3 - s)/2, \quad b = t(1 + s)/2, \quad d = s - 1.$$

The selected root is unique in $(0, 1)$: the polynomial is strictly decreasing there and changes sign. For the circumradius-one pentagon with vertex $(1, 0)$, the edge normals have arguments $(2k + 1)\pi/5$ and lengths $\sec(\pi/5) = d$. Rotating the second factor by $-\pi/2$ gives the following ten rows $a_{0,i}$, in (q_1, q_2, p_1, p_2) order:

$$\begin{pmatrix} 1 & t & 0 & 0 \\ -\alpha & b & 0 & 0 \\ -d & 0 & 0 & 0 \\ -\alpha & -b & 0 & 0 \\ 1 & -t & 0 & 0 \\ 0 & 0 & t & -1 \\ 0 & 0 & b & \alpha \\ 0 & 0 & 0 & d \\ 0 & 0 & -b & \alpha \\ 0 & 0 & -t & -1 \end{pmatrix}.$$

These rows give the normalized facet presentation of K_0 . Writing $\theta = \pi/5$, the area of one pentagon is the sum of five triangles with central angle 2θ , hence $B = \frac{5}{2} \sin(2\theta) = 5 \sin \theta \cos \theta$. Fubini therefore gives

$$V_0 = B^2 = 25 \sin^2 \theta \cos^2 \theta = \frac{25(5+s)}{32} = \frac{25(15-t^2)}{64}.$$

To specialize Lemma 7.3, observe that F_i is an edge times the other centered pentagon. Its centroid is $\bar{x}_i = a_{0,i}/\|a_{0,i}\|^2$, its three-volume is $2 \sin \theta B$, and $\|a_{0,i}\| = \sec \theta$. Consequently

$$DV_0[h] = -\mu \sum_{i=1}^{10} a_{0,i} \cdot h_i, \quad \mu = 10 \sin^2 \theta \cos^4 \theta = \frac{25}{32} + \frac{5s}{16}.$$

This gives the volume derivative for every row perturbation at the base point.

7.3 The touching upper bounds

For an ordered facet word $\sigma = (\sigma_1, \dots, \sigma_m)$, write

$$C_\sigma(a) = \begin{pmatrix} a_{\sigma_1} & \cdots & a_{\sigma_m} \\ 1 & \cdots & 1 \end{pmatrix}, \quad e = (0, 0, 0, 0, 1)^T.$$

The equations $C_\sigma(a)\beta = e$ impose closure and normalization. For positive feasible weights let

$$Q_\sigma(a, \beta) = \sum_{j < i} \beta_i \beta_j \omega_0(a_{\sigma_j}, a_{\sigma_i}), \quad A_\sigma(a) = \frac{1}{2Q_\sigma(a, \beta(a))}, \quad U_\sigma(a) = \frac{A_\sigma(a)^2}{2V(a)}.$$

When $Q_\sigma > 0$, the variational capacity formula gives $\text{sys}(K(a)) \leq U_\sigma(a)$.

Let T be the tangent space at a_0 to the orbit generated by translations, positive dilations and linear symplectic maps. The following lemma records the finite calculation that will be used in the rest of the proof.

Lemma 7.4 (Exact upper bounds and symmetry directions). *The space T has dimension fifteen. On a neighborhood of a_0 there are twenty-six smooth functions U_j such that*

$$\text{sys}(K(a)) \leq U_j(a), \quad U_j(a_0) = \text{sys}(K_0).$$

Their derivative rows $r_j = DU_j(a_0)$ vanish on T , have rank twenty-five, and satisfy

$$\sum_{j=1}^{26} \lambda_j r_j = 0, \quad \lambda_j > 0, \quad \sum_{j=1}^{26} \lambda_j = 1.$$

The construction and computer-algebra verification of this lemma occupy the following calculations. Holding selected weights fixed and solving the five closure and normalization equations for the others produces the smooth sections: an invertible five-column minor gives smoothness, and strict positivity at the base point persists nearby.

One of the twenty-six upper bounds

We now work through one of the twenty-six upper bounds to illustrate the construction and differentiation. The other twenty-five are constructed in the same way. The exact computations for all twenty-six were verified using a computer algebra system. Appendix C.1 gives the complete program with explanations of its checks.

Consider the seven-facet word and position sets

$$\sigma = (2, 3, 8, 9, 5, 4, 6), \quad I = (1, 2, 3, 4, 5), \quad J = (6, 7).$$

We solve closure and normalization for the weights at positions I and hold the weights at positions J fixed. These are positions in the word, not facet labels. Set

$$r = \frac{6909830056250513}{10^{17}}, \quad c = \frac{t^2 + 5}{40}, \quad z = \frac{5 - t^2}{20} = \frac{s}{10}, \quad k = \frac{3 - t^2}{4}.$$

Fix $\beta_6 = r$ and $\beta_7 = z$; the decimal-looking rational is an exact rational choice, not a rounding assertion. The complete base vector is

$$\beta_0 = (c + kr, c - r, c, c, z - kr, r, z)^T.$$

At this word the closure matrix's selected minor is

$$M = C_\sigma(a_0)_I = \begin{pmatrix} -\alpha & -d & 0 & 0 & 1 \\ b & 0 & 0 & 0 & -t \\ 0 & 0 & 0 & -b & 0 \\ 0 & 0 & d & \alpha & 0 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad \det M = 10(1 - t^2) > 0.$$

The full matrix is obtained by adjoining columns $(a_4, 1)^T$ and $(a_6, 1)^T$. Writing $e = (0, 0, 0, 0, 1)^T$, define the nearby section by

$$\beta_J(a) = (r, z)^T, \quad \beta_I(a) = C_\sigma(a)_I^{-1}(e - C_\sigma(a)_J(r, z)^T).$$

Direct substitution gives $C_\sigma(a_0)\beta_0 = e$ and $\beta_0 > 0$. For example $0.726 < t < 0.727$ and $0.069 < r < 0.070$ suffice to check all seven signs from the displayed formulas. Invertibility and positivity persist in a neighborhood.

Use $\omega(x, y) = x_{q_1}y_{p_1} + x_{q_2}y_{p_2} - x_{p_1}y_{q_1} - x_{p_2}y_{q_2}$ and

$$Q_\sigma(a, \beta) = \sum_{j < i} \beta_i \beta_j \omega(a_{\sigma_j}, a_{\sigma_i}).$$

At the fixed base geometry, solving closure with arbitrary $(\beta_6, \beta_7) = (u, v)$ gives the particularly simple identity

$$Q_\sigma = -4tv^2 + (2t - 2t^3/5)v.$$

It is independent of u and has its maximum at $v = z$; its value is $Q_0 = t/5$. Thus its Hessian on the two-dimensional closure affine space is singular (in the (u, v) coordinates it is $\text{diag}(0, -8t)$). The five-by-five closure minor is nevertheless invertible, so the displayed feasible section continues smoothly and supplies an upper bound nearby.

$$A_0 = \frac{1}{2Q_0} = \frac{5}{2t} = 5t - \frac{t^3}{2}, \quad U_0 = \frac{A_0^2}{2V_0} = \frac{11 - t^2}{10} > 1.$$

The HKO capacity formula gives $c_{\text{EHZ}}(K(a_0)) = 2 \cos(\pi/10)(1 + \cos(\pi/5)) = A_0$. Thus the constructed upper bound agrees with the systolic ratio at the base point.

A worked coordinate outside product-preserving perturbations

We now choose a perturbation that varies the first momentum component of the second q -facet's normal:

$$h_2 = (0, 0, 1, 0), \quad h_i = 0 \quad (i \neq 2).$$

Put $w = c + kr = \beta_{0,1}$. Differentiated closure gives

$$D\beta_J[h] = 0, \quad D\beta_I[h] = -wM^{-1}(0, 0, 1, 0, 0)^T = w \begin{pmatrix} t^3/20 - t/4 \\ t^3/20 - t/4 \\ t^3/40 - 3t/8 \\ -3t^3/20 + 5t/4 \\ t^3/40 - 3t/8 \end{pmatrix}.$$

Inserting these values into the product rule for Q gives the exact scalar

$$DQ[h] = \frac{3-t^2}{80} + \left(\frac{3t^2}{20} - \frac{1}{4} \right) r - \frac{t^2+5}{4} r^2 =: q_h.$$

Since a_2 has zero momentum coordinates, $DV_0[h] = 0$, while

$$DA[h] = -\frac{25}{2t^2}q_h, \quad DU[h] = -\frac{125}{4t^3V_0}q_h.$$

This coefficient of the upper-bound derivative is approximately -0.1801707325 . Changing only the two polygon factors keeps every q -facet normal in the q -plane. This perturbation adds a p_1 -component to one of those normals, so the derivative describes a change of the upper bound outside that product family.

7.4 Computing the twenty-six derivatives

The accompanying table gives the exact fixed values for every section, including the worked entry. Except at entry 22, the minor positions are $I = (1, 2, 3, 4, 5)$ and $J = (6, \dots, m)$. Entry 22 instead has $I = (1, 2, 3, 4, 6)$ and $J = (5, 7)$. The table is ordered by the witness's zero-based certificate index. For each table row set $\beta_J = u$ and compute

$$\beta_I = C_I^{-1}(e - C_J u), \quad (D\beta)_J[h] = 0, \quad (D\beta)_I[h] = -C_I^{-1}(DC[h])\beta.$$

For each of the forty coordinate directions, compute

$$\begin{aligned} DQ[h] &= \sum_{j < i} ((D\beta_i[h])\beta_j + \beta_i(D\beta_j[h]))\omega(a_{\sigma_j}, a_{\sigma_i}) \\ &\quad + \sum_{j < i} \beta_i\beta_j(\omega(h_{\sigma_j}, a_{\sigma_i}) + \omega(a_{\sigma_j}, h_{\sigma_i})), \\ R[h] &= -\frac{A_0}{2V_0Q_0^2}DQ[h] - \frac{A_0^2}{2V_0^2}DV_0[h]. \end{aligned}$$

Together with the explicit table these formulas specify the entire 26×40 matrix R exactly.

The finite predicates are: every word has distinct valid labels; $\det C_I \neq 0$; $C\beta = e$; all $\beta_i > 0$; $Q = t/5$; the differentiated closure equations hold; the rows annihilate the fifteen symmetry columns; $\text{rank } R = 25$; and the one-dimensional kernel of R^T has a vector whose entries have one strict sign. Normalize that vector to sum to one, producing $\lambda > 0$, $\lambda^T R = 0$. The symmetry columns are explicitly $h_i = -a_{i,j}a_i$ for $j = 1, \dots, 4$, $h_i = -a_i$, and $h_i = -X^T a_i$ for the ten generators

$$X = \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix}, \quad B = B^T, \quad C = C^T,$$

using the four matrix units for A and the three symmetric matrix units for each of B, C . Their column rank is required to be fifteen. These exact arithmetic checks establish the asserted independence, rank and positive relation. The constructions above give smooth feasible sections on a common neighborhood, and the HKO capacity formula gives the touching identities. This proves Lemma 7.4.

7.5 The transverse slope estimate

The exact calculation is now summarized by Lemma 7.4. We use only its rank and positivity statements in the remaining argument.

Lemma 7.5 (Uniformly negative transverse slopes). *Choose a linear complement S to the symmetry tangent space T , so that $(\mathbb{R}^4)^{10} = T \oplus S$ and $\dim S = 25$. There is $\eta > 0$ such that*

$$\min_{1 \leq j \leq 26} r_j(v) \leq -\eta \|v\| \quad \text{for every } v \in S.$$

Proof. The rows have rank twenty-five and vanish on the fifteen-dimensional space T . Their common kernel is therefore exactly T , and their restrictions span S^* . For $v \in S \setminus \{0\}$, suppose that every $r_j(v)$ were nonnegative. Their positively weighted sum is zero, so each would have to vanish. Then $v \in T \cap S = \{0\}$, a contradiction. Thus $\min_j r_j(v) < 0$ for every nonzero transverse direction.

The function $v \mapsto \min_j r_j(v)$ is continuous on the unit sphere of S . Its maximum there is strictly negative by compactness. Write that maximum as $-\eta$ and use homogeneity to obtain the assertion. $\square$

Figure 7 illustrates the slope condition in one and two transverse dimensions.

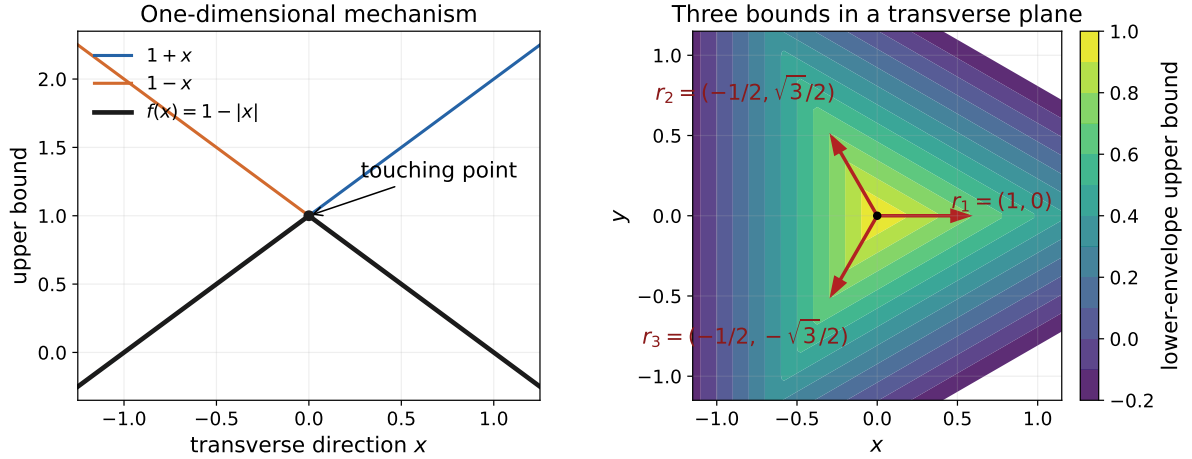


Figure 7: Toy models of touching upper bounds. Left: the lower envelope of $1+x$ and $1-x$ is $1-|x|$. Right: level sets of the lower envelope of three affine functions with gradients $(1,0)$, $(-1/2, \sqrt{3}/2)$ and $(-1/2, -\sqrt{3}/2)$. The gradients span the plane and sum to zero, and the envelope decreases in every nonzero direction. In the proof the systolic ratio lies below the envelope and agrees with it at the base point.

7.6 From slopes to a neighborhood

We now pass from the derivative estimate to nearby bodies. The uniform constant η lets us choose one neighborhood for all transverse directions.

Lemma 7.6 (Strict decrease on the slice). *For all sufficiently small nonzero $v \in S$,*

$$\text{sys}(K(a_0 + v)) \leq \text{sys}(K_0) - \frac{\eta}{2} \|v\| < \text{sys}(K_0).$$

Proof. There are only twenty-six smooth upper bounds. Their differentiability at a_0 therefore supplies a common neighborhood in which

$$|U_j(a_0 + v) - U_j(a_0) - r_j(v)| \leq \frac{\eta}{2} \|v\| \quad (j = 1, \dots, 26).$$

Shrink it so that all bounds are defined and dominate the ratio there. For each $v \in S$, Lemma 7.5 supplies an index with $r_j(v) \leq -\eta\|v\|$. Since $U_j(a_0) = \text{sys}(K_0)$,

$$\text{sys}(K(a_0 + v)) \leq U_j(a_0 + v) \leq \text{sys}(K_0) - \frac{\eta}{2}\|v\|. \quad \square$$

Lemma 7.7 (Local symmetry decomposition). *Every row list sufficiently close to a_0 is the image of some $a_0 + v$, with $v \in S$ small, under a symmetry close to the identity. The symmetry and v can be chosen smoothly in local group coordinates.*

Proof. Translation by y , dilation by $\rho > 0$ and a linear symplectic map M act on the rows, respectively, by

$$a_i \mapsto \frac{a_i}{1 + a_i \cdot y}, \quad a_i \mapsto \rho^{-1}a_i, \quad a_i \mapsto M^{-T}a_i.$$

These actions are smooth near the identity. Choose fifteen local coordinates θ for this symmetry group and write its action as $g(\theta)$. The map

$$\Phi(\theta, v) = g(\theta) \cdot (a_0 + v), \quad v \in S,$$

has derivative at $(0, 0)$ with image T in the group variables and image S in the slice variables. The fifteen symmetry derivatives are independent by Lemma 7.4. Thus $D\Phi(0, 0)$ is an isomorphism, and the inverse function theorem gives the decomposition. $\square$

The toy model in Figure 8 shows how a strict transverse maximum becomes a family of equal maxima along symmetry.

Symmetry extension of a strict transverse maximum

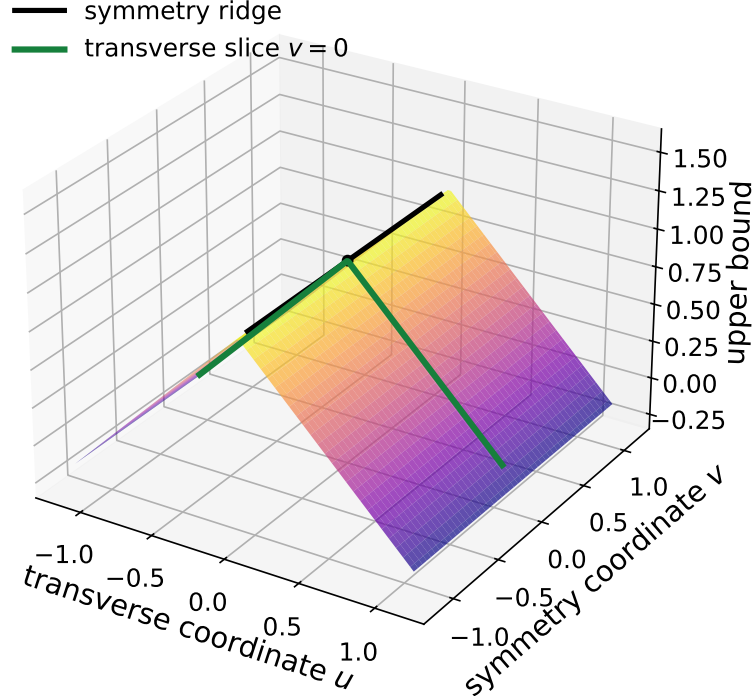


Figure 8: The toy function $f(u, v) = 1 - |u|$ has a strict maximum on the transverse slice $v = 0$ and is constant along the symmetry coordinate v . Its maxima form the ridge $u = 0$. In the HKO proof the symmetry tangent space T has dimension fifteen and the transverse space S has dimension twenty-five. The picture illustrates their roles in the local decomposition.

Proof of Theorem 7.1. By Lemma 7.2, a sufficiently Hausdorff-close ten-facet body has a row list in the neighborhood of Lemma 7.7. It is therefore a symmetry image of $K(a_0 + v)$ for some small $v \in S$. The ratio is unchanged by that symmetry. Lemma 7.6 gives a strict decrease if $v \neq 0$, and equality if $v = 0$. Equality consequently forces the body to be a symmetry image of K_0 . Conversely, every such image has the same ratio as K_0 . $\square$

The twenty-six exact choices

The constants $c = (t^2 + 5)/40$ and $z = (5 - t^2)/20$ are as above. Facet labels and positions start at one; entry numbers start at zero to identify the rows in the accompanying computational witness. Every finite decimal in the table denotes its exact rational value.

Entry	Word σ	Fixed positions J	Exact fixed values
0	(1, 2, 8, 4, 10, 6)	(6)	(c)
1	(1, 2, 8, 7, 4, 10)	(6)	(z)
2	(2, 3, 8, 9, 5, 4, 6)	(6, 7)	(0.06909830056250513, z)
3	(1, 8, 7, 3, 4, 10)	(6)	(z)
4	(2, 9, 8, 5, 4, 6)	(6)	(z)
5	(2, 8, 4, 5, 6, 10)	(6)	(c)
6	(1, 6, 2, 8, 5, 4, 10)	(6, 7)	(0.180901699437495, c)
7	(1, 7, 8, 4, 3, 9, 10)	(6, 7)	(0.0202254248593736825 t^2 + 0.0584220259843842225, 0.18090169943749473)
8	(1, 5, 7, 6, 3, 10, 9)	(6, 7)	(0.020225424859373655 t^2 + 0.058422025984384415, 0.18090169943749462)
9	(2, 8, 4, 3, 9, 5, 6)	(6, 7)	(0.1809016994374947, z)
10	(2, 3, 9, 8, 4, 5, 6)	(6, 7)	(0.18090169943749512, z)
11	(2, 8, 9, 5, 7, 6, 3)	(6, 7)	(0.18090169943749473, c)
12	(1, 7, 8, 4, 10, 2)	(6)	(c)
13	(1, 7, 4, 3, 9, 10)	(6)	(c)
14	(2, 9, 5, 6, 7, 3)	(6)	(c)
15	(2, 8, 3, 9, 5, 6)	(6)	(z)
16	(2, 8, 4, 5, 10, 6)	(6)	(c)
17	(1, 7, 3, 2, 8, 4, 10)	(6, 7)	(0.18090169943749507, z)
18	(1, 5, 6, 7, 3, 2, 9)	(6, 7)	(0.06909830056250547, z)
19	(1, 7, 3, 9, 5, 6, 2)	(6, 7)	(c , 0.06909830056250552)
20	(1, 8, 5, 4, 6, 10, 2)	(6, 7)	(c , 0.14926352961534042)
21	(1, 6, 2, 8, 7, 4, 10)	(6, 7)	(z , 0.1809016994374947)
22	(1, 5, 7, 4, 3, 9, 10)	(5, 7)	(0.18090169943749482, c)
23	(1, 7, 3, 4, 10, 9, 5)	(6, 7)	(c , 0.06909830056250518)
24	(1, 7, 6, 3, 10, 9, 5)	(6, 7)	(0.1809016994374948, c)
25	(2, 8, 9, 5, 4, 6, 3)	(6, 7)	(z , 0.12028993467659169)

8 Finding mathematical patterns with data science

We use data science to look for geometric relationships involving symplectic capacity. We sample random polytopes and compute a broad range of readily available geometric quantities, including symplectic invariants and measurements that can change under symplectic transformations. The sampling distributions are specified in Section 8.1.

We then look for simple statistical patterns without requiring a geometric explanation in advance. When a pattern emerges, we return to the definitions and try to understand why the quantities might be related. This can suggest a tentative conjecture, which further examples may help sharpen or refute. The aim is to formulate a precise statement and find a proof that explains the observed relationship.

The experiments reported below carried out parts of this program. They found a strong association between symplectic ridge area and the numerical systolic ratio. Examining the descriptor geometrically clarified what it measures, but did not establish a general relation to capacity. The pattern also served a complementary aim: finding bodies with large ratios. Selection rules based on it improved the results of fresh sampling before we had an explanation for their success. The chapter follows this signal from its statistical discovery through geometric interpretation and search tests, then considers search methods that change the bodies themselves.

8.1 The sampled bodies and the measured ratio

For a convex body $K \subset \mathbb{R}^4$, the target is

$$\text{sys}(K) = \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)}.$$

We sought values above one. Multiplying all coordinates by $t > 0$ multiplies capacity by t^2 and volume by t^4 , so a common dilation leaves the ratio unchanged. The search must change shape or its symplectic alignment.

A polytope was specified by inequalities

$$K = \{x : \langle n_i, x \rangle \leq h_i, \quad i = 1, \dots, F\}.$$

For the generic family, each n_i was independent and uniform on the unit sphere in $\mathbb{R}^4$: we normalized a vector of four independent standard Gaussian coordinates. The heights were independent and uniform on $[0.8, 1.2)$. A proposed body was retained only if it was bounded and every proposed inequality defined a facet. The construction also rejected zero or near-duplicate normals. These conditions are part of the sampling law; “random polytope” here does not mean a uniform choice from all convex bodies.

The second family consisted of Lagrangian products $P \times Q$. In coordinates (q_1, q_2, p_1, p_2) , each factor was a planar polygon constructed in the same way, using independent uniform angles for its unit normals. A $k \times m$ product has k sides in its first factor and m in its second. We call a fixed choice of these side counts, or of F in the generic family, a sample group. The retained exploratory data contained the following groups.

Family	Groups	Bodies per group	Total
Generic, $F = 5, \dots, 12$	8	512	4,096
Products, $3 \leq k \leq m \leq 6$	10	1,024	10,240

The numerical observation associated with a body is

$$s_{\text{num}} = \frac{c_{\text{num}}^2}{2V_{\text{num}}},$$

where the two inputs are the computed capacity and volume. The largest value in these 14,336 records was 0.862586; no stored value exceeded one. The historical records can include reused cached evaluations, and their retained source information does not certify every capacity or volume. Accordingly, this and the historical comparisons below concern recorded numerical ratios. The newer three pilots use an evaluator with certified capacity bounds, but still use floating-point volume. Those bounds do not make the complete ratio certified, or change the status of the older data.

8.2 A collection of geometric measurements

The first collection combined face structure with geometry. Face counts, vertex degrees and incidences record how a polytope is assembled. They can be computed without capacity and are unchanged by invertible linear maps. Ridge areas bring the symplectic form into the measurements. We define them here before asking which quantities were informative; the statistical analysis was free to find that some contributed very little.

A two-dimensional face of a four-dimensional polytope is called a ridge. For a ridge R with cyclically ordered vertices $v_0, \dots, v_{r-1}$, define its absolute symplectic area by

$$A_\omega(R) = \frac{1}{2} \left| \sum_{i=0}^{r-1} \omega_0(v_i, v_{i+1}) \right|, \quad v_r = v_0, \quad \omega_0 = dq_1 \wedge dp_1 + dq_2 \wedge dp_2.$$

This is the polygonal area formula with the symplectic pairing in place of the planar determinant. It measures the area seen by ω_0 ; a ridge can have positive Euclidean area and zero symplectic area. Two summaries proved useful:

$$S(K) = \frac{\sum_R A_\omega(R)}{\sqrt{\text{vol}_4(K)}}, \quad C(K) = \frac{\max_R A_\omega(R)}{\sum_R A_\omega(R)}.$$

The first records total symplectic ridge area relative to volume. The second records the fraction concentrated on the largest ridge, when the total is positive. Both are unchanged by translation, common dilation and linear symplectic maps. For S , the numerator and denominator both scale by t^2 . The area formula is translation invariant because the terms involving the translation vector cancel around the closed polygon.

Later analyses added ordinary Euclidean ridge area $E(R)$ and the ratio $\kappa(R) = A_\omega(R)/E(R)$ for a ridge of positive area. These separate its size from the alignment of its plane with the symplectic form, relative to the Euclidean metric. They need not survive a nonorthogonal symplectic transformation. Normalizing $E(R)$ by the square root of volume removes common scaling, but does not give symplectic invariance. Thus the broader collection contained both invariants and non-invariants. These later measurements were used to interpret the ridge signal; they were not inputs to the original 39-feature prediction experiment below.

8.3 Looking for patterns in the measurements

We first treated the descriptors as columns of data and asked which ones ordered or predicted the numerical ratio. Smaller S usually accompanied larger recorded ratios. The Spearman correlation, which compares the ranks of the two quantities, was -0.938 . The ordinary linear correlation was only -0.205 : the strong relationship was in the ordering, without being well described by a straight line. Because the data mix bodies with different facet counts, we also compared ranks within each of the eighteen sample groups. The association was negative in every group, with correlations from -0.997 to -0.841 . This reuses the same data; it checks a possible explanation of the pooled association rather than supplying an independent repetition.

Prediction experiments gave another view of the signal. A random forest [Bre01], which averages predictions from many decision trees, used 39 descriptors: 27 describing face counts and incidences, and 12 summarizing ridge areas. The test split held out groups defined by capacity-source label and facet count, leaving 8,192 training rows and 6,144 test rows. Its test coefficient of determination was $R^2 = 0.885$. Here

$$R^2 = 1 - \frac{\sum_{i \in \text{test}} (s_i - \hat{s}_i)^2}{\sum_{i \in \text{test}} (s_i - \bar{s}_{\text{test}})^2}$$

compares squared prediction error with prediction by the test-set mean. The mean absolute error was 0.0495. This was one group split, not repeated cross-validation across independent datasets.

A separate feature-family comparison used histogram-based gradient-boosted regression [Sci], which successively fits trees to improve the prediction. Under its common training/test split, ridge-area features alone gave $R^2 = 0.887$, whereas combinatorial counts alone gave 0.043. The comparison points to information in the ridge areas beyond those counts. Its split and feature list differ from the random-forest experiment, so the two scores do not rank the learning algorithms against each other.

8.4 Returning from a pattern to geometry

The ridge result suggests a first, deliberately imprecise idea: reducing symplectic area on the two-dimensional faces might favor a large systolic ratio. This is a starting point for interpretation. It leaves open what should be held fixed, whether a statement concerns individual bodies or a sampling distribution, and how face areas could constrain capacity.

The HKO body supplies a useful reference. Its normalized ridge sum is 8.944272, close to the low end of the retained sample. However, the sample minimum is smaller, 8.925918. Thus even this particularly successful body does not give a separating rule of the form “smaller ridge sum than every ordinary example.” The observed association motivates selection; it does not establish a monotone geometric law relating S to capacity.

Products allow us to unpack the descriptor further. Let u_i and v_j be the oriented edge vectors of P and Q . A ridge lying entirely in a q - or p -plane has zero symplectic area. A mixed ridge is the rectangle formed from one edge in each factor, and its symplectic area is $|u_i \cdot v_j|$. Consequently

$$S(P \times Q) = \frac{\sum_{i,j} |u_i \cdot v_j|}{\sqrt{\text{area}(P) \text{area}(Q)}}.$$

For a fixed u_i , the sum over the edges of Q is the total variation of the projection of its boundary onto the direction of u_i . Convexity gives one increasing and one decreasing chain between the extreme projections, so

$$\sum_j |u_i \cdot v_j| = 2\|u_i\| \text{width}_Q(u_i/\|u_i\|).$$

The descriptor therefore combines the edge lengths of one factor with the widths of the other in the edge directions. This gives concrete geometry to investigate instead of treating the successful column as an unexplained score.

It also helps test the scope of the tentative idea. For two aligned unit squares, eight edge pairs have absolute scalar product one and the rest zero, so $S = 8$. Their product is the unit four-dimensional cube, whose capacity is one and whose systolic ratio is $1/2$.¹ HKO has the larger ridge score 8.944272 and a ratio greater than one. Thus a universal claim that a smaller ridge score forces a larger systolic ratio is false. The strong negative association in the sampled distributions needs a more specific explanation.

An existing analysis returned to the individual faces through

$$A_\omega(R) = E(R)\kappa(R).$$

In the generic ten-facet selection described below, the low-ridge panel had smaller mean normalized Euclidean ridge area than controls (0.917 versus 1.365), and smaller Euclidean-area-weighted alignment factor (0.377 versus 0.503). Both components therefore moved with the coarse filter. Within the selected panel, however, the ten most extreme ridge scores had slightly larger ordinary area and smaller alignment factor than the other ninety, without a larger mean

¹For the centered unit cube the dual rows are $\pm 2e_i$. In the variational formula, closure forces equal weights x_j on the two q_j rows and equal weights y_j on the two p_j rows, with $\sum_{j=1}^2 (x_j + y_j) = 1/2$. The order of the four rows in a coordinate pair gives a contribution at most $8x_j y_j$ to Q . Thus $Q \leq 8 \sum_j x_j y_j \leq 1/2$. A cyclic word on one coordinate pair with its four weights equal to $1/4$ attains $Q = 1/2$, giving $c = 1/(2Q) = 1$.

ratio. This decomposition does not isolate a cause. It explains why the next question cannot simply be which component to drive lower without considering the rest of the shape.

These observations return us to a sharper mathematical task: identify a restricted family or controlled change for which the edge-and-width quantity has a predictable relation to capacity. The current evidence does not supply that relation. Fresh selection tests establish practical use of the signal, while the later experiments vary support heights and symplectic alignment. Those pilots measure changes in ratio; they do not establish which ridge-area component explains a capacity change.

8.5 Testing whether a pattern can guide fresh selection

Alongside geometric interpretation, we tested the signal as a search rule. A predictor can describe an existing dataset without helping on new bodies. We therefore changed the order of computation: generate fresh geometry, compute its descriptors, fix the selected bodies and comparison bodies, and only then evaluate their ratios. Selection cannot use the target values at this stage.

An initial pool of 100,000 product candidates tested thirty scalar selection sets. These included low ridge sums and other ridge summaries, as well as high face counts, selecting globally or separately within side-count groups. For example, the per-group low-sum rule selected the ten smallest values of S in each group. The union of all selections and their baselines required 1,675 evaluations, not 100,000. Its maximum recorded ratio was 0.867546. Several rules selected the same bodies; their outcomes cannot be added as independent repetitions. The next experiments isolated more specific rules.

The concentration experiment generated a new pool of 100,000 candidates. Within each side-count group, it first retained the lowest one percent by S . It then divided this tail into two equally sized parts according to C . The question was whether spreading the already small total symplectic area more evenly would select better bodies. The half with smaller C had a mean recorded ratio larger by 0.0340 than the other half. The difference was positive in eight of the ten groups. This met the descriptive criterion fixed before evaluation: a positive aggregate difference and positive differences in at least seven groups. The run evaluated 2,490 bodies and reached a maximum of 0.902384. The criterion was a continuation rule for this experiment, not a statistical significance test.

A different descriptor came from the primal vertices. Let Γ be their population covariance matrix: subtract the mean vertex and average the outer products of the centered vectors. Its two Williamson eigenvalues $0 < \nu_1 \leq \nu_2$ describe its symplectic scales. Equivalently, the eigenvalues of $J\Gamma$ are $\pm i\nu_1, \pm i\nu_2$, where $J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$. We used

$$\rho(K) = \nu_2/\nu_1$$

to favor bodies whose vertex covariance has similar scales in the two symplectic directions. This uses geometry alone. It is unchanged by common dilation and linear symplectic transformations, as well as translation because the covariance is centered.

Two independently seeded pools of 50,000 products compared three selections in every side-count group: the lowest 0.5% by ρ , the ridge rule above (lowest 1% by S , then its lower half by C), and 25 disjoint controls. There were 500 memberships in each selection across the two pools. Because 64 bodies belonged to both geometric selections, 1,436 distinct targets were evaluated.

The mean difference between the covariance selection and controls was 0.333, with an interval $[0.310, 0.356]$. Against the ridge selection, the difference was 0.0145, with interval $[-0.0149, 0.0439]$. These intervals use the twenty seed-by-group differences and a Student- t calculation with nineteen degrees of freedom. They do not represent twenty independent seeds.

The covariance rule selected much larger recorded ratios than controls, while this comparison did not establish an advantage over ridge selection. No recorded value above one was found in these selection experiments.

8.6 Transfer and the limit of a stronger filter

The ridge observation first came from a mixture of generic polytopes and products. To test a concrete selection rule on generic ten-facet bodies, we generated 10,000 fresh candidates and selected the lowest one percent by *mean* normalized ridge area, $S(K)$ divided by the ridge count. Generic bodies with the same facet count can have different ridge counts, so this differs from selecting by the sum. The 100 selected bodies had mean recorded ratio 0.626, compared with 0.339 for 100 disjoint controls. The difference was 0.287, with a percentile-bootstrap interval $[0.250, 0.324]$ from 20,000 resamples of the realized panels.

The same experiment asked whether a more extreme filter would help further. The ten bodies in the lowest 0.1% were compared with the other ninety selected bodies. Their mean difference was -0.0318 , with interval $[-0.0946, 0.0331]$. This failed the criterion for expanding that particular search to a larger pool. It did not establish that stronger filtering is harmful: the interval permits effects of either sign. The useful contrast is that a coarse geometric filter worked well while the next restriction did not show an additional gain.

We also changed the product sampling law. Starting from the original random normals and heights, we replaced every support height in both factors by one. Each side of the resulting polygon lies on a tangent line to the unit circle. We required both the original factors and the changed factors to be admissible, and then separately normalized the two changed factors to area one. This produces a conditioned family of tangential polygons.

On one fresh pool with 3,200 eligible bodies in each of the 4×6 and 6×6 groups, ridge and covariance selection were compared with shared, disjoint controls. Each selection had sixteen memberships per group; five selected bodies overlapped, giving 91 actual evaluations. Averaging the two group effects equally, relative to controls, covariance selection gained 0.259 and ridge selection gained 0.225. Their respective resampling intervals were $[0.192, 0.329]$ and $[0.160, 0.296]$. These intervals condition on the realized panels; one seed and two side-count groups do not measure variation across arbitrary source laws. The shared controls also make the comparisons dependent. The experiment nevertheless gives a concrete instance in which both rules remained useful after changing how bodies were generated.

8.7 Changing support heights and symplectic alignment

The transfer experiment selected among tangential products. It did not isolate the effect of making a particular polygon tangential. For that question we kept the normals fixed and compared the original product with three changes: equalize the first factor's heights, equalize the second factor's heights, or equalize both. Sixteen independently generated pairs, eight of type 4×4 and eight of type 4×6 , supplied 64 bodies. We accepted a pair only if all four constructions were valid, before evaluating any ratio. The paired result therefore concerns this jointly admissible source.

Each factor was then scaled to area one. Separate factor dilations preserve the mathematical ratio, since

$$\text{diag}(aI_2, bI_2) = \sqrt{ab} \text{diag}(\sqrt{a/b} I_2, \sqrt{b/a} I_2)$$

is a common dilation followed by a symplectic map. Replacing unequal support heights is the shape change being tested. Changing both factors raised the mean numerical ratio by 0.0153: eleven pairs improved and five declined. The mean changes were 0.0040 for 4×4 and 0.0266 for 4×6 . All 64 evaluations succeeded. An average improvement therefore did not give a rule that equalizing the heights helps every body.

Rotation changes a different aspect of the geometry. An orthogonal map preserves Euclidean shape but need not preserve ω_0 . A rotated Lagrangian product can consequently have a different capacity and need no longer be a Lagrangian product for the fixed symplectic form.

We compared two allocations of sixteen evaluations. One evaluated a fresh product, a symplectic rotation as a control, and fourteen prescribed orthogonal rotations. The other evaluated sixteen fresh products, sharing the first body with the rotation allocation. We repeated the comparison four times, twice for 3×3 and twice for 4×4 . The rotation schedule and all source bodies were fixed before target evaluation. Shared starts gave 124 distinct requests and 128 charged memberships. The symplectic controls preserved the numerical ratios within $4 \cdot 10^{-16}$.

Rotation improved three of the four starting bodies. Fresh sampling still produced the larger maximum in every comparison. In one 4×4 case, rotation raised the ratio from 0.328 to 0.529, while fresh sampling reached 0.774. The rotation evaluations also took longer. For these four starts and this prescribed schedule, the same number of evaluations was more effective when spent on fresh bodies.

8.8 Adapting the sampling distribution

Fresh sampling won the rotation comparison, but it repeatedly draws from the same law. We next tested whether the evaluated bodies could improve that law. The cross-entropy method [Boe+05] used here repeatedly fits a simple distribution to the best part of a population, then samples another population from it.

This experiment used 5×5 products. We described each pentagon by the cyclic gaps between its five normal angles and the relative lengths of its five dual normals. There are four independent angular-gap coordinates and four independent logarithmic lengths per factor. One relative angle between the factors gives seventeen coordinates altogether. Separate factor scales and a simultaneous planar rotation do not change the mathematical ratio, which allows their removal from this description. The precise coordinates and update are given below so that adapting the distribution has a concrete meaning.

Both arms began with the same 64 independent products from the original angle-and-height source. The adaptive arm fitted a diagonal Gaussian in these coordinates, then moved its mean and coordinate variances halfway toward those of the sixteen bodies with largest numerical ratio. It drew another 64 bodies, evaluated them, and repeated this update. Each arm had four populations including initialization, for 256 charged evaluations. The comparison arm continued independent sampling throughout. A variance floor kept the adaptive sampler from eliminating a coordinate’s variation entirely.

Three seeds produced the following results. The top-eight statistic is the median of the eight highest ratios after removing repeated, exactly equal dual-vertex arrays in an arm; different bodies with equal scores remain. It tests whether improvement extends beyond a single best observation.

Seed	Maximum		Top-eight median	
	IID	Adaptive	IID	Adaptive
1	.759574	.798513	.728815	.782274
2	.810904	.831348	.753788	.776334
3	.760068	.855655	.722310	.813742

The adaptive arm was better on both statistics for all three seeds. The shared initial populations account for 1,344 actual evaluations versus 1,536 charged arm memberships. All actual requests succeeded and none gave a numerical ratio above one. Three paired runs give a useful small comparison, without establishing superiority across other generators or choices of population size and update rule.

The source restriction needs particular care. An initial version stopped when one body exceeded the evaluator’s admitted primal-vertex infinity-norm range. The completed version applied the evaluator’s existing admission conditions to both proposal arms before accepting a body for evaluation. This was a procedural revision after seeing the failure. Both arms reject unsupported inputs, but the admission test acts on the actual coordinate representation: IID bodies retain their original representation and adaptive bodies use the normalized reconstruction. It is not a symmetry-invariant condition on all pentagon products. The result is therefore for this implemented, admission-conditioned comparison.

The complete run took 215.8 seconds: 182.2 seconds for construction and admission checks, 32.9 seconds in evaluator processes, and 0.053 seconds for distribution updates. Including construction and charging initialization to each arm, the adaptive runs used about 35–36 seconds per seed and IID about 46–51 seconds. Equal evaluation counts did not imply equal cost. These measured costs favor the adaptive arm in this run; they are not the outcome of a separate equal-wall-time experiment.

8.9 Local search from matched starting bodies

The adaptive sampler changes a distribution over products. A separate, earlier experiment changed the dual normals of generic ten-facet bodies directly. Seven fixed implementations started from the same 64 bodies. Each run allowed at most 128 charged full evaluations and a one-second accumulated evaluator-plus-optimizer time ceiling, excluding the initial evaluation. Invalid proposals still consumed calls. Work begun below the time ceiling was allowed to finish, so terminal results can include a small overrun; they are not values available at exactly one second.

The capacity calculation suggests local branch functions, each associated with a particular candidate in the variational problem. When several smooth functions compete through their minimum, following the gradient of one selected function can fail to increase the minimum. For example, the minimum of x and $-x$ decreases in both directions away from zero, although either individual function has a direction of increase. This motivates methods that include more than the currently selected branch.

The literal gradient implementation took a fixed step along that branch’s gradient and accepted usable evaluations even when the ratio decreased. The safeguarded version removed the symmetry directions, normalized the gradient, and accepted only improvements; its relative step length expanded by 1.25 after success and contracted by 0.5 after failure. A finite-gap model included branches whose actions were within ten percent of the minimum and maximized the minimum of their affine predictions in a small ball. Its proposal still required a full evaluation and a strict improvement. The finite branch list is a local model, not a complete formula for all branches near the body.

Two history methods retained candidates from the current and up to three previous accepted anchors, then used this larger list for two inner model steps before full evaluation. One also predicted additional candidates from branch transitions for sufficiently long steps. Their outer rule accepted any usable evaluation, so a move could reduce the objective even while the best value seen so far stayed unchanged. The remaining methods were a Gaussian population search with covariance adaptation (CMA-ES) [Han16], and a poll of both signs of a transverse coordinate basis, keeping the best improvement.

Most methods used the complement of the fifteen translation, scaling and linear symplectic tangent directions. This leaves twenty-five directions at a generic ten-facet body. The coordinate poll and local model methods rebuilt their transverse space at their anchors; CMA-ES used its initial space, and the literal gradient used the ambient coordinates. These are choices in the tested implementations, not interchangeable realizations of one global quotient chart.

Implementation	Median best value	Across-start 10–90% range
Four-anchor history	.984783	.957033–.998515
History with transition prediction	.981647	.955934–.993962
Finite-gap model	.977159	.948177–.994597
Literal gradient	.881344	.789542–.923898
Safeguarded gradient	.855083	.780427–.916023
CMA-ES	.818309	.718359–.869249
Coordinate poll	.601030	.358319–.786578

Each entry summarizes 64 terminal best-so-far recorded values. The ranges show variation across starts, not confidence intervals for the medians. All 448 values remained below one. Comparing methods within the same start, the median history-minus-finite-gap difference was 0.00883, with an existing percentile-bootstrap interval $[0.00335, 0.01533]$. The bootstrap resampled the 64 matched starts 4,000 times. This paired median is not the difference of the two medians in the table.

The table ranks these implementations under the recorded allocation and historical evaluator. Its resampling does not account for configuration selection, untested starting distributions or evaluator error. The original producer tree and the individual proposal/evaluation trace tables are not locally available; retained run records and analysis identities support the reported aggregates, rather than a replay of the executable. The historical objective used floating-point geometry and volume. These limits prevent interpreting the table as a certified ordering of mathematical capacities or as a comparison of optimally tuned algorithm families.

8.10 What a failed local probe tells us

A large terminal value also raises a geometric question: has search arrived at a local maximum? Evaluating a finite set of nearby points can discover an improvement, but failing to find one does not prove local maximality. The endpoint diagnostics gave concrete examples of this distinction.

Sixteen endpoints of the history method were chosen across the observed performance range of the common starts. At each, we evaluated both signs of twenty-five transverse basis directions at relative radii 10^{-3} , 10^{-4} and 10^{-5} : 150 probes per endpoint. Seven endpoints had at least one raw positive change and nine had none. The sixteen endpoints were selected by outcome ranks, so these fractions do not estimate a population convergence probability.

After allowing a larger search budget on those starts, a further continuation experiment examined three selected optimizer endpoints and HKO. It tested branch-model and selected-gradient proposals at five decreasing step lengths, falling back to a signed basis poll, and allowed at most ten accepted improvements. All three optimizer endpoints took ten improving steps. One had previously decreased along every tested signed basis direction, yet improved by 0.00223 along a sequence of oblique moves with total normalized path length 0.01. HKO accepted no move. The theorem establishes HKO’s local status; the finite search supplies a control observation consistent with it.

Another fixed suite probed a regular triangle–hexagon product, a rotated square–square product and the CH2021 example, together with a regular pentagon product chosen on a known increasing rotation path. It combined signed transverse basis directions, projected random directions and prescribed rotations at three radii. All 1,482 probes were valid. None of the three target bodies had a nominal gain above 10^{-12} , while the pentagon control had twelve, all in the rotation probes. Its generic basis and random probes found none. Even this control therefore shows that a finite collection of generic directions can miss an available increase. The target misses remain numerical diagnostics; they do not exclude untested directions or changes in facet count. The historical run also has incomplete source provenance, and its broad capacity-derived bounds do not turn the nominal target misses into a proof.

8.11 What the experiments contribute

The exploratory approach found a strong ridge-area pattern, then gave it more concrete meaning through edge widths and the area–alignment decomposition. That interpretation ruled out a universal monotonicity claim and left a more restricted capacity relation to be formulated. The search approach found geometric information useful before capacity evaluation. Low ridge area and balanced vertex covariance selected larger recorded ratios in fresh pools, and some of that usefulness survived changes in the generating law. More severe ridge filtering did not automatically improve selection. Changing individual bodies also had varied effects: equalizing heights helped on average but harmed some pairs, while prescribed rotations lost to fresh sampling under the tested call allocation. Adapting the product distribution improved on IID in three small paired runs.

The local-search comparison obtained larger recorded values from models that retained several candidate branches. Its endpoint checks also exposed improving directions missed by a basis poll. These are distinct experimental results, with different sources and evaluation contracts. Together they suggest useful ways to continue searching, while leaving the central challenge of reaching new high-ratio bodies unresolved. The initial sample, prospective selections, three new pilots and matched-start optimizer runs produced no stored value above one. This observation excludes the deliberately supplied HKO and pentagon controls; it neither proves high-ratio bodies absent from the searched families nor shows that the exploration has exhausted useful ideas.

Computational details for the adaptive product sampler

For reproducibility, let $g_1, \dots, g_5 > 0$ be the cyclic normal-angle gaps of a factor and let r_i be its dual-normal lengths. Set

$$\ell_i = \log(g_i/g_5) \quad (i = 1, \dots, 4), \quad z_i = \log r_i - \frac{1}{5} \sum_{j=1}^5 \log r_j.$$

We store the first four z_i and recover $z_5 = -\sum_{i=1}^4 z_i$. Reconstruction uses

$$g_i = \frac{2\pi e^{\ell_i}}{1 + \sum_{j=1}^4 e^{\ell_j}}, \quad g_5 = \frac{2\pi}{1 + \sum_{j=1}^4 e^{\ell_j}}, \quad h_i = e^{-z_i}.$$

The first normal angle is zero in the first factor and ϕ in the second. The initial IID encoding chooses a cyclic origin by lexicographically minimizing the sequence of gap/length pairs. Sampled adaptive coordinates are retained for the updates without another cyclic relabelling.

For an ordinary coordinate j , write $\hat{\mu}_{E,j}$ and $\hat{v}_{E,j}$ for the mean and population variance of the sixteen elites. The update is

$$\mu_j^+ = \frac{1}{2}\mu_j + \frac{1}{2}\hat{\mu}_{E,j}, \quad v_j^+ = \max\{\frac{1}{2}v_j + \frac{1}{2}\hat{v}_{E,j}, 0.05v_{0,j}\}.$$

Initialization uses the population moments of the shared 64 bodies. Coordinates are drawn independently with these Gaussian moments. The phase uses circular means, a unit-vector blend of old and elite means, and variance of angular deviations wrapped into $[-\pi, \pi)$. Samples are wrapped modulo 2π . A vanishing resultant retains the previous mean; initial fallback uses the smallest-ID candidate’s phase. Ties in elite selection are broken by candidate ID.

Each population permits at most 640 construction attempts for 64 accepted bodies. Failed geometry and unsupported evaluator inputs are rejected before target evaluation; an incomplete population does not trigger an update. The evaluator admission includes finite, nonredundant bounded geometry and primal and dual infinity-norm bounds between 10^{-3} and 10^3 . The three seeds are 202609140301, 202609140302 and 202609140303. The retained report records the stopped first version and the subsequent supported version separately.

9 The Rotated-Pentagon Profile

We now isolate one structured Lagrangian-product family for which the capacity can be determined exactly. Let $P_5 \subset \mathbb{R}^2$ be a centered regular pentagon of circumradius 1, let $R(\theta)$ denote counterclockwise planar rotation, and set

$$K_\theta = P_5 \times_L R(\theta)P_5 \subset \mathbb{R}_q^2 \oplus \mathbb{R}_p^2.$$

Only the relative rotation of the factors matters.

This family contains the Haim–Kislev–Ostrover counterexample. Their paper uses the symplectic form $\sum_i dp_i \wedge dq_i = -\omega_0$, and writes the second pentagon as $R(-\pi/2)P_5$. The convention-changing map $(q, p) \mapsto (q, -p)$ carries this to $K_{\pi/2}$ in our convention. The simultaneous-reflection symmetry used below also identifies $K_{\pi/2}$ with $K_{-\pi/2}$. Thus the result below recovers their value $(3 + \sqrt{5})/5 > 1$ [HO26, Proposition 1.4], and shows that their relative angle maximizes the systolic ratio within the entire one-parameter family.

9.1 From sampled values to a formula

The formula was first suggested by a finite sweep of rotation angles. At each sampled angle we evaluated the Haim–Kislev quadratic programs from Section 4 in floating-point arithmetic. The samples trace a symmetric curve with its maximum at the midpoint angle $\pi/10$; see Figure 9. This was useful for finding the candidate branch and formula, but it cannot exclude an unsampled crossing with another branch. The theorem below instead uses an exhaustive finite reduction and algebraic comparisons over an ordered number field.

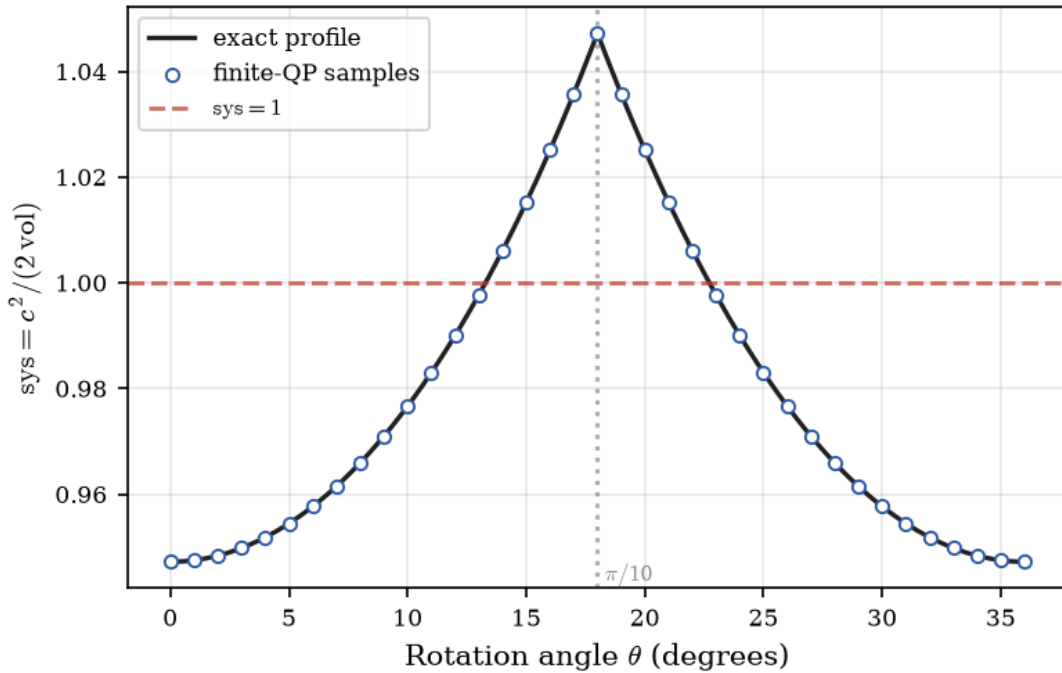


Figure 9: Floating-point finite-QP samples for K_θ , shown as blue markers, together with the profile proved below. The horizontal line marks the conjectured Viterbo bound. The agreement of the samples with the formula curve is an illustration and discovery check, not an input to the proof.

9.2 Exact profile

For $\theta \in \mathbb{R}$, define its distance from the nearest multiple of $\pi/5$ by

$$d(\theta) = \min_{k \in \mathbb{Z}} \left| \theta - \frac{k\pi}{5} \right|.$$

Thus $0 \leq d(\theta) \leq \pi/10$.

Theorem 9.1 (Rotated-pentagon profile). *For every $\theta \in \mathbb{R}$,*

$$\text{sys}(K_\theta) = \frac{5 + 2\sqrt{5}}{10} \sec^2(d(\theta)).$$

Consequently, the maximum within this family is attained exactly when $d(\theta) = \pi/10$, and its value is

$$\max_{\theta \in \mathbb{R}} \text{sys}(K_\theta) = \frac{3 + \sqrt{5}}{5} > 1.$$

Proof. Symmetry reduction. The profile has period $2\pi/5$, because $R(\theta + 2\pi/5)P_5 = R(\theta)P_5$. Simultaneous reflection of both factors gives invariance under $\theta \mapsto -\theta$, hence a first reduction to $0 \leq \theta \leq \pi/5$.

There is a further symmetry because the factors are equal and the pentagon is odd-sided. The symplectic map $-J_0$ sends

$$P_5 \times_L R(\theta)P_5 \quad \text{to} \quad R(\theta)P_5 \times_L (-P_5).$$

After rotating both factors by $-\theta$, and using $-P_5 = R(\pi)P_5 = R(\pi/5)P_5$ modulo the pentagon's $2\pi/5$ -symmetry, we obtain $K_{\pi/5-\theta}$. It is therefore enough to prove

$$\text{sys}(K_\theta) = \frac{5 + 2\sqrt{5}}{10} \sec^2 \theta, \quad 0 \leq \theta \leq \frac{\pi}{10}. \quad (35)$$

The candidate branch. We use the facet convention of the executable certificate. Put

$$h = \cos \frac{\pi}{5}, \quad n_i = \left(\cos \left(\frac{\pi}{2} + \frac{2\pi i}{5} \right), \sin \left(\frac{\pi}{2} + \frac{2\pi i}{5} \right) \right).$$

In the coordinate order (q_1, q_2, p_1, p_2) , the dual vertices of the ten facets are

$$a_i = \frac{1}{h}(n_i, 0), \quad a_{i+5} = \frac{1}{h}(0, R(\theta)n_i), \quad 0 \leq i < 5.$$

Consider the alternating two-bounce word with q -blocks (3), (1, 0) and p -blocks (8), (5, 6); its flattened word is

$$\sigma_{\text{act}} = (3, 8, 1, 0, 5, 6).$$

In this order, set

$$\beta_{\text{act}} = \left(\frac{\sqrt{5}}{10}, \frac{\sqrt{5}}{10}, \frac{5 - \sqrt{5}}{20}, \frac{5 - \sqrt{5}}{20}, \frac{5 - \sqrt{5}}{20}, \frac{5 - \sqrt{5}}{20} \right).$$

Direct substitution gives

$$\beta_{\text{act}} \in M_{\sigma_{\text{act}}}, \quad Q_{\sigma_{\text{act}}}(\beta_{\text{act}}) = \frac{\cos \theta}{2(1 + \cos(\pi/5))^2}.$$

All weights are positive. Since the capacity is $1/(2Q_{\max})$, this single feasible word gives

$$c_{\text{EHZ}}(K_\theta) \leq c_{\text{act}}(\theta) := \left(1 + \cos \frac{\pi}{5}\right)^2 \sec \theta.$$

Rotation preserves the second factor's area, so

$$\text{vol}_4(K_\theta) = \text{area}(P_5)^2, \quad \text{area}(P_5) = \frac{5}{2} \sin \frac{2\pi}{5}.$$

It follows that

$$\frac{c_{\text{act}}(\theta)^2}{2 \text{vol}_4(K_\theta)} = \frac{5 + 2\sqrt{5}}{10} \sec^2 \theta.$$

This is the required upper bound in (35).

Why the competitor list is exhaustive. It remains to prove that no branch has smaller action. By Theorem 4.4, for every θ there is an active word realizing the maximum in the capacity formula, with two or three blocks of each type, alternating between q and p . Each block contains either one facet or an ordered pair of adjacent facets, and distinct blocks of one type use disjoint facets. This is an existence statement for the capacity value; it does not claim that all minimizing orbits have two or three bounces.

The certificate generates a raw superset of this family. For $k = 2, 3$, it chooses k disjoint q -blocks and k disjoint p -blocks, fixes one selected q -block as the cyclic starting block, and enumerates all relative orders of the others. Fixing the start loses nothing: cyclic shift preserves the closure constraints and Q_σ . The generator does not quotient by word reversal or by pentagon dihedral symmetry, so those duplicates remain as separate checks. It produces 7200 two-bounce and 43200 three-bounce cyclic representatives before pruning.

Consecutive same-factor facets of an active word must intersect. A mixed transition must also have the positive directed symplectic sign. On $0 < \theta < \pi/10$, the certificate proves exactly that every such mixed sign is either strictly positive throughout or strictly negative throughout. Removing the negative transitions therefore removes only impossible active words. Exactly 3340 distinct raw words survive.

Exact classification of a fixed word. The computation uses

$$t = \tan(\theta/2), \quad 0 < t < \tan(\pi/20).$$

All pentagon constants lie in the maximal real subfield K of $\mathbb{Q}(\zeta_{20})$, and

$$\cos \theta = \frac{1 - t^2}{1 + t^2}, \quad \sin \theta = \frac{2t}{1 + t^2}.$$

Thus every KKT matrix entry, multiplier, quadratic value, and action gap is a rational function in $K(t)$.

For each surviving word σ , SageMath solves the equality-constrained KKT system over $K(t)$. An active word realizing that maximum has positive weights, and hence must occur among these stationary branches. For a returned branch the proof predicates are

$$\beta_i > 0, \quad Q_\sigma > 0, \quad \frac{1}{2Q_\sigma} - c_{\text{act}}(\theta) \geq 0.$$

The roots and poles of the finitely many rational functions divide the open half-domain into cells. SageMath isolates these algebraic numbers exactly in its real algebraic field and checks one exact sample in each cell, where every sign is constant. No floating-point sign decision enters the certificate.

One branch by hand. To make the repeated calculation concrete, consider the word with three blocks per factor

$$\tau = (0, 5, 3, 8, 1, 7).$$

Write $s = \sqrt{5}$. Exact row reduction of the KKT system gives, in the order of the facets in τ ,

$$\beta_\tau = \left(\frac{5-s}{20}, \frac{s}{10}, \frac{s}{10}, \frac{5-s}{20}, \frac{5-s}{20}, \frac{5-s}{20} \right).$$

Thus all six weights are positive, independently of θ . Substitution in the quadratic form gives

$$Q_\tau(\theta) = \frac{3(s-2)}{5} \cos \theta + \frac{(5s-11) \cos(\pi/10)}{5} \sin \theta.$$

This is positive on the entire half-domain. The active word above has $Q_{\text{act}} = (3-s) \cos \theta/5$, and direct simplification yields the particularly transparent comparison

$$Q_{\text{act}}(\theta) - Q_\tau(\theta) = \frac{5s-11}{5} \sin\left(\frac{\pi}{10} - \theta\right). \quad (36)$$

Since $5s-11 > 0$, the competitor has smaller Q , and hence larger action, for $0 \leq \theta < \pi/10$; it ties at the right endpoint. In action form the same branch is

$$A_\tau(\theta) = \frac{5(2 + \sqrt{5})}{2(\tan(\pi/5) \sin \theta + 3 \cos \theta)}, \quad (37)$$

and the action gap is

$$A_\tau(\theta) - c_{\text{act}}(\theta) = \frac{10 \sin(\pi/10 - \theta)}{(4 \tan(\pi/5) \sin \theta + 12 \cos \theta) \cos \theta}.$$

This example also illustrates why it is useful to retain whole branch formulas: A_τ has its own strict minimum at

$$\tan \theta = \frac{1}{3} \tan \frac{\pi}{5}, \quad \theta \approx 13.61^\circ,$$

although it never determines the capacity on the open half-domain. This is the regular positive- Q path through the executable's classifier: solve the KKT system, test positivity, and determine the sign of the action gap. Many words instead reach the inconsistent, zero- Q , forced-zero, or no-feasible-cell outcomes; the remaining feasible words either tie identically or have strictly positive gap, as Table 2 records.

When the stationary system is singular, its solutions form an affine space. The calculation following (20) shows that Q_σ is constant on that space. The remaining question is whether a solution has all active weights positive. The finite classifier checks the affine equations and the resulting positivity conditions; it does not assume that a singular system has a unique optimizer.

The classifier is fail-closed. Each word must reach one of the six conclusions in Table 2; every other case is labelled for manual review, and that label is not accepted by the full run.

The source and retained full transcript are in

experiments/regular-products/
pentagon-rotation-formula-proof/

The unlimited run checks that all 3340 generated words are distinct and classified, then prints its final success marker. Its header records the SageMath version, assertion state, source digest, and arguments; the executable aborts if Python assertions are disabled. The Haim–Kislev formula, finite-enumeration theorem, symmetries, and continuity close are mathematical results proved above or imported by citation. The theorem-specific implementation obligation is instead to check that the Sage source encodes the displayed facets, KKT system, enumeration,

pruning, and classification predicates. Section B.1 shows cleaned excerpts for that audit. After this correspondence is checked, the remaining software-level reliance is the ordinary use of SageMath’s standard exact number-field arithmetic, linear algebra, and real-root isolation; the thesis does not attempt to verify SageMath itself. The empirical sweep and the Rust search code are not proof inputs.

certificate outcome	count	implication for a positive capacity competitor
inconsistent KKT system	25	no stationary branch exists generically
zero quadratic value	1680	the generic stationary value cannot give positive finite action
forced zero weight	470	every solution of a singular system has a zero active weight
no feasible cell	735	the generic branch is nowhere strictly positive
identical zero gap	20	the branch ties the displayed branch wherever feasible
strictly positive gap	410	the branch has larger action on every feasible cell

Table 2: Exhaustive outcomes of the exact classification. The counts sum to 3340; they record the run rather than provide an additional mathematical assumption.

Exceptional parameters and endpoints. The rational-function computation proves the comparison away from a finite set: besides the roots and poles used in the sign-cell decomposition, the coefficient and augmented KKT matrices can drop below their generic ranks only at zeros of finitely many nonzero minors. Taking the union of these exceptional sets over the finite word list leaves a dense open subset of $0 < \theta < \pi/10$. On that subset no feasible branch lies below $c_{\text{act}}(\theta)$, while the active word attains this value. Hence (35) holds there.

The path $\theta \mapsto K_\theta$ is Hausdorff-continuous and its volume is constant. Proposition 2.4 therefore makes $\text{sys}(K_\theta)$ continuous. Equality extends from the dense open subset to every exceptional interior parameter and to the endpoints 0 and $\pi/10$. This proves (35); the initial symmetries prove the theorem for all real θ . $\square$

10 Three-Dimensional Visualization

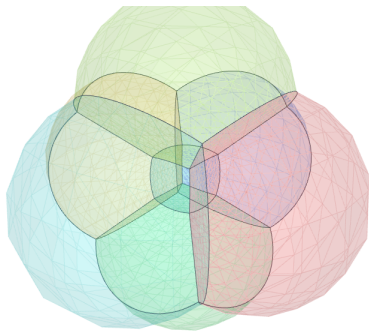
A four-dimensional polytope cannot be drawn directly, but its boundary can be viewed in three dimensions after choosing a projection. For a polytope $K \subset \mathbb{R}^4$ with the origin in its interior, we first apply the radial map

$$\partial K \longrightarrow S^3, \quad x \longmapsto \frac{x}{\|x\|},$$

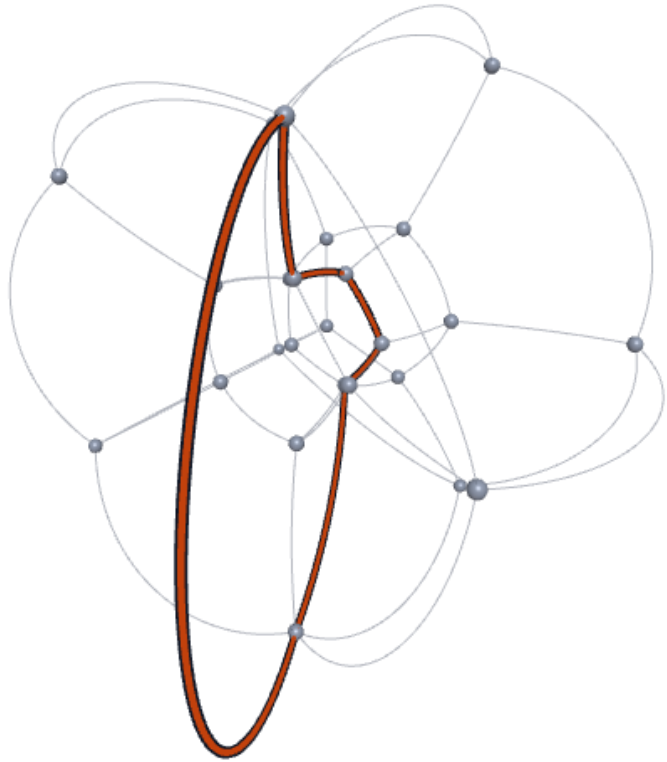
and then stereographically project S^3 from the north pole $e_4 = (0, 0, 0, 1)$ to $\mathbb{R}^3$. The viewer samples projected edges and Reeb segments along great-circle arcs. Its coloured two-face interiors are interpolated spherical meshes, included to make the boundary structure easier to see rather than to preserve metric geometry. The stereographic projection also distorts scale and clips arcs close to its pole.

Figure 10 shows projected two-faces of the four-dimensional cube and the projected one-skeleton of the HKO polytope K_{HKO} from Section 7 with one numerically recovered minimum-action generalized Reeb orbit. The producer exports only recovered orbits passing numerical closure and containment checks within fixed tolerances. The orbit shown is therefore a computed example, not a proof of a capacity statement.

The interactive views were also explored as a possible source of geometric patterns for our exploratory search. In practice, passing from four dimensions to a three-dimensional projection discarded too much visible Reeb-geometric information, and the impressions changed with



(a) Projected two-face meshes of the four-dimensional cube.



(b) The HKO one-skeleton and a recovered minimum-action orbit.

Figure 10: Radial and stereographic views of four-dimensional polytopes. In (a), colour separates projected two-face meshes of the cube. In (b), the grey one-skeleton belongs to the Lagrangian product of two regular pentagons and the dark-outlined orange curve is a recovered six-segment closed Reeb orbit. Both panels use the pole e_4 and omit parts mapped beyond the viewer's clipping radius. They are qualitative explanatory views, not symplectic invariants or proof evidence.

the projection pole. We found no visual pattern that gave a reliable hypothesis or candidate-selection rule. For this exploration, the useful outcome is therefore limited to the explanatory role illustrated above.

11 Numerics

The computations in this thesis support two different kinds of claims. Statements used inside the HKO and rotated-pentagon proofs are checked by separate SageMath packets over rational or ordered algebraic fields. The ordinary Rust implementation instead computes the EHZ capacity of a four-dimensional polytope supplied in binary64 coordinates. It uses floating-point arithmetic for speed, but it does not accept an unverified floating-point sign or ordering decision: every such decision is enclosed by an outward error bound or resolved in exact rational arithmetic.

11.1 Input and output contracts

A finite binary64 number is a dyadic rational. The numerical capacity contract regards the supplied numbers as these exact rationals; it does not claim to recover unavailable source coordinates from which they may have been rounded. Before the QP search, exact geometry checks establish that

$$K = \{x \in \mathbb{R}^4 : \langle a_i, x \rangle \leq 1\}$$

is bounded, nonempty, full-dimensional, and described without redundant facets. The exact primal vertices and vertex–facet incidence are then constructed from the dyadic input. Floating-point filters may discard a subset only when a one-sided bound proves the exact conclusion; an indeterminate filter result is resolved exactly.

The production route accepts at most sixteen facets and requires every primal and dual vertex to have infinity norm in

$$[10^{-3}, 10^3].$$

These bounds are a deliberately narrow applicability policy, not a theorem about all polytopes. They exclude scales for which the current binary64 enclosures have not been justified and tested. The general candidate stream is additionally limited to 100 000 cyclic words. Violating any of these conditions returns an error rather than an uncertified capacity.

For a general polytope the result is an outward interval $[c_L, c_U]$ containing the exact capacity of the dyadic input. A scalar wrapper returns the midpoint only if this interval certifies the relative error requested by the caller. For a recognized Lagrangian product the specialized route returns the exact rational capacity of the dyadic input, together with outward binary64 endpoints for convenient downstream use.

11.2 Certified fixed-word decisions

For a word σ , write the equality-constrained stationary system in the symmetric form

$$\mathcal{K} \begin{pmatrix} \beta \\ \lambda \end{pmatrix} = \begin{pmatrix} H & C^\top \\ C & 0 \end{pmatrix} \begin{pmatrix} \beta \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ d \end{pmatrix}, \quad d = (0, 0, 0, 0, 1)^\top.$$

Here $C\beta = d$ is closure plus normalization and H is twice the quadratic form in (19). A numerical factorization supplies an approximate solution $\hat{x}$ and approximate inverse R . The following standard inverse-defect estimate is the central certificate.

Lemma 11.1 (Verified inverse defect). *Suppose outward arithmetic establishes*

$$\bar{r} \geq \|\mathcal{K}\hat{x} - b\|_\infty, \quad \bar{e} \geq \|I - \mathcal{K}R\|_\infty, \quad \bar{e} < 1.$$

Then $\mathcal{K}$ is invertible, and its exact solution x^ satisfies*

$$\|\hat{x} - x^*\|_\infty \leq \rho := \frac{\|R\|_\infty \bar{r}}{1 - \bar{e}}.$$

Proof. The Banach lemma applied to $\mathcal{K}R = I - (I - \mathcal{K}R)$ gives $\|\mathcal{K}^{-1}\|_\infty \leq \|R\|_\infty / (1 - \bar{\epsilon})$. Apply this bound to the residual $\mathcal{K}\hat{x} - b$. $\square$

Thus $\hat{\beta}_i - \rho > 0$ for every i certifies positive weights, while $\hat{\beta}_i + \rho \leq 0$ for some i certifies failure. If ξ is the multiplier for normalization, stationarity gives

$$2Q(\beta^*) + \xi^* = 0.$$

Consequently the same radius yields

$$Q(\beta^*) \in \left[-\frac{\hat{\xi} + \rho}{2}, -\frac{\hat{\xi} - \rho}{2} \right].$$

For $0 < q_L \leq Q \leq q_U$, monotonicity then gives the action enclosure

$$\frac{1}{2q_U} \leq \mathcal{A} \leq \frac{1}{2q_L}.$$

The implementation computes the residual and inverse-defect bounds with explicit roundoff and subnormal-error terms. It first uses a cheap induced norm bound and retries an indeterminate system with a tighter entrywise bound. The proof assumes round-to-nearest binary64 arithmetic with gradual underflow and the inspected matrix-multiplication evaluation order. The runtime checks gradual underflow; if the contract is unavailable, or if either bound remains indeterminate, the decision is made by exact rational arithmetic. A compiler, dependency, target, or arithmetic-contract change therefore requires the floating-point certificate tests to be repeated.

11.3 Complete general and product routes

The general search constructs the exact transition graph, pruned by facet incidence and symplectic signs, then enumerates its simple cyclic words. Candidate correctness is not obtained merely by solving every displayed KKT system. A fixed-word maximum must have nonpositive curvature in every tangent direction. Conversely, a certified vector $z \in \ker C_\tau$ with $z^\top H_\tau z > 0$ rules out τ . If τ injects into a longer word σ while preserving cyclic order, zero-extending z preserves both its tangent constraints and positive curvature. Hence σ is ruled out as well. Processing words by length allows these proved obstructions to skip many later factorizations.

For every word not excluded in this way, the route uses the verified fixed-word decision above, with exact fallback for every indeterminate case. The resulting objective intervals are aggregated outwards over the complete candidate family, and the reciprocal reverses the endpoints to enclose the capacity. Exact arithmetic is therefore a selective correctness fallback, not the default arithmetic for every word.

For a Lagrangian product, the inverse-defect route is correct but frequently indeterminate and unnecessarily expensive. The specialized implementation uses Algorithm 4.7. It encloses factor-closure weights and bilinear objectives in floating-point arithmetic, rejects a support or order only when the intervals prove it cannot win, and resolves ambiguous supports and every possible winner exactly. Because the final comparison is rational, this route returns the exact capacity and every tied sparse winner in the closure-vertex candidate family. It intentionally does not enumerate all minimizing geometric branches.

11.4 Empirical verification and limitations

The general numerical audit compared the floating-point predicates with exact dyadic-rational arithmetic on 249 KKT systems: ordinary generated systems, scales from 10^{-2} through 10^3 , and systems approaching singularity down to binary64 coincidence. It found no incorrect determinate decision and no violation of a β - or Q -radius. Fifteen cases were indeterminate

and were resolved exactly. The largest observed ratios of actual error to certified radius were 0.00149 for β and 0.00081 for Q .

Complete-route comparisons used 88 words from three generated general polytopes and 1271 words from three product controls. Exact aggregation found no missed capacity and no missed minimizer class; every exact capacity lay in the returned interval. On the general cases, the largest difference between the reported midpoint and exact capacity was $2.31 \cdot 10^{-14}$. These finite comparisons test the implementation and exercise fallback, but the soundness claim rests on the preceding error bounds and complete-candidate arguments.

The product audit used 88 ordinary and adversarial products. Across 1835 closure weights and 173 496 objective intervals it found no interval violation, no incorrect determinate sign, and no disagreement between the hybrid and complete exact closure-vertex computations. Raw comparison with zero disagreed with the exact objective sign 23 559 times, mostly at exact zero or tiny values; the certified predicate returned indeterminate instead. A separate retained population of 10 240 random products had a literal capacity winner of length six in every case. This supports the implementation and stress-tests Theorem 4.6; it is not its proof and does not exclude longer tied minimizers.

The guarantees above concern the exact dyadic polytope represented by the input. They do not propagate measurement error or prior rounding uncertainty in the coordinates, and they apply only within the stated facet, norm, and candidate-count limits. Proof-facing claims about algebraic parameter families therefore continue to use the separate exact SageMath certificates.

12 Availability of Code and Data

At the time of writing, the public development repository for this thesis is

<https://github.com/JoernStoehler/msc-math>.

This is a living Git repository rather than a frozen archival release, and its contents may change after submission. The claims below therefore describe material tracked in the repository at submission time; the URL is not a permanent snapshot or a DOI.

The strongest computation-bearing results have proof-facing packets in the tracked tree. The HKO packet in `experiments/hko-local-maximum/theorem/` contains the finite witness, the exact SageMath verifier, its recorded verification summary, and the Rust programs used to select the witness. Rust supplies finite choices; it is not trusted for theorem correctness. SageMath reconstructs the algebraic objects and checks the exact predicates stated in Section 7.4. Those predicates imply the theorem only together with the mathematical upper-function and slice argument given there.

The rotated-pentagon packet is stored under

`experiments/regular-products/
pentagon-rotation-formula-proof/`.

It contains the exact SageMath classifier and the recorded full-run transcript for all 3340 finite words left after the mathematical reduction. Here too, the program does not replace the proof: the enumeration theorem, the translation from checked signs to action comparisons, and the continuity argument remain mathematical inputs. The repository exposes the finite computation and its output for inspection, with SageMath’s exact arithmetic and root isolation inside the software trust boundary.

Selected non-proof evidence is also tracked as ordinary repository files. In particular, the product-capacity implementation audit retains its 88-row numerical packet, and the historical first-order experiment retains its 12-run finite-step panel. The bounded generated-candidate experiment retains 1675 rows selected before the systolic ratio was computed, the corresponding 1675 evaluated rows, its frozen configuration, and compact reports. These records support only

the finite experiments described in the corresponding chapters. They neither prove a population statement nor supply evidence for local maximality.

The repository is not a self-contained copy of every dataset used during the project. Its artifact registry identifies twelve external snapshots through twenty-one intended release paths, but those files are not present at those paths in a plain checkout at the time of writing. For example, the 14336-row random/product provenance table is tracked, while the source random/product datasets and the method-facing invariant table are not. Consequently, cloning the repository alone does not provide every input needed to repeat all empirical analyses, regenerate every figure, or reproduce every reported run. Packet-local instructions describe available producers and checks, but compatible software, dependencies, compute resources, and untracked inputs may still be required; timing and other environment-dependent fields need not reproduce byte for byte.

No frozen release, public Zenodo record, or DOI for this thesis is claimed here. Any archival deposit created later would be a separate, immutable object whose own public record, inventory, and hashes would have to establish its contents. Until such an object exists, the GitHub URL above provides access only to the living repository and to the finite material actually tracked there.

13 AI in the research process

The use of AI in this project changed as the available models became more capable. My experience was that they supplied increasingly useful research ideas and could carry out longer tasks with fewer interruptions. This changed what I could delegate. A request could describe a mathematical or experimental outcome and leave the agent to choose intermediate steps, write code, inspect the results and revise its approach. The separate disclosure records AI use; here I discuss what this experience suggests about organizing mathematical work with agents.

13.1 Delegating an incomplete proof strategy

The local maximality proof is the clearest example of how much of a research task could be delegated. I asked a GPT-5-series model for a proof that the HKO body is a local maximum. I suggested looking for several upper bounds and somehow showing that together they control a neighborhood after quotienting out the symmetries. This was only the overall shape of a strategy. I had not specified a finite certificate or worked out how the upper bounds should be constructed.

The model developed essentially the entire argument from that starting point. It identified a sufficient configuration of twenty-six upper bounds in the twenty-five transverse dimensions. An initial derivative-based approach had provided only twenty-five nonsingular cases. The model then used the weights β in the quadratic objective to construct feasible sections: an invertible 5×5 closure minor lets five weights depend smoothly on the facet parameters while the remaining weights stay fixed. This supplied upper bounds even where the initial approach failed. It also completed proof steps involving the volume and wrote the SageMath code for the exact computations.

The assignment was therefore a conjecture and an incomplete proof idea; the result was a developed proof together with computational verification code. The derivative conditions and the form of the finite certificate were part of the model's contribution, not conditions I supplied in the request. The useful specification was the mathematical outcome—a proof of local maximality—while the model had substantial freedom to discover what would establish it.

13.2 Matching the check to the claim

The different parts of this thesis require different evidence. In the HKO argument, arithmetic checks establish a finite statement about explicitly defined matrices and weights. The written proof still has to identify those matrices with derivatives of upper bounds for the geometric quantity. It also has to pass from labelled normals to a neighborhood of convex bodies. Omitting either connection would leave a mathematical gap even if every computed identity were correct.

For the search experiments, a high observed value answers a different question from a proof of a local maximum. The comparison of prescribed rotations with fresh sampling makes the distinction concrete: rotations improved several starting bodies, but fresh sampling reached the larger maximum in each of the four comparisons. An instruction merely to find an improvement could have returned a successful example while leaving the allocation question unanswered. Here the control determines what we learn from the computation.

The same issue occurs in more routine work. A retained code-repair example included a regression test that passed even when the original sign defect was restored. Testing the test against that defect revealed what an ordinary successful run did not establish. Such checks are useful because they target a particular possible mistake, rather than adding another undifferentiated approval of the result.

13.3 What still requires a reader

Mathematical exposition introduces a further difficulty. An agent can recover the meaning of a dense paragraph and judge it correct while a human reader has to stop repeatedly to decipher it. The intended reader of this thesis knows smooth symplectic geometry, but still needs an introduction to the nonsmooth variational problem and its finite reduction. A later definition does not remove the effort spent reading an earlier sentence that needed it.

This makes reading feedback valuable even when individual calculations have been checked. The useful feedback identifies where an explanation ceases to work: a missing implication, an unfamiliar object used before its definition, or a paragraph whose contribution remains unclear. Whether an agent can predict that feedback is itself an empirical question. Its ability to summarize a chapter does not establish that the chapter is comfortable to read. Nor does an automated check decide whether a mathematically correct side result deserves the space it occupies. These judgments concern the reader and the purpose of the thesis.

13.4 A connection with training on verifiable outcomes

There is a suggestive connection to reinforcement learning with verifiable rewards. Guo and collaborators trained DeepSeek-R1-Zero using rule-based accuracy and format rewards; mathematical answers and code test outcomes supplied checkable feedback. Their later DeepSeek-R1 combined reinforcement learning with supervised fine-tuning and also used model-based rewards for general tasks [GYZ+25]. This is evidence that outcome feedback can shape learned problem-solving behavior, not that every useful judgment has a reliable automatic reward.

An outcome supplied in an agent’s task context is not a training reward: the task does not thereby update the model’s parameters. The practical connection is that both require a criterion aligned with the intended achievement. In the HKO example, the finite conditions that could be checked were themselves discovered during the research task. They were not a test suite supplied in advance. My experience motivates using agents for work with clear outcomes and checking whether the returned evidence actually supports them. It does not isolate the contribution of a particular training method or measure the time this project would have taken without AI.

14 Conclusion

Is the HKO body a local maximum of the systolic ratio among convex bodies? Our main theorem gives a partial answer: it is a local maximum among ten-facet polytopes. Every sufficiently nearby body has ratio at most $(3 + \sqrt{5})/5$, and equality occurs precisely under translations, positive dilations and linear symplectic maps. The restriction to ten facets is essential to the coordinate neighborhood used in the proof.

The argument also gives a useful way to work at a singular optimization problem. One can continue feasible choices and use their values as upper bounds, without continuing stationary optimizers. At HKO, finitely many such bounds suffice: after the symmetry directions are removed, their derivatives force a strict decrease in every direction. The algebraic calculation establishes the finite conditions, while the geometric proof identifies the upper bounds and relates their coordinates to nearby bodies.

The empirical work approached the problem from another direction. It used geometric measurements and standard data-science methods to look for patterns that might lead back to mathematical explanations. We found patterns below systolic ratio one and confirmed some of their usefulness on fresh samples, but did not establish an extension to ratios above one. A geometric calculation expresses the ridge-area measurement on products through edge lengths and widths of the factors. It clarifies what the descriptor measures without explaining its association with capacity. Turning that association into a theorem remains unfinished, and the exploratory samples produced no new ratio above one.

We implemented, optimized and extended two methods for computing capacity: the Haim–Kislev quadratic program and the Chaidez–Hutchings flow graph. The expanded variational proof explains why the first problem is finite; the second method gives an alternative for cross-checking computations. For products of planar polygons, we proved reductions that make the finite search smaller. An exhaustive exact calculation then proves the separate theorem giving the capacity throughout the relative-rotation family of two regular pentagons, with the HKO body a global maximizer of the systolic ratio within that family.

Two directions remain particularly close to the questions studied here. First, the local theorem could be tested against deformations that change the facet count. Adding or splitting facets leaves the forty-dimensional chart used in the proof, so it requires a new argument. Second, the statistical associations invite geometric interpretation. A useful next conjecture would explain an association on a clearly specified family and survive attempts to break it by changing the sampling distribution.

The results therefore give a local theorem around the known counterexample, finite methods for investigating nearby and sampled polytopes, and measured successes and failures of several search ideas. The larger question of how capacity and volume constrain one another remains open beyond these families and neighborhoods.

A Data-Science Experiment Results

This appendix gives the optimizer steps and additional failure diagnostics for the local-search comparison in Section 8.9. Sampling definitions and statistical results are presented with their interpretation in Section 8.

A.1 Finite-budget local optimizer comparison

The run configuration labels the 64 starting bodies as held out. The saved hashes identify the seven configurations used, but do not establish that these bodies were absent from earlier development or tuning. A later development configuration reuses 16 starts. Their earlier

selection history is therefore unresolved; this does not show that the reported outcomes were used to select the configurations.

Every run used the same allocation rule: a nominal 1000 ms start-new-work cutoff for charged full-evaluator plus optimizer time and a cap of 128 full historical-evaluator calls. In-progress work could complete atomically past the cutoff, and methods could terminate early, so realized compute differs across runs. The 448 traces contain 28824 full calls. Using the historical numerical objective described in Section 8.9, four-anchor branch history has the largest comparison-block median best value, 0.9848, followed by transition prediction at 0.9816 and the finite-gap affine model at 0.9772. No run reaches a recorded numerical ratio of one.

The implemented optimizer steps. Except for the literal gradient, the methods remove the fifteen infinitesimal symmetry directions from the forty dual-normal coordinates. The local models and coordinate poll recompute this transverse space at their anchors; CMA-ES keeps its initial space. The literal gradient uses the ambient coordinates. Every outer proposal is checked by a fresh geometry reconstruction and the configured full orbit-word search. The implementations below are frozen comparison points; their parameters are not claimed to optimize the corresponding algorithm families.

Table 3: One outer iteration of each optimizer in the recorded comparison.

method	implemented step
literal and safeguarded branch gradients	Differentiate one currently minimizing branch. The literal method takes $a^+ = a + 0.01D_a s_\sigma(a)$ even after a decrease. The safeguarded method projects and normalizes this gradient, accepts only a fully evaluated improvement, and expands or halves its distance after acceptance or rejection.
finite-gap affine model	Keep the current branch gaps in a 10% action window, linearize their values, and maximize the minimum affine value on a local ball. Fully evaluate the proposal and use the same expand-or-halve distance rule.
four-anchor branch history	Union candidate words from the four most recent full searches, using a 30% action window and normalized beta allowance 0.3. At a nearby point, re-solve the named KKT branches, discard transition-infeasible and beta-inadmissible solutions, linearize the remaining values, and maximize their affine minimum. Take at most two internal moves before the next full evaluation.
history with transition prediction	Use the preceding method, but for a normalized internal move of length at least 0.08, also predict newly feasible transitions in the move direction and evaluate the words thereby generated.
CMA-ES	Fully evaluate a population of eight Gaussian perturbations, then update the mean, covariance, and normalized scale from their ranking.
coordinate direct search	Poll both signs of the 25 transverse basis axes. Move to the best improvement and double the radius, or halve the radius if no poll point improves.

The companion endpoint packet does not test local maximality. Signed polls along the transverse basis find an improving direction at seven of 16 one-second branch-history endpoints and two of the corresponding five-second endpoints. Separate validated continuation paths show that even an endpoint with no improving basis direction can retain appreciable ascent between the tested axes. These observations diagnose insufficient finite budgets; they do not estimate the distance to a local maximum.

Detailed HKO calibration. On 16 known perturbations along four development directions from HKO, the multi-branch continuation accepted no move at HKO and improved every

perturbation. One step recovered 93.4%–100% of the known evaluator gap and removed 77.9%–100% of the distance to HKO. In contrast, the gradient of one current minimizing branch recovered -22.1% – 41.4% of the gap.

Selected finite-step failure audit. At the eight highest four-anchor tuning endpoints, five admitted one validated improvement, by $1.57 \cdot 10^{-6}$ to $4.61 \cdot 10^{-5}$. Across the 80 multi-branch proposals, 52 decreased the recomputed evaluator value despite a represented target winner and positive affine prediction; 40 of these losses retained determinate, unchanged geometry. All 40 current-branch gradient proposals decreased.

The three-proposal audit compared two such failures with one positive control. At normalized radius 10^{-5} , the failures perturbed their named KKT matrices by 218 and 67 times the smallest base eigenvalue magnitude and crossed zero; the control had ratio 1.53, did not cross zero, and kept the correct sign. At the smallest tested radius, 10^{-8} , central differences approximated the implemented analytic derivatives with relative errors below 1% for the named action and 1.9% for the named branch ratio. Binary64 and exact-arithmetic geometry reconstruction agreed at all 39 audited points, and relative volume discrepancies were below 10^{-15} . For these selected cases the observed failure is finite-distance nonlinearity of an ill-conditioned KKT branch.

These diagnostics use selected controls rather than population samples. They do not provide a convergence theorem, a population error rate, or a universal conditioning cutoff.

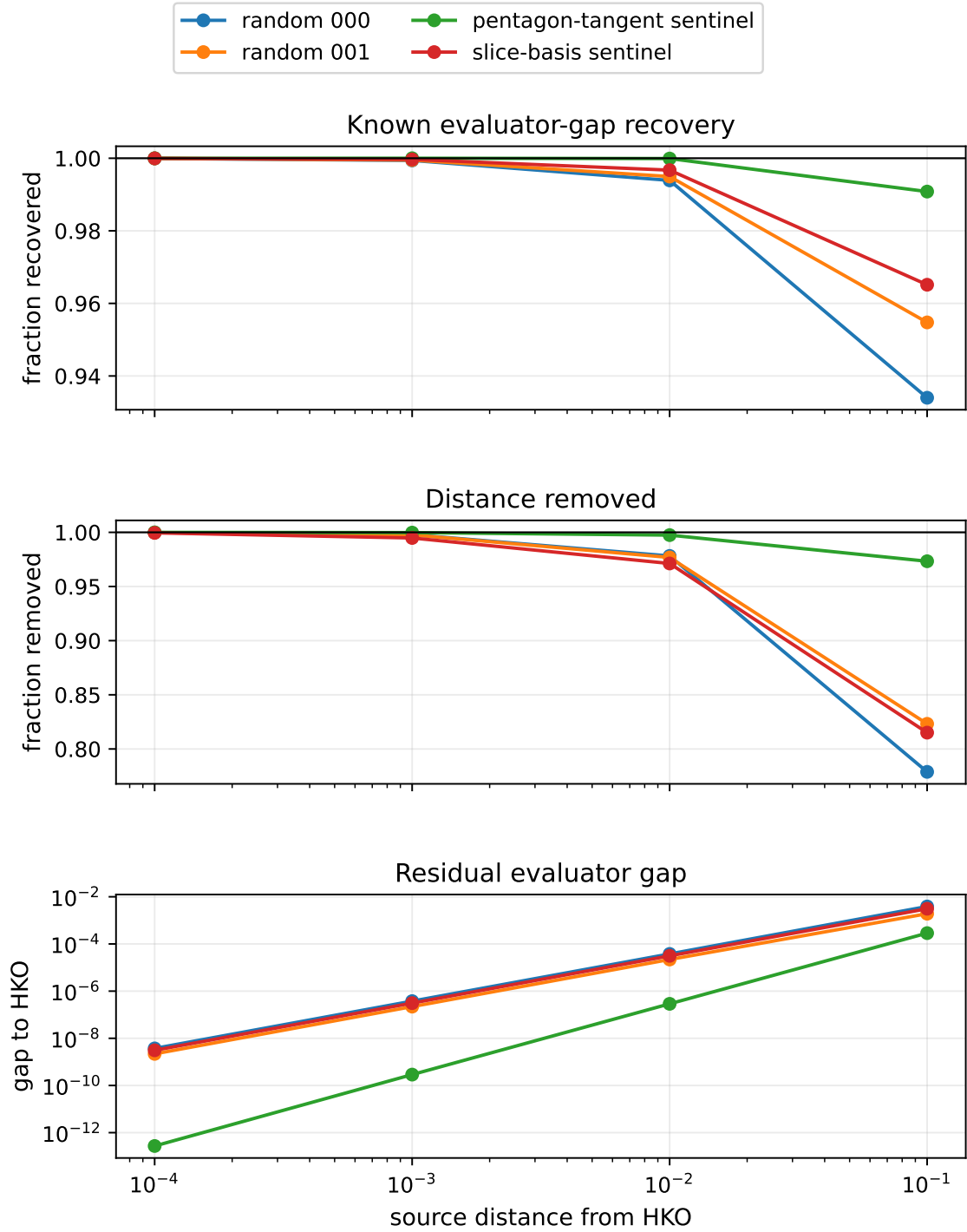


Figure 11: One-step calibration on known perturbations of HKO. From top to bottom, the panels show recovered evaluator gap, removed distance, and residual evaluator gap.

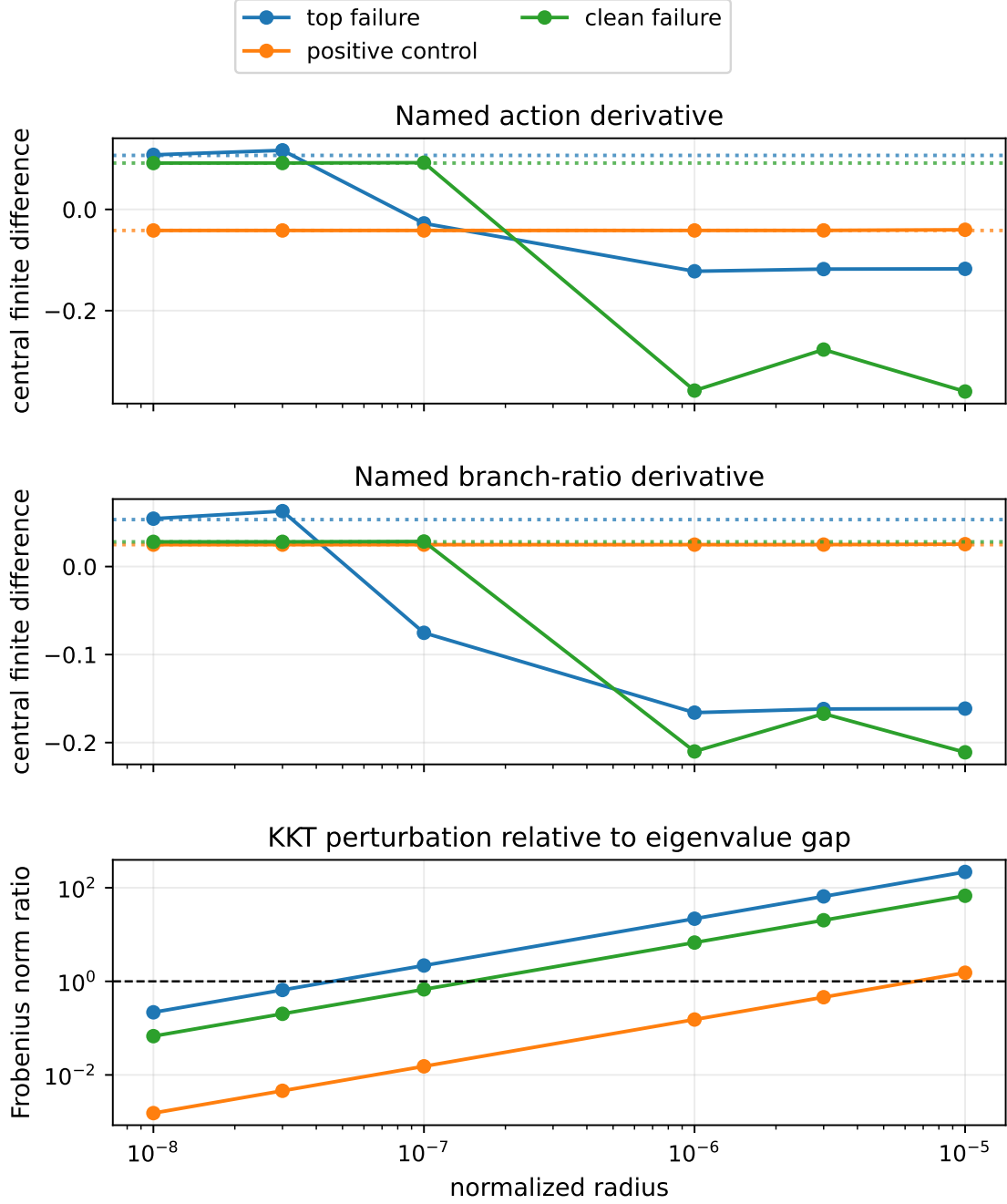


Figure 12: Finite differences for two failed affine proposals and one positive control. Prediction changes regime when the KKT perturbation becomes large relative to the smallest source eigenvalue magnitude (bottom). Dotted lines in the top two panels are the analytic derivatives.

B Pentagon computation: source and exploratory plots

B.1 How Sage implements the finite predicates

Section 9 reduces the rotation theorem to exact finite predicates. This appendix makes the correspondence between those predicates and the Sage source directly inspectable. The excerpts are copied in cleaned form from the executable identified in that section: diagnostics, return-value bookkeeping, and repeated type conversions are omitted, short local names are used, and long expressions are line-broken. Named utility constructors and diagnostics are not repeated. The source identified by the digest in the retained transcript remains the runnable owner.

Facet duals and the fixed-word quadratic program

The first excerpt constructs the ten dual vertices in the coordinate order (q_1, q_2, p_1, p_2) , implements ω_0 , and forms the KKT system. The last four rows impose $\sum_i \beta_i a_i = 0$, while the final row imposes $\sum_i \beta_i = 1$. The function `q_value` is exactly the ordered quadratic form Q_σ used above.

```
def cos_theta():
    return lift(1 - t**2) / lift(1 + t**2)

def sin_theta():
    return lift(2 * t) / lift(1 + t**2)

def rotate(point):
    x, y = point
    c, s = cos_theta(), sin_theta()
    return (c * x - s * y, s * x + c * y)

def pentagon_normals():
    return [(lift(cos_units(5 + 4*k)),
              lift(sin_units(5 + 4*k))) for k in range(5)]

def dual_vertices():
    height = lift(cos_units(2)) # cos(pi/5)
    normals = pentagon_normals()
    vertices = []
    for x, y in normals:
        vertices.append((x / height, y / height, 0, 0))
    for normal in normals:
        x, y = rotate(normal)
        vertices.append((0, 0, x / height, y / height))
    return vertices

def omega(u, v):
    return (u[0] * v[2] - u[2] * v[0]
            + u[1] * v[3] - u[3] * v[1])

def kkt_matrix(sigma):
    m = len(sigma)
    mat = Matrix(F, m + 5, m + 5, 0)
    rhs = vector(F, [0] * (m + 5))
    for i in range(m):
        for j in range(i + 1, m):
            value = omega(DUALS[sigma[i]], DUALS[sigma[j]])
            mat[i, j] = value
            mat[j, i] = value
```

```

for i in range(m):
    for d in range(4):
        value = DUALS[sigma[i]][d]
        mat[i, m + d] = value
        mat[m + d, i] = value
    mat[i, m + 4] = 1
    mat[m + 4, i] = 1
rhs[m + 4] = 1
return mat, rhs

def q_value(sigma, beta):
    q = 0
    for i in range(1, len(sigma)):
        for j in range(i):
            q += (beta[i] * beta[j]
                  * omega(DUALS[sigma[j]], DUALS[sigma[i]]))
    return reduced(q)

```

The finite block family

The next excerpt is the implementation of $([Q \mid QQ][P \mid PP])^k$, $k = 2, 3$. A block is a singleton or an ordered adjacent pair. The selections are disjoint. The first selected q -block fixes the cyclic start, while the remaining relative orders are all enumerated.

```

def adjacent_same_factor(a, b):
    return (a - b) % 5 in {1, 4}

def blocks(facets):
    result = [(i,) for i in facets]
    for i, j in combinations(facets, 2):
        if adjacent_same_factor(i % 5, j % 5):
            result.append((i, j))
            result.append((j, i))
    return result

def non_overlapping(selection):
    used = set()
    for block in selection:
        if used & set(block):
            return False
        used.update(block)
    return True

def non_overlapping_selections(block_list, k):
    for selection in combinations(block_list, k):
        if non_overlapping(selection):
            yield selection

def enumerate_k_bounce_sigmas(k):
    q_blocks = blocks(Q_FACETS)
    p_blocks = blocks(P_FACETS)
    for qs in non_overlapping_selections(q_blocks, k):
        for ps in non_overlapping_selections(p_blocks, k):
            for q_order in permutations(range(k - 1)):
                for p_order in permutations(range(k)):
                    sigma = list(qs[0]) + list(ps[p_order[0]])
                    for r in range(1, k):
                        sigma.extend(qs[1 + q_order[r - 1]])

```

```

        sigma.extend(ps[p_order[r]])
    yield tuple(sigma)

```

Exact sign cells and transition pruning

For a rational function in $K(t)$, Sage isolates every numerator and denominator root in the open half-domain. The resulting cells have constant sign, so one exact algebraic midpoint decides each cell. The endpoint label retained in each certificate is diagnostic only; endpoints are handled by the continuity argument above. Transition pruning uses the open-cell status and checks that every mixed transition has no interior sign cut.

```

def real_roots_in_open_half_domain(poly):
    roots = []
    for root, multiplicity in poly.change_ring(AA).roots():
        root = AA(root)
        if AA(0) < root < HALF_DOMAIN_ENDPOINT:
            roots.append(root)
    return sorted(set(roots))

def sign_certificate(expr):
    expr = reduced(expr)
    numerator = expr.numerator()
    denominator = expr.denominator()
    if numerator == 0:
        return SignCertificate("zero", 0, "zero")
    cuts = sorted(set(
        real_roots_in_open_half_domain(numerator)
        + real_roots_in_open_half_domain(denominator)))
    points = [AA(0)] + cuts + [HALF_DOMAIN_ENDPOINT]
    num_aa = aa_polynomial(numerator)
    den_aa = aa_polynomial(denominator)
    signs = []
    for left, right in zip(points, points[1:]):
        sample = (left + right) / 2
        signs.append((num_aa(sample) / den_aa(sample)).sign())
    endpoint = endpoint_label(num_aa, den_aa)
    if all(sign > 0 for sign in signs):
        return SignCertificate("positive_open", len(cuts), endpoint)
    if all(sign < 0 for sign in signs):
        return SignCertificate("negative_open", len(cuts), endpoint)
    return SignCertificate("mixed_open", len(cuts), endpoint)

def open_domain_cells(expressions):
    cuts = []
    for expr in expressions:
        expr = reduced(expr)
        cuts += real_roots_in_open_half_domain(expr.numerator())
        cuts += real_roots_in_open_half_domain(expr.denominator())
    points = [AA(0)] + sorted(set(cuts))
    points += [HALF_DOMAIN_ENDPOINT]
    return [(a + b) / 2 for a, b in zip(points, points[1:])]

def facet_intersection_nonempty(i, j):
    if i == j:
        return True
    if (i < 5) != (j < 5):
        return True

```

```

    return adjacent_same_factor(i % 5, j % 5)

def transition_table_open():
    table = {}
    for i in range(10):
        for j in range(10):
            cert = sign_certificate(omega(DUALS[i], DUALS[j]))
            if (i < 5) == (j < 5):
                assert cert.status == "zero"
                table[i, j] = facet_intersection_nonempty(i, j)
            else:
                assert cert.status in {
                    "positive_open", "negative_open"
                }
                assert cert.interior_cut_count == 0
                table[i, j] = (facet_intersection_nonempty(i, j)
                               and cert.status == "positive_open")

    return table

def transition_pruned_sigmas_open():
    table = transition_table_open()
    sigmas = []
    for k in (2, 3):
        for sigma in enumerate_k_bounce_sigmas(k):
            edges = zip(sigma, sigma[1:] + sigma[:1])
            if all(table[i, j] for i, j in edges):
                sigmas.append(sigma)
    return sigmas

```

Fail-closed branch classification

The classifier solves the displayed KKT equations over $K(t)$. It accepts an identically zero action gap as a tie; otherwise, on every cell where all weights and Q_σ are positive, it requires a strictly positive action gap. A singular or nonpositive-gap case not covered by a proved exit is sent to manual review. The full loop rejects that status. For readability, `status(...)` below abbreviates the executable's result constructor.

```

ACCEPTED_STATUSES = {
    "no_kkt_solution", "singular_kkt_forced_zero_beta",
    "zero_q_identity", "zero_gap_identity",
    "not_feasible_on_open_domain",
    "strict_gap_positive_on_feasible_open_domain",
}

def forced_zero_weight(beta, kernel):
    for i in range(len(beta)):
        if beta[i] == 0 and all(reduced(v[i]) == 0 for v in kernel):
            return True
    return False

def classify_sigma(sigma, min_action):
    mat, rhs = kkt_matrix(sigma)
    try:
        solution = mat.solve_right(rhs)
    except ValueError:
        augmented = mat.augment(rhs.column())
        if augmented.rank() > mat.rank():

```

```

        return status("no_kkt_solution")
    return status("requires_manual_review")

beta = [reduced(solution[i]) for i in range(len(sigma))]
q = q_value(sigma, beta)
kernel = None
if q == 0:
    kernel = mat.right_kernel().basis()
    if kernel:
        if forced_zero_weight(beta, kernel):
            return status("singular_kkt_forced_zero_beta")
        return status("requires_manual_review")
    return status("zero_q_identity")

gap = reduced(1 / (2 * q) - min_action)
if gap == 0:
    return status("zero_gap_identity")

feasible_cells = 0
for sample in open_domain_cells(tuple(beta) + (q, gap)):
    feasible = all(sign_at(b, sample) > 0 for b in beta)
    feasible = feasible and sign_at(q, sample) > 0
    if feasible:
        feasible_cells += 1
        if sign_at(gap, sample) <= 0:
            return status("requires_manual_review")

if feasible_cells == 0:
    if kernel is None:
        kernel = mat.right_kernel().basis()
    if kernel:
        if forced_zero_weight(beta, kernel):
            return status("singular_kkt_forced_zero_beta")
        return status("requires_manual_review")
    return status("not_feasible_on_open_domain")
return status("strict_gap_positive_on_feasible_open_domain")

for sigma in transition_pruned_sigmas_open():
    classification = classify_sigma(sigma, minimum_action())
    assert classification.status in ACCEPTED_STATUSES

```

In the runnable source every unresolved singular case becomes manual review. Hence the last assertion passes only when every word reaches one of the six exits in Table 2.

This comparison of formulas and source is the theorem-specific implementation audit. At the software-foundation level, the calculation relies on SageMath to perform its standard exact number-field arithmetic, linear algebra, and real-root isolation correctly. SageMath [The25] is routinely used for such exact computer-algebra calculations; verifying SageMath itself is outside the scope of this thesis.

B.2 Sampled stationary branches

Figure 13 places this example competitor in the larger stationary-branch problem. An independent floating-point producer freezes 3340 raw two- and three-block words at the generic angle $\theta = \pi/20$, then solves the same words at 73 angles from 0 to $\pi/10$. Its word count agrees with the exact certificate's count; the two word sets were not compared independently. This is an empirical explanation of the finite classification, not another input to the proof.

Many raw words give numerically indistinguishable data on this grid. To avoid drawing the same curve on top of itself hundreds of times, the plot collapses two words only when their statuses agree at all 73 angles and their sampled actions agree after rounding to ten decimal places. It then draws one complete 73-angle sequence for that group. Thus every curve follows one word across the whole grid; it is never assembled by selecting different words at different angles.

Sampled enumerated KKT-branch landscape

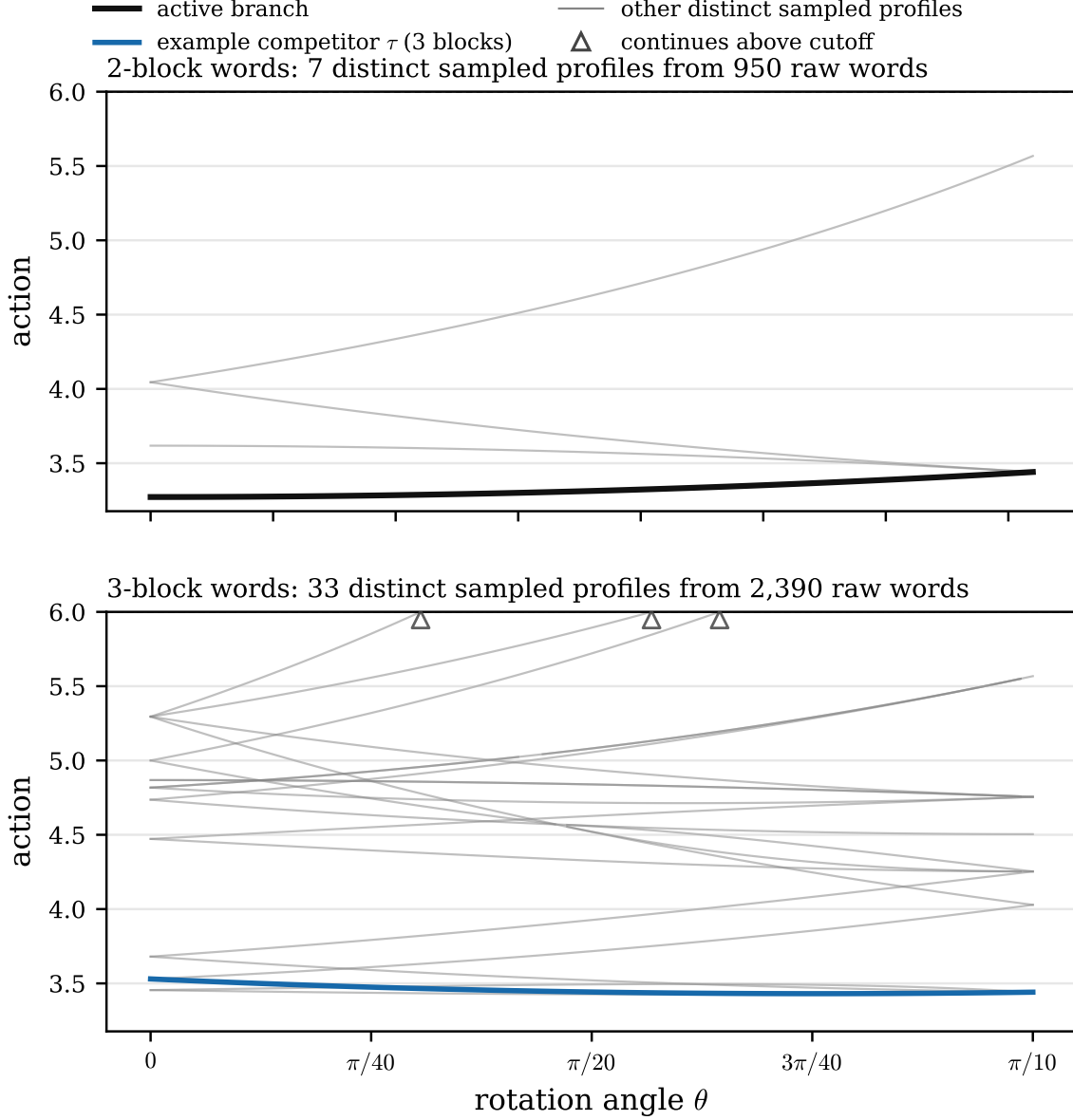


Figure 13: Sampled enumerated KKT-branch landscape. Black marks the active branch and blue the example competitor τ , whose formula was calculated above. Each gray curve is one of the other distinct sampled profiles defined in the preceding paragraph; the panel headings report the total number of distinct sampled profiles and the number of raw words they summarize. A profile with no admissible sample produces no visible curve. Only consecutive admissible samples are joined. Triangles just below the upper boundary mark curves continuing above the fixed cutoff 6; a gap means that the corresponding word was not numerically admissible at the missing samples. The grid neither certifies feasibility sets nor proves the lower envelope.

Admissibility over the full 73-angle grid			
sampled presence	two-block	three-block	all
no admissible sample	795	1820	2615
admissible at every sample	50	155	205
one contiguous sampled run	105	414	519
multiple sampled runs	0	1	1

Status at the right endpoint $\theta = \pi/10$			
sampled status	two-block	three-block	all
admissible	110	425	535
numerically inadmissible	770	1746	2516
indeterminate	70	219	289
solve failure	0	0	0

Table 4: Raw-word classification on the sampled grid. The upper part distinguishes words that are nowhere admissible on the grid, admissible throughout, or admissible on one or more sampled runs; these classes are not claims about feasibility-set topology. The sole multiple-run word has one numerically inadmissible grid point separating its two sampled admissible runs. The lower part records the producer’s four-way status at $\theta = \pi/10$. No solve failure occurred at any sampled angle.

C HKO computation: an annotated exact program

C.1 Executable exact verification of the HKO bounds

This appendix isolates the finite certificate used in Lemma 7.4. The listings below form a complete SageMath program. Facet labels and positions in each word are displayed one-based; the executable converts them to zero-based indices at the section boundary.

The geometric lemmas in the main text establish that these ten labelled dual vertices form the relevant chart, that volume is smooth there, and that each positive feasible section supplies a touching upper bound. The finite CAS task is to instantiate 26 such sections, differentiate their upper bounds, and check the quotient certificate.

Exact field, geometry, and quadratic form. The first source block defines the ordered number field, the ten dual vertices, the symplectic form, the closure matrix, and the quadratic form.

```
#!/usr/bin/env sage -python
from sage.all import *
R = PolynomialRing(QQ, "x")
x = R.gen()
f = x**4 - 10*x**2 + 5
F = NumberField(f, 't', embedding=AA.polynomial_root(f, RIF(0, 1)))
t = F.gen(); s = (5 - t**2)/2; al = (3 - s)/2; be = t*(1 + s)/2; d = s - 1
a = [vector(F, v) for v in [(1, t, 0, 0), (-al, be, 0, 0), (-d, 0, 0, 0), (-al, -be, 0, 0), (1, -t, 0, 0), (0, 0, t, -1),
    (0, 0, be, al), (0, 0, 0, d), (0, 0, -be, al), (0, 0, -t, -1)]]
def om(u, v): return u[0]*v[2] + u[1]*v[3] - u[2]*v[0] - u[3]*v[1]
def C(w): return matrix(F, [[a[i][j] for i in w] for j in range(4)] + [[F(1)]*len(w)])
def q(w, b): return sum(b[i]*b[j]*om(a[w[j]], a[w[i]]) for i in range(1, len(w)) for j in range(i)
    )
```

All 26 exact assignments. The following is the complete `rows` object, including every word, solved coordinate set, fixed coordinate set, and exact fixed value. Values written with the caret character are strings parsed by Sage's exact evaluator; they are data, not Python exponentiation.

```
# (one-based word, solved positions I, fixed positions J, exact fixed values)
rows = [
    ([1, 2, 8, 4, 10, 6], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([1, 2, 8, 7, 4, 10], [1, 2, 3, 4, 5], [6], ['-t^2/20+1/4']),
    ([2, 3, 8, 9, 5, 4, 6], [1, 2, 3, 4, 5], [6, 7], ['6909830056250513/10^17', '-t^2/20+1/4']),
    ([1, 8, 7, 3, 4, 10], [1, 2, 3, 4, 5], [6], ['-t^2/20+1/4']),
    ([2, 9, 8, 5, 4, 6], [1, 2, 3, 4, 5], [6], ['-t^2/20+1/4']),
    ([2, 8, 4, 5, 6, 10], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([1, 6, 2, 8, 5, 4, 10], [1, 2, 3, 4, 5], [6, 7], ['36180339887499/(2*10^14)', 't^2/40+1/8']),
    ([1, 7, 8, 4, 3, 9, 10], [1, 2, 3, 4, 5], [6, 7], ['(8090169943749473*t^2+23368810393753689)/(4*10^17)', '18090169943749473/10^17']),
    ([1, 5, 7, 6, 3, 10, 9], [1, 2, 3, 4, 5], [6, 7], ['(4045084971874731*t^2+11684405196876883)/(2*10^17)', '9045084971874731/(5*10^16)']),
    ([2, 8, 4, 3, 9, 5, 6], [1, 2, 3, 4, 5], [6, 7], ['1809016994374947/10^16', '-t^2/20+1/4']),
    ([2, 3, 9, 8, 4, 5, 6], [1, 2, 3, 4, 5], [6, 7], ['2261271242968689/(125*10^14)', '-t^2/20+1/4']),
    ([2, 8, 9, 5, 7, 6, 3], [1, 2, 3, 4, 5], [6, 7], ['18090169943749473/10^17', 't^2/40+1/8']),
    ([1, 7, 8, 4, 10, 2], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([1, 7, 4, 3, 9, 10], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([2, 9, 5, 6, 7, 3], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([2, 8, 3, 9, 5, 6], [1, 2, 3, 4, 5], [6], ['-t^2/20+1/4']),
    ([2, 8, 4, 5, 10, 6], [1, 2, 3, 4, 5], [6], ['t^2/40+1/8']),
    ([1, 7, 3, 2, 8, 4, 10], [1, 2, 3, 4, 5], [6, 7], ['18090169943749507/10^17', '-t^2/20+1/4']),
    ([1, 5, 6, 7, 3, 2, 9], [1, 2, 3, 4, 5], [6, 7], ['6909830056250547/10^17', '-t^2/20+1/4']),
    ([1, 7, 3, 9, 5, 6, 2], [1, 2, 3, 4, 5], [6, 7], ['t^2/40+1/8', '863728757031319/(125*10^14)']),
    ([1, 8, 5, 4, 6, 10, 2], [1, 2, 3, 4, 5], [6, 7], ['t^2/40+1/8', '7463176480767021/(5*10^16)']),
]
```

```
([1,6,2,8,7,4,10],[1,2,3,4,5],[6,7],[-t^2/20+1/4','1809016994374947/10^16']),
([1,5,7,4,3,9,10],[1,2,3,4,6],[5,7],[9045084971874741/(5*10^16)',t^2/40+1/8']),
([1,7,3,4,10,9,5],[1,2,3,4,5],[6,7],[t^2/40+1/8','3454915028125259/(5*10^16)']),
([1,7,6,3,10,9,5],[1,2,3,4,5],[6,7],[452254248593737/(25*10^14)',t^2/40+1/8']),
([2,8,9,5,4,6,3],[1,2,3,4,5],[6,7],[-t^2/20+1/4','12028993467659169/10^17'])]
```

Feasible sections and derivative rows. The next block is the actual section solver. It explicitly converts the one-based assignment data, solves $C_I \beta_I = e - C_J u$, restores $\beta_J = u$, and checks the minor's invertibility, closure, normalization, and positivity. The constants V , A , and vr are respectively the base volume, the known HKO capacity, and the volume derivative from Lemma 7.3. The derivative block computes $D\beta$, then differentiates Q and forms $r_j = DU_j(a_0)$ by the product rule for $U_j = A_j^2/(2V)$.

```
def val(z): return F(sage_eval(z, locals={'t': t}))
def section(W):
    w,I,J,u=W; w=[i-1 for i in w]; I=[i-1 for i in I]; J=[i-1 for i in J]
    assert len(set(w))==len(w) and all(0<=i<10 for i in w); assert sorted(I+J)==list(range(len(w)))
    cm=C(w); assert cm[:,I].det()!=0; U=vector(F,[val(z) for z in u]); b=[None]*len(w)
    x=cm[:,I].solve_right(vector(F,[0,0,0,0,1])-cm[:,J]*U)
    for k,i in enumerate(I): b[i]=x[k]
    for k,j in enumerate(J): b[j]=U[k]
    b=vector(F,b); assert cm*b==vector(F,[0,0,0,0,1]) and all(z>0 for z in b)
    return w,I,J,b
V=QQ(25)*(5+s)/QQ(32); A=5*t-t**3/qq(2); qm=1/(2*A)
vr=vector(F,sum((list(-(QQ(25)/QQ(32)+QQ(5)*s/qq(16))*v) for v in a),[]))
def drow(w,I,J,b):
    cm=C(w); db=[[F(0)]]*40 for _ in w]
    for z in range(40):
        facet,coord=divmod(z,4); rhs=vector(F,[0]*5)
        if facet in w: rhs[coord]=-b[w.index(facet)]
        xi=cm[:,I].solve_right(rhs); assert cm[:,I]*xi==rhs
        for k,i in enumerate(I): db[i][z]=xi[k]
    for j in J: assert all(x==0 for x in db[j])
    dq=[]
    for z in range(40):
        facet,coord=divmod(z,4); u=vector(F,[1 if k==coord else 0 for k in range(4)]); x=F(0)
        for i in range(1,len(w)):
            for j in range(i):
                x+=(db[i][z]*b[j]+b[i]*db[j][z])*om(a[w[j]],a[w[i]])
                if w[j]==facet: x+=b[i]*b[j]*om(u,a[w[i]])
                if w[i]==facet: x+=b[i]*b[j]*om(a[w[j]],u)
            dq.append(x)
    return vector(F,[A/V*(-x/(2*qm**2))-A**2/(2*V**2)*vr[i] for i,x in enumerate(dq)])
```

Symmetry and finite conclusion. The final source block constructs the four translations, dilation, and ten $\mathfrak{sp}(4)$ columns, then checks symmetry annihilation, row rank 25, and a strictly positive normalized left-kernel relation. The check $q(w,b)==qm$ establishes that each section gives the known HKO action at the base point, so its upper bound touches the systolic ratio there. Thus the explicitly assigned rows, together with the computed rr and lam , witness the existence assertions of Lemma 7.4. The rank and positive relation supply the hypotheses of Lemma 7.5.

```
# Build the fifteen symmetry columns, then verify all rows, rank, and kernel.
Jm=matrix(F,[[0,0,-1,0],[0,0,0,-1],[1,0,0,0],[0,1,0,0]])
T=[vector(F,sum((-v[k]*v) for v in a),[])) for k in range(4)]
T += [vector(F,sum((list(-v) for v in a),[]))]
G=[matrix(F,g) for g in [
    [[1,0,0,0],[0,0,0,0],[0,0,-1,0],[0,0,0,0]], [[0,1,0,0],[0,0,0,0],[0,0,0,0],[0,0,-1,0]],
    [[0,0,0,0],[1,0,0,0],[0,0,0,-1],[0,0,0,0]], [[0,0,0,0],[0,1,0,0],[0,0,0,0],[0,0,0,-1]],
```

```

[[0,0,1,0],[0,0,0,0],[0,0,0,0],[0,0,0,0]],[[0,0,0,1],[0,0,1,0],[0,0,0,0],[0,0,0,0]],
[[0,0,0,0],[0,0,0,1],[0,0,0,0],[0,0,0,0]],[[0,0,0,0],[0,0,0,0],[1,0,0,0],[0,0,0,0]],
[[0,0,0,0],[0,0,0,0],[0,1,0,0],[1,0,0,0]],[[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,1,0,0]]]
for X in G:
    assert X.transpose()*Jm+Jm*X==0
    T.append(vector(F,sum((list(-X.transpose()*v) for v in a),[])))
rr=[]
assert len(rows)==26
for W in rows:
    w,I,J,b=section(W); assert q(w,b)==qm; rr.append(drow(w,I,J,b))
M=matrix(F,[list(r) for r in rr]); assert M.rank()==25 and matrix(F,[list(c) for c in T]).rank
()==15
for r in rr:
    for c in T: assert r.dot_product(c)==0
kernel=M.transpose().right_kernel().basis(); assert len(kernel)==1
lam=kernel[0]
if all(x<0 for x in lam): lam=-lam
assert all(x>0 for x in lam); lam=lam/sum(lam); assert lam*M==0
print('HK0 appendix core passed: rows=26 rank=25 symmetry-rank=15')

```

Running the included program with `sage -python` returns `HK0 appendix core passed: rows=26 rank=25 symmetry-rank=15`. These checks establish the finite hypotheses used in the main proof. The geometric chart and volume formula are supplied by Lemmas 7.2 and 7.3; Lemmas 7.5–7.7 then pass from the computed derivatives to local maximality.

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